Bioprocess Engineering Principles

Pauline M. Doran

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Preface

Recent developments in genetic and molecular biology have excited world-wide interest in biotechnology. The ability to manipulate DNA has already changed our perceptions of medicine, agriculture and environmental management. Scientific breakthroughs in gene expression, protein engineering and cell fusion are being translated by a strengthening biotechnology industry into revolutionary new products and services.

Many a student has been enticed by the promise of biotechnology and the excitement of being near the cutting edge of scientific advancement. However, the value of biotechnology is more likely to be assessed by business, government and consumers alike in terms of commercial applications, impact on the marketplace and financial success. Graduates trained in molecular biology and cell manipulation soon realise that these techniques are only part of the complete picture; bringing about the full benefits of biotechnology requires substantial manufacturing capability involving large-scale processing of biological material. For the most part, chemical engineers have assumed the responsibility for bioprocess development. However, increasingly, biotechnologists are being employed by companies to work in co-operation with biochemical engineers to achieve pragmatic commercial goals. Yet, while aspects of biochemistry, microbiology and molecular genetics have for many years been included in chemical-engineering curricula, there has been relatively little attempt to teach biotechnologists even those qualitative aspects of engineering applicable to process design.

The primary aim of this book is to present the principles of bioprocess engineering in a way that is accessible to biological scientists. It does not seek to make biologists into bioprocess engineers, but to expose them to engineering concepts and ways of thinking. The material included in the book has been used to teach graduate students with diverse backgrounds in biology, chemistry and medical science. While several excellent texts on bioprocess engineering are currently available, these generally assume the reader already has engineering training. On the other hand, standard chemical-engineering texts do not often consider examples from bioprocessing and are written almost exclusively with the petroleum and chemical industries in mind. There was a need for a textbook which explains the engineering approach to process analysis while providing worked examples and problems about biological systems. In this book, more than 170 problems and calculations encompass a wide range of bioprocess applications involving recombinant cells, plant- and animal-cell cultures and immobilised biocatalysts as well as traditional fermentation systems. It is assumed that the reader has an adequate background in biology.

One of the biggest challenges in preparing the text was determining the appropriate level of mathematics. In general, biologists do not often encounter detailed mathematical analysis. However, as a great deal of engineering involves formulation and solution of mathematical models, and many important conclusions about process behaviour are best explained using mathematical relationships, it is neither easy nor desirable to eliminate all mathematics from a textbook such as this. Mathematical treatment is necessary to show how design equations depend on crucial assumptions; in other cases the equations are so simple and their application so useful that non-engineering scientists should be familiar with them. Derivation of most mathematical models is fully explained in an attempt to counter the tendency of many students to memorise rather than understand the meaning of equations. Nevertheless, in fitting with its principal aim, much more of this book is descriptive compared with standard chemical-engineering texts.

The chapters are organised around broad engineering subdisciplines such as mass and energy balances, fluid dynamics, transport phenomena and reaction theory, rather than around particular applications of bioprocessing. That the same fundamental engineering principle can be readily applied to a variety of bioprocess industries is illustrated in the worked examples and problems. Although this textbook is written primarily for senior students and graduates of biotechnology, it should also be useful in food-, environmental- and civil-engineering courses. Because the qualitative treatment of selected topics is at a relatively advanced level, the book is appropriate for chemical-engineering graduates, undergraduates and industrial practitioners.

I would like to acknowledge several colleagues whose advice I sought at various stages of manuscript preparation. Jay Bailey, Russell Cail, David DiBiasio, Noel Dunn and Peter Rogers each reviewed sections of the text. Sections 3.3 and 11.2 on analysis of experimental data owe much to Robert J. Hall who provided lecture notes on this topic. Thanks are also due to Jacqui Quennell whose computer drawing skills are evident in most of the book's illustrations.

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January 1994

Part 1
Introduction

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1

Bioprocess Development: An Interdisciplinary Challenge

Bioprocessing is an essential part of many food, chemical and pharmaceutical industries. Bioprocess operations make use of microbial, animal and plant cells and components of cells such as enzymes to manufacture new products and destroy harmful wastes.

Use of microorganisms to transform biological materials for production of fermented foods has its origins in antiquity. Since then, bioprocesses have been developed for an enormous range of commercial products, from relatively cheap materials such as industrial alcohol and organic solvents, to expensive specialty chemicals such as antibiotics, therapeutic proteins and vaccines. Industrially-useful enzymes and living cells such as bakers' and brewers' yeast are also commercial products of bioprocessing.

Table 1.1 gives examples of bioprocesses employing whole cells. Typical organisms used and the approximate market size for the products are also listed. The table is by no means exhaustive; not included are processes for wastewater treatment, bioremediation, microbial mineral recovery and manufacture of traditional foods and beverages such as yoghurt, bread, vinegar, soy sauce, beer and wine. Industrial processes employing enzymes are also not listed in Table 1.1; these include brewing, baking, confectionery manufacture, fruit-juice clarification and antibiotic transformation. Large quantities of enzymes are used commercially to convert starch into fermentable sugars which serve as starting materials for other bioprocesses.

Our ability to harness the capabilities of cells and enzymes has been closely related to advancements in microbiology, biochemistry and cell physiology. Knowledge in these areas is expanding rapidly; tools of modern biotechnology such as recombinant DNA, gene probes, cell fusion and tissue culture offer new opportunities to develop novel products or improve bioprocessing methods. Visions of sophisticated medicines, cultured human tissues and organs, biochips for new-age computers, environmentally-compatible pesticides and powerful pollution-degrading microbes herald a revolution in the role of biology in industry.

Although new products and processes can be conceived and partially developed in the laboratory, bringing modern biotechnology to industrial fruition requires engineering skills and know-how. Biological systems can be complex and difficult to control; nevertheless, they obey the laws of chemistry and physics and are therefore amenable to engineering analysis. Substantial engineering input is essential in many aspects of bioprocessing, including design and operation of bioreactors, sterilisers and product-recovery equipment, development of systems for process automation and control, and efficient and safe layout of fermentation factories. The subject of this book, *bioprocess engineering*, is the study of engineering principles applied to processes involving cell or enzyme catalysts.

1.1 Steps in Bioprocess Development: A Typical New Product From Recombinant DNA

The interdisciplinary nature of bioprocessing is evident if we look at the stages of development required for a complete industrial process. As an example, consider manufacture of a new recombinant-DNA-derived product such as insulin, growth hormone or interferon. As shown in Figure 1.1, several steps are required to convert the idea of the product into commercial reality; these stages involve different types of scientific expertise.

The first stages of bioprocess development (Steps 1–11) are concerned with genetic manipulation of the host organism; in this case, a gene from animal DNA is cloned into *Escherichia coli*. Genetic engineering is done in laboratories on a small scale by scientists trained in molecular biology and biochemistry. Tools of the trade include Petri dishes, micropipettes, microcentrifuges, nano- or microgram quantities of restriction enzymes, and electrophoresis gels for DNA and protein fractionation. In terms of bioprocess development, parameters of major importance are stability of the constructed strains and level of expression of the desired product.

After cloning, the growth and production characteristics of

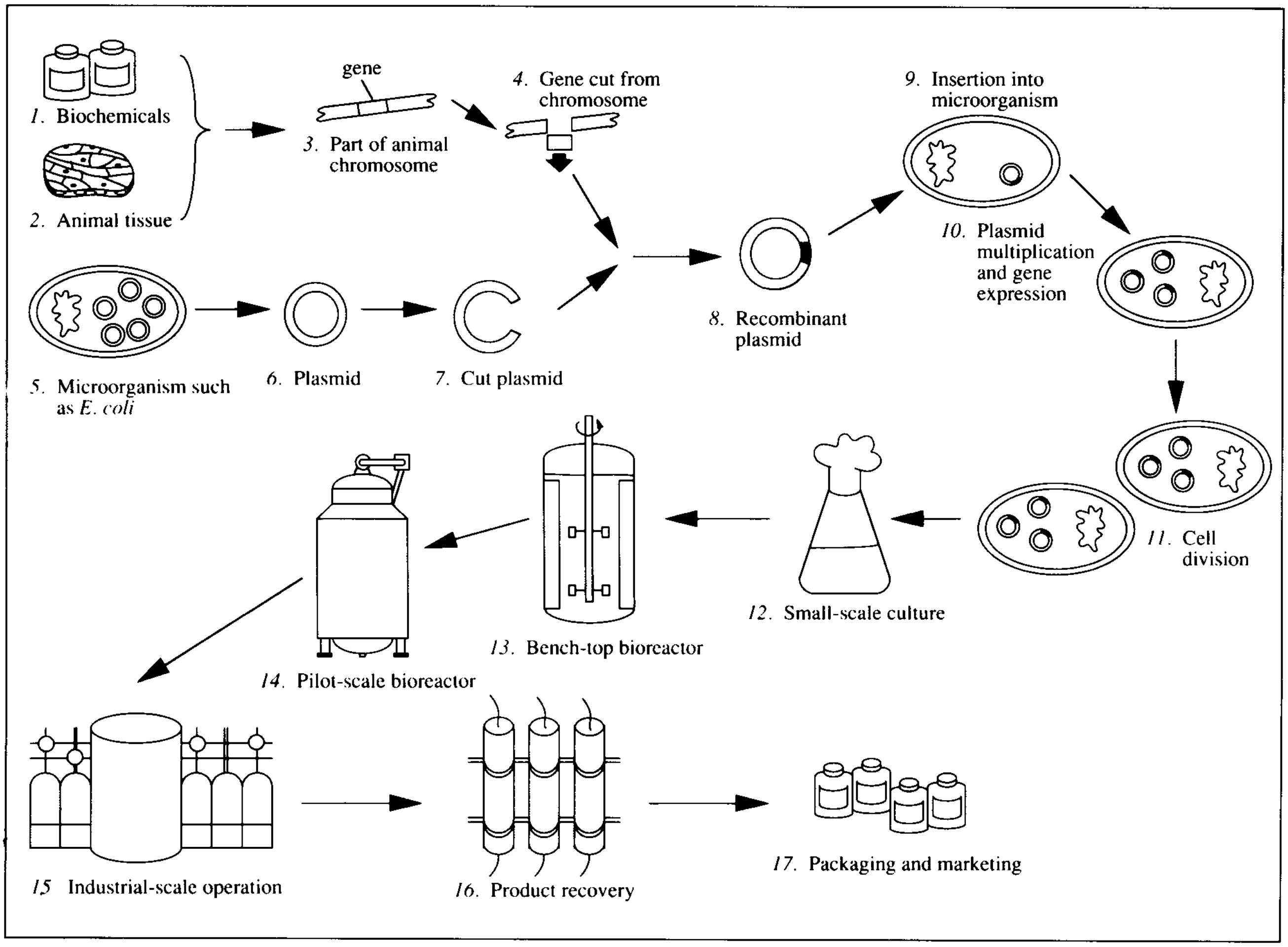
Table 1.1 Major products of biological processing

(Adapted from M.L. Shuler, 1987, Bioprocess engineering. In: Encyclopedia of Physical Science and Technology, *vol 2, R.A. Meyers, Ed., Academic Press, Orlando)*

Fermentation product	*Typical organism used*	*Approximate world market size* (kg yr^{-1})
Bulk organics		
Ethanol (non-beverage)	*Saccharomyces cerevisiae*	2×10^{10}
Acetone/butanol	*Clostridium acetobutylicum*	2×10^{6} (butanol)
Biomass		
Starter cultures and yeasts for food and agriculture	Lactic acid bacteria or bakers' yeast	5×10^{8}
Single-cell protein	*Pseudomonas methylotrophus* or *Candida utilis*	$0.5–1 \times 10^{8}$
Organic acids		
Citric acid	*Aspergillus niger*	$2–3 \times 10^{8}$
Gluconic acid	*Aspergillus niger*	5×10^{7}
Lactic acid	*Lactobacillus delbrueckii*	2×10^{7}
Itaconic acid	*Aspergillus itaconicus*	
Amino acids		
L-glutamic acid	*Corynebacterium glutamicum*	3×10^{8}
L-lysine	*Brevibacterium flavum*	3×10^{7}
L-phenylalanine	*Corynebacterium glutamicum*	2×10^{6}
L-arginine	*Brevibacterium flavum*	2×10^{6}
Others	*Corynebacterium* spp.	1×10^{6}
Microbial transformations		
Steroids	*Rhizopus arrhizus*	
D-sorbitol to L-sorbose (in vitamin C production)	*Acetobacter suboxydans*	4×10^{7}
Antibiotics		
Penicillins	*Penicillium chrysogenum*	$3–4 \times 10^{7}$
Cephalosporins	*Cephalosporium acremonium*	1×10^{7}
Tetracyclines (e.g. 7-chlortetracycline)	*Streptomyces aureofaciens*	1×10^{7}
Macrolide antibiotics (e.g. erythromycin)	*Streptomyces erythreus*	2×10^{6}
Polypeptide antibiotics (e.g. gramicidin)	*Bacillus brevis*	1×10^{6}
Aminoglycoside antibiotics (e.g. streptomycin)	*Streptomyces griseus*	
Aromatic antibiotics (e.g. griseofulvin)	*Penicillium griseofulvum*	
Extracellular polysaccharides		
Xanthan gum	*Xanthomonas campestris*	5×10^{6}
Dextran	*Leuconostoc mesenteroides*	small

Nucleotides		
5′-guanosine monophosphate	*Brevibacterium ammoniagenes*	1×10^5
Enzymes		
Proteases	*Bacillus* spp.	6×10^5
α-amylase	*Bacillus amyloliquefaciens*	4×10^5
Glucoamylase	*Aspergillus niger*	4×10^5
Glucose isomerase	*Bacillus coagulans*	1×10^5
Pectinase	*Aspergillus niger*	1×10^4
Rennin	*Mucor miehei* or recombinant yeast	1×10^4
All others		5×10^4
Vitamins		
B_{12}	*Propionibacterium shermanii* or *Pseudomonas denitrificans*	1×10^4
Riboflavin	*Eremothecium ashbyii*	
Ergot alkaloids	*Claviceps paspali*	5×10^3
Pigments		
Shikonin	*Lithospermum erythrorhizon* (plant-cell culture)	60
β-carotene	*Blakeslea trispora*	
Vaccines		
Diphtheria	*Corynebacterium diphtheriae*	< 50
Tetanus	*Clostridium tetani*	
Pertussis (whooping cough)	*Bordetella pertussis*	
Poliomyelitis virus	Live attenuated viruses grown in monkey kidney or human diploid cells	
Rubella	Live attenuated viruses grown in baby-hamster kidney cells	
Hepatitis B	Surface antigen expressed in recombinant yeast	
Therapeutic proteins		< 20
Insulin	Recombinant *Escherichia coli*	
Growth hormone	Recombinant *Escherichia coli* or recombinant mammalian cells	
Erythropoietin	Recombinant mammalian cells	
Factor VIII-C	Recombinant mammalian cells	
Tissue plasminogen activator	Recombinant mammalian cells	
Interferon-α_2	Recombinant *Escherichia coli*	
Monoclonal antibodies	Hybridoma cells	< 20
Insecticides		
Bacterial spores	*Bacillus thuringiensis*	
Fungal spores	*Hirsutella thompsonii*	

Figure 1.1 Steps in development of a complete bioprocess for commercial manufacture of a new recombinant-DNA-derived product.



the cells must be measured as a function of culture environment (Step 12). Practical skills in microbiology and kinetic analysis are required; small-scale culture is mostly carried out using shake flasks of 250-ml to 1-litre capacity. Medium composition, pH, temperature and other environmental conditions allowing optimal growth and productivity are determined. Calculated parameters such as cell growth rate, specific productivity and product yield are used to describe performance of the organism.

Once the culture conditions for production are known, scale-up of the process starts. The first stage may be a 1- or 2-litre *bench-top bioreactor* equipped with instruments for measuring and adjusting temperature, pH, dissolved-oxygen concentration, stirrer speed and other process variables (Step 13). Cultures can be more closely monitored in bioreactors than in shake flasks so better control over the process is possible. Information is collected about the oxygen requirements of the cells, their shear sensitivity, foaming characteristics and other parameters. Limitations imposed by the reactor on activity of the organism must be identified. For example, if the bioreactor cannot provide dissolved oxygen to an aerobic culture at a sufficiently high rate, the culture will become oxygen-starved. Similarly, in mixing the broth to expose the cells to nutrients in the medium, the stirrer in the reactor may cause cell damage. Whether or not the reactor can provide conditions for optimal activity of the cells is of prime concern. The situation is assessed using measured and calculated parameters such as mass-transfer coefficients, mixing time, gas hold-up, rate of oxygen uptake, power number, impeller shear-rate, and many others. It must also be decided whether the culture is best operated as a batch, semi-batch or continuous process; experimental results for culture performance under various modes of reactor operation may be examined. The viability of the process as a commercial venture is of great interest; information about activity of the cells is used in further calculations to determine economic feasibility.

Following this stage of process development, the system is scaled up again to a *pilot-scale bioreactor* (Step 14). Engineers trained in bioprocessing are normally involved in pilot-scale operations. A vessel of capacity 100–1000 litres is built according to specifications determined from the bench-scale prototype. The design is usually similar to that which worked best on the smaller scale. The aim of pilot-scale studies is to examine the response of cells to scale-up. Changing the size of the equipment seems relatively trivial; however, loss or variation of performance often occurs. Even though the geometry of the reactor, method of aeration and mixing, impeller design and other features may be similar in small and large fermenters, the effect on activity of cells can be great. Loss of productivity following scale-up may or may not be recovered; economic projections often need to be re-assessed as a result of pilot-scale findings.

If the scale-up step is completed successfully, design of the *industrial-scale operation* commences (Step 15). This part of process development is clearly in the territory of bioprocess engineering. As well as the reactor itself, all of the auxiliary service facilities must be designed and tested. These include air supply and sterilisation equipment, steam generator and supply lines, medium preparation and sterilisation facilities, cooling-water supply and process-control network. Particular attention is required to ensure the fermentation can be carried out aseptically. When recombinant cells or pathogenic organisms are involved, design of the process must also reflect containment and safety requirements.

An important part of the total process is *product recovery* (Step 16), also known as *downstream processing*. After leaving the fermenter, raw broth is treated in a series of steps to produce the final product. Product recovery is often difficult and expensive; for some recombinant-DNA-derived products, purification accounts for 80–90% of the total processing cost. Actual procedures used for downstream processing depend on the nature of the product and the broth; physical, chemical or biological methods may be employed. Many operations which are standard in the laboratory become uneconomic or impractical on an industrial scale. Commercial procedures include filtration, centrifugation and flotation for separation of cells from the liquid, mechanical disruption of the cells if the product is intracellular, solvent extraction, chromatography, membrane filtration, adsorption, crystallisation and drying. Disposal of effluent after removal of the desired product must also be considered. Like bioreactor design, techniques applied industrially for downstream processing are first developed and tested using small-scale apparatus. Scientists trained in chemistry, biochemistry, chemical engineering and industrial chemistry play important roles in designing product recovery and purification systems.

After the product has been isolated in sufficient purity it is packaged and marketed (Step 17). For new pharmaceuticals such as recombinant human growth hormone or insulin, medical and clinical trials are required to test the efficacy of the product. Animals are used first, then humans. Only after these trials are carried out and the safety of the product established can it be released for general health-care application. Other tests are required for food products. Bioprocess engineers with a detailed knowledge of the production process are often involved in documenting manufacturing procedures for submission to regulatory authorities. Manufacturing standards must be met; this is particularly the case for recombinant products where a greater number of safety and precautionary measures is required.

As shown in this example, a broad range of disciplines is involved in bioprocessing. Scientists working in this area are constantly confronted with biological, chemical, physical, engineering and sometimes medical questions.

1.2 A Quantitative Approach

The biological characteristics of cells and enzymes often impose constraints on bioprocessing; knowledge of them is therefore an important prerequisite for rational engineering design. For instance, thermostability properties must be taken into account when choosing the operating temperature of an enzyme reactor, while susceptibility of an organism to substrate inhibition will determine whether substrate is fed to the fermenter all at once or intermittently. It is equally true, however, that biologists working in biotechnology must consider the engineering aspects of bioprocessing; selection or manipulation of organisms should be carried out to achieve the best results in production-scale operations. It would be disappointing, for example, to spend a year or two manipulating an organism to express a foreign gene if the cells in culture produce a highly viscous broth that cannot be adequately mixed or supplied with oxygen in large-scale vessels. Similarly, improving cell permeability to facilitate product excretion has limited utility if the new organism is too fragile to withstand the mechanical forces developed during fermenter operation.

Another area requiring cooperation and understanding between engineers and laboratory scientists is medium formation. For example, addition of serum may be beneficial to growth of animal cells, but can significantly reduce product yields during recovery operations and, in large-scale processes, requires special sterilisation and handling procedures.

All areas of bioprocess development—the cell or enzyme used, the culture conditions provided, the fermentation equipment and product-recovery operations—are interdependent. Because improvement in one area can be disadvantageous to another, ideally, bioprocess development should proceed using an integrated approach. In practice, combining the skills of engineers with those of biologists can be difficult owing to the very different ways in which biologists and engineers are trained. Biological scientists generally have strong experimental technique and are good at testing qualitative models; however, because calculations and equations are not a prominent feature of the life sciences, biologists are usually less familiar with mathematics. On the other hand, as calculations are important in all areas of equipment design and process analysis, quantitative methods, physics and mathematical theories play a central role in engineering. There is also a difference in the way biologists and biochemical engineers think about complex processes such as cell and enzyme function. Fascinating as the minutiae of these biological systems may be, in order to build working reactors and other equipment, engineers must take a simplified and pragmatic approach. It is often disappointing for the biology-trained scientist that engineers seem to ignore the wonder, intricacy and complexity of life to focus only on those aspects which have significant quantitative effect on the final outcome of the process.

Given the importance of interaction between biology and engineering in bioprocessing, these differences in outlook between engineers and biologists must be overcome. Although it is unrealistic to expect all biotechnologists to undertake full engineering training, there are many advantages in understanding the practical principles of bioprocess engineering if not the full theoretical detail. The principal objective of this book is to teach scientists trained in biology those aspects of engineering science which are relevant to bioprocessing. An adequate background in biology is assumed. At the end of this study, you will have gained a heightened appreciation for bioprocess engineering. You will be able to communicate on a professional level with bioprocess engineers and know how to analyse and critically evaluate new processing proposals. You will be able to carry out routine calculations and checks on processes; in many cases these calculations are not difficult and can be of great value. You will also know what type of expertise a bioprocess engineer can offer and when it is necessary to consult an expert in the field. In the laboratory, your awareness of engineering methods will help avoid common mistakes in data analysis and design of experimental apparatus.

As our exploitation of biology continues, there is an increasing demand for scientists trained in bioprocess technology who can translate new discoveries into industrial-scale production. As a biotechnologist, you could be expected to work at the interface of biology and engineering science. This textbook on bioprocess engineering is designed to prepare you for this challenge.

2

Introduction to Engineering Calculations

Calculations used in bioprocess engineering require a systematic approach with well-defined methods and rules. Conventions and definitions which form the backbone of engineering analysis are presented in this chapter. Many of these you will use over and over again as you progress through this text. In laying the foundation for calculations and problem-solving, this chapter will be a useful reference which you may need to review from time to time.

The first step in quantitative analysis of systems is to express the system properties using mathematical language. This chapter begins by considering how physical and chemical processes are translated into mathematics. The nature of physical variables, dimensions and units are discussed, and formalised procedures for unit conversions outlined. You will have already encountered many of the concepts used in measurement, such as concentration, density, pressure, temperature, etc.; rules for quantifying these variables are summarised here in preparation for Chapters 4–6 where they are first applied to solve processing problems. The occurrence of reactions in biological systems is of particular importance; terminology involved in stoichiometric analysis is considered in this chapter. Finally, since equations representing biological processes often involve physical or chemical properties of materials, references for handbooks containing this information are provided.

Worked examples and problems are used to illustrate and reinforce the material described in the text. Although the terminology and engineering concepts used in these examples may be unfamiliar, solutions to each problem can be obtained using techniques fully explained within this chapter. Many of the equations introduced as problems and examples are explained in more detail in later sections of this book; the emphasis in this chapter is on use of basic mathematical principles irrespective of the particular application. At the end of the chapter is a check-list so you can be sure you have assimilated all the important points.

2.1 Physical Variables, Dimensions and Units

Engineering calculations involve manipulation of numbers. Most of these numbers represent the magnitude of measurable *physical variables,* such as mass, length, time, velocity, area, viscosity, temperature, density, and so on. Other observable characteristics of nature, such as taste or aroma, cannot at present be described completely using appropriate numbers; we cannot, therefore, include these in calculations.

From all the physical variables in the world, the seven quantities listed in Table 2.1 have been chosen by international

Table 2.1 Base quantities

Base quantity	*Dimensional symbol*	*Base SI unit*	*Unit symbol*
Length	L	metre	m
Mass	M	kilogram	kg
Time	T	second	s
Electric current	I	ampere	A
Temperature	Θ	kelvin	K
Amount of substance	N	gram-mole	mol or gmol
Luminous intensity	J	candela	cd
Supplementary units			
Plane angle	–	radian	rad
Solid angle	–	steradian	sr

agreement as a basis for measurement [1]. Two further supplementary units are used to express angular quantities. The base quantities are called *dimensions*, and it is from these that the dimensions of other physical variables are derived. For example, the dimensions of velocity, defined as distance travelled per unit time, are LT^{-1}; the dimensions of force, being mass × acceleration, are LMT^{-2}. A list of useful derived dimensional quantities is given in Table 2.2.

Physical variables can be classified into two groups: *substantial variables* and *natural variables*.

2.1.1 Substantial Variables

Examples of substantial variables are mass, length, volume, viscosity and temperature. Expression of the magnitude of substantial variables requires a precise physical standard against which measurement is made. These standards are called *units*. You are already familiar with many units, e.g. metre, foot and mile are units of length; hour and second are units of time. Statements about the magnitude of substantial variables must contain two parts: the number and the unit

Table 2.2 Dimensional quantities (dimensionless quantities have dimension 1)

Quantity	*Dimensions*	*Quantity*	*Dimensions*
Acceleration	LT^{-2}	Osmotic pressure	$L^{-1}MT^{-2}$
Angular velocity	T^{-1}	Partition coefficient	1
Area	L^2	Period	T
Atomic weight	1	Power	L^2MT^{-3}
('relative atomic mass')		Pressure	$L^{-1}MT^{-2}$
Concentration	$L^{-3}N$	Rotational frequency	T^{-1}
Conductivity	$L^{-3}M^{-1}T^3I^2$	Shear rate	T^{-1}
Density	$L^{-3}M$	Shear stress	$L^{-1}MT^{-2}$
Diffusion coefficient	L^2T^{-1}	Specific death constant	T^{-1}
Distribution coefficient	1	Specific gravity	1
Effectiveness factor	1	Specific growth rate	T^{-1}
Efficiency	1	Specific heat capacity	$L^2T^{-2}\Theta^{-1}$
Energy	L^2MT^{-2}	Specific interfacial area	L^{-1}
Enthalpy	L^2MT^{-2}	Specific latent heat	L^2T^{-2}
Entropy	$L^2MT^{-2}\Theta^{-1}$	Specific production rate	T^{-1}
Equilibrium constant	1	Specific volume	L^3M^{-1}
Force	LMT^{-2}	Shear strain	1
Fouling factor	$MT^{-3}\Theta^{-1}$	Stress	$L^{-1}MT^{-2}$
Frequency	T^{-1}	Surface tension	MT^{-2}
Friction coefficient	1	Thermal conductivity	$LMT^{-3}\Theta^{-1}$
Gas hold-up	1	Thermal resistance	$L^{-2}M^{-1}T^3\Theta$
Half life	T	Torque	L^2MT^{-2}
Heat	L^2MT^{-2}	Velocity	LT^{-1}
Heat flux	MT^{-3}	Viscosity (dynamic)	$L^{-1}MT^{-1}$
Heat-transfer coefficient	$MT^{-3}\Theta^{-1}$	Viscosity (kinematic)	L^2T^{-1}
Illuminance	$L^{-2}J$	Void faction	1
Maintenance coefficient	T^{-1}	Volume	L^3
Mass flux	$L^{-2}MT^{-1}$	Weight	LMT^{-2}
Mass-transfer coefficient	LT^{-1}	Work	L^2MT^{-2}
Momentum	LMT^{-1}	Yield coefficient	1
Molar mass	MN^{-1}		
Molecular weight	1		
('relative molecular mass')			

used for measurement. Clearly, reporting the speed of a moving car as 20 has no meaning unless information about the units, say km h^{-1}, is also included.

As numbers representing substantial variables are multiplied, subtracted, divided or added, their units must also be combined. The values of two or more substantial variables may be added or subtracted only if their units are the same, e.g.:

$$5.0\ \text{kg} + 2.2\ \text{kg} = 7.2\ \text{kg}.$$

On the other hand, the values and units of *any* substantial variables can be combined by multiplication or division, e.g.:

$$\frac{1500\ \text{km}}{12.5\ \text{h}} = 120\ \text{km h}^{-1}.$$

The way in which units are carried along during calculations has important consequences. Not only is proper treatment of units essential if the final answer is to have the correct units, units and dimensions can also be used as a guide when deducing how physical variables are related in scientific theories and equations.

2.1.2 Natural Variables

The second group of physical variables are natural variables. Specification of the magnitude of these variables does not require units or any other standard of measurement. Natural variables are also referred to as *dimensionless variables, dimensionless groups* or *dimensionless numbers.* The simplest natural variables are ratios of substantial variables. For example, the aspect ratio of a cylinder is its length divided by its diameter; the result is a dimensionless number.

Other natural variables are not as obvious as this, and involve combinations of substantial variables that do not have the same dimensions. Engineers make frequent use of dimensionless numbers for succinct representation of physical phenomena. For example, a common dimensionless group in fluid mechanics is the Reynolds number, *Re.* For flow in a pipe, the Reynolds number is given by the equation:

$$Re = \frac{Du\rho}{\mu} \tag{2.1}$$

where ρ is fluid density, u is fluid velocity, D is pipe diameter and μ is fluid viscosity. When the dimensions of these variables are combined according to Eq. (2.1), the dimensions of the numerator exactly cancel those of the denominator. Other dimensionless variables relevant to bioprocess engineering are the Schmidt number, Prandtl number, Sherwood number, Peclet number, Nusselt number, Grashof number, power number and many others. Definitions and applications of these natural variables are given in later chapters of this book.

In calculations involving rotational phenomena, rotation is described using number of revolutions or radians:

$$\text{number of radians} = \frac{\text{length of arc}}{\text{radius}} \tag{2.2}$$

$$\text{number of revolutions} = \frac{\text{length of arc}}{\text{circumference}} = \frac{\text{length of arc}}{2\pi r} \tag{2.3}$$

where r is radius. One revolution is equal to 2π radians. Radians and revolutions are non-dimensional because the dimensions of length for arc, radius and circumference in Eqs (2.2) and (2.3) cancel. Consequently, rotational speed (e.g. number of revolutions per second) and angular velocity (e.g. number of radians per second), as well as frequency (e.g. number of vibrations per second), all have dimensions T^{-1}. Degrees, which are subdivisions of a revolution, are converted into revolutions or radians before application in engineering calculations.

2.1.3 Dimensional Homogeneity in Equations

Rules about dimensions determine how equations are formulated. 'Properly constructed' equations representing general relationships between physical variables must be dimensionally homogeneous. For dimensional homogeneity, the dimensions of terms which are added or subtracted must be the same, and the dimensions of the right-hand side of the equation must be the same as the left-hand side. As a simple example, consider the Margules equation for evaluating fluid viscosity from experimental measurements:

$$\mu = \frac{M}{4\pi h\Omega}\left(\frac{1}{R_o^2} - \frac{1}{R_i^2}\right). \tag{2.4}$$

The terms and dimensions in this equation are listed in Table 2.3. Numbers such as 4 have no dimensions; the symbol π represents the number 3.1415926536 which is also dimensionless. A quick check shows that Eq. (2.4) is dimensionally homogeneous since both sides of the equation have dimensions

$L^{-1}MT^{-1}$ and all terms added or subtracted have the same dimensions. Note that when a term such as R_o is raised to a power such as 2, the units and dimensions of R_o must also be raised to that power.

For dimensional homogeneity, the argument of any transcendental function, such as a logarithmic, trigonometric or exponential function, must be dimensionless. The following examples illustrate this principle.

(i) An expression for cell growth is:

$$\ln \frac{x}{x_0} = \mu t \tag{2.5}$$

where x is cell concentration at time t, x_0 is initial cell concentration, and μ is the specific growth rate. The argument of the logarithm, the ratio of cell concentrations, is dimensionless.

(ii) The displacement y due to action of a progressive wave with amplitude A, frequency $\omega/_{2\pi}$ and velocity v is given by the equation:

$$y = A \sin\left[\omega\left(t - \frac{x}{v}\right)\right] \tag{2.6}$$

where t is time and x is distance from the origin. The argument of the sine function, $\omega\ (t - \frac{x}{v})$, is dimensionless.

(iii) The relationship between α, the mutation rate of *Escherichia coli*, and temperature T, can be described using an Arrhenius-type equation:

$$\alpha = \alpha_0 e^{-E/_{RT}} \tag{2.7}$$

where α_0 is the mutation reaction constant, E is specific activation energy and R is the ideal gas constant (see Section 2.5). The dimensions of RT are the same as those of E, so the exponent is as it should be: dimensionless.

Dimensional homogeneity of equations can sometimes be masked by mathematical manipulation. As an example, Eq. (2.5) might be written:

$$\ln x = \ln x_0 + \mu t. \tag{2.8}$$

Inspection of this equation shows that rearrangement of the terms to group $\ln x$ and $\ln x_0$ together recovers dimensional homogeneity by providing a dimensionless argument for the logarithm.

Table 2.3 Terms and dimensions of Eq. (2.4)

Term	*Dimensions*	*SI Units*
μ (dynamic viscosity)	$L^{-1}MT^{-1}$	pascal second (Pa s)
M (torque)	L^2MT^{-2}	newton metre (N m)
h (cylinder height)	L	metre (m)
Ω (angular velocity)	T^{-1}	radian per second (rad s^{-1})
R_o (outer radius)	L	metre (m)
R_i (inner radius)	L	metre (m)

Integration and differentiation of terms affect dimensionality. Integration of a function with respect to x increases the dimensions of that function by the dimensions of x. Conversely, differentiation with respect to x results in the dimensions being reduced by the dimensions of x. For example, if C is the concentration of a particular compound expressed as mass per unit volume and x is distance, $^{dC}/_{dx}$ has dimensions $L^{-4}M$, while $^{d^2C}/_{dx^2}$ has dimensions $L^{-5}M$. On the other hand, if μ is the specific growth rate of an organism with dimensions T^{-1}, then $\int \mu\, dt$ is dimensionless where t is time.

2.1.4 Equations Without Dimensional Homogeneity

For repetitive calculations or when an equation is derived from observation rather than from theoretical principles, it is sometimes convenient to present the equation in a non-homogeneous form. Such equations are called *equations in numerics* or *empirical equations*. In empirical equations, the units associated with each variable must be stated explicitly. An example is Richards' correlation for the dimensionless gas hold-up ϵ in a stirred fermenter [2]:

$$\left(\frac{P}{V}\right)^{0.4} u^{1/2} = 30\epsilon + 1.33 \tag{2.9}$$

where P is power in units of horsepower, V is ungassed liquid volume in units of ft^3, u is linear gas velocity in units of ft s^{-1} and ϵ is fractional gas hold-up, a dimensionless variable. The dimensions of each side of Eq. (2.9) are certainly not the same. For direct application of Eq. (2.9), only those units stated above can be used.

2.2 Units

Several systems of units for expressing the magnitude of physical variables have been devised through the ages. The metric system of units originated from the National Assembly of France in 1790. In 1960 this system was rationalised, and the SI or Système International d'Unités was adopted as the international standard. Unit names and their abbreviations have been standardised; according to SI convention, unit abbreviations are the same for both singular and plural and are not followed by a period. SI prefixes used to indicate multiples and sub-multiples of units are listed in Table 2.4. Despite widespread use of SI units, no single system of units has universal application. In particular, engineers in the USA continue to apply British or imperial units. In addition, many physical property data collected before 1960 are published in lists and tables using non-standard units.

Familiarity with both metric and non-metric units is necessary. Many units used in engineering such as the slug (1 slug = 14.5939 kilograms), dram (1 dram = 1.77185 grams), stoke (a unit of kinematic viscosity), poundal (a unit of force) and erg (a unit of energy), are probably not known to you. Although no longer commonly applied, these are legitimate units which may appear in engineering reports and tables of data.

In calculations it is often necessary to convert units. Units are changed using *conversion factors*. Some conversion factors, such as 1 inch = 2.54 cm and 2.20 lb = 1 kg, you probably already know. Tables of common conversion factors are given in Appendix A at the back of this book. Unit conversions are not only necessary to convert imperial units to metric; some physical variables have several metric units in common use. For example, viscosity may be reported as centipoise or kg h^{-1} m^{-1}; pressure may be given in standard atmospheres, pascals, or millimetres of mercury. Conversion of units seems simple enough; however difficulties can arise when several variables are being converted in a single equation. Accordingly, an organised mathematical approach is needed.

For each conversion factor, a *unity bracket* can be derived. The value of the unity bracket, as the name suggests, is unity. As an example,

$$1 \text{ lb} = 453.6 \text{ g} \tag{2.10}$$

can be converted by division of both sides of the equation by 1 lb to give a unity bracket denoted by | |:

$$1 = \left| \frac{453.6 \text{ g}}{1 \text{ lb}} \right| . \tag{2.11}$$

Similarly, division of both sides of Eq. (2.10) by 453.6 g gives another unity bracket:

$$1 = \left| \frac{1 \text{ lb}}{453.6 \text{ g}} \right| . \tag{2.12}$$

To calculate how many pounds are in 200 g, we can multiply 200 g by the unity bracket in Eq. (2.12) or divide 200 g by the unity bracket in Eq. (2.11). This is permissible since the value

Table 2.4 SI prefixes

(from J. V. Drazil, 1983, Quantities and Units of Measurement, *Mansell, London)*

Factor	*Prefix*	*Symbol*	*Factor*	*Prefix*	*Symbol*
10^{-1}	deci*	d	10^{18}	exa	E
10^{-2}	centi*	c	10^{15}	peta	P
10^{-3}	milli	m	10^{12}	tera	T
10^{-6}	micro	μ	10^{9}	giga	G
10^{-9}	nano	n	10^{6}	mega	M
10^{-12}	pico	p	10^{3}	kilo	k
10^{-15}	femto	f	10^{2}	hecto*	h
10^{-18}	atto	a	10^{1}	deka*	da

* Used for areas and volumes.

of both unity brackets is unity, and multiplication or division by 1 does not change the value of 200 g. Using the option of multiplying by Eq. (2.12):

$$200\ \text{g} = 200\ \cancel{\text{g}} \cdot \left| \frac{1\ \text{lb}}{453.6\ \cancel{\text{g}}} \right| \tag{2.13}$$

On the right-hand side, cancelling the old units leaves the desired unit, lb. Dividing the numbers gives:

$$200\ \text{g} = 0.441\ \text{lb}. \tag{2.14}$$

A more complicated calculation involving a complete equation is given in Example 2.1.

Example 2.1 Unit conversion

Air is pumped through an orifice immersed in liquid. The size of the bubbles leaving the orifice depends on the diameter of the orifice and the properties of the liquid. The equation representing this situation is:

$$\frac{g(\rho_L - \rho_G)\,D_b^3}{\sigma D_o} = 6$$

where g = gravitational acceleration = 32.174 ft s^{-2}; ρ_L = liquid density = 1 g cm^{-3}; ρ_G = gas density = 0.081 lb ft^{-3}; D_b = bubble diameter; σ = gas–liquid surface tension = 70.8 dyn cm^{-1}; and D_o = orifice diameter = 1 mm.

Calculate the bubble diameter D_b.

Solution:

Convert the data to a consistent set of units, e.g. g, cm, s. From Appendix A, the conversion factors required are:

1 ft = 0.3048 m;
1 lb = 453.6 g; and
1 dyn cm^{-1} = 1 g s^{-2}.

Also: 1 m = 100 cm; and
10 mm = 1 cm.

Converting units:

$$g = 32.174\ \frac{\text{ft}}{\text{s}^2} \cdot \left| \frac{0.3048\ \text{m}}{1\ \text{ft}} \right| \cdot \left| \frac{100\ \text{cm}}{1\ \text{m}} \right| = 980.7\ \text{cm s}^{-2}$$

$$\rho_G = 0.081\ \frac{\text{lb}}{\text{ft}^3} \cdot \left| \frac{453.6\ \text{g}}{1\ \text{lb}} \right| \cdot \left| \frac{1\ \text{ft}}{0.3048\ \text{m}} \right|^3 \cdot \left| \frac{1\ \text{m}}{100\ \text{cm}} \right|^3 = 1.30 \times 10^{-3}\ \text{g cm}^{-3}$$

$$\sigma = 70.8\ \text{dyn cm}^{-1} \cdot \left| \frac{1\ \text{g s}^{-2}}{1\ \text{dyn cm}^{-1}} \right| = 70.8\ \text{g s}^{-2}$$

$$D_o = 1\ \text{mm} \cdot \left| \frac{1\ \text{cm}}{10\ \text{mm}} \right| = 0.1\ \text{cm}.$$

Rearranging the equation to give an expression for D_b^3:

$$D_b^3 = \frac{6\,\sigma D_o}{g(\rho_L - \rho_G)}.$$

Substituting values gives:

$$D_b^3 = \frac{6\,(70.8\text{ g s}^{-2})\,(0.1\text{ cm})}{980.7\text{ cm s}^{-2}\,(1\text{ g cm}^{-3} - 1.30\times10^{-3}\text{ g cm}^{-3})} = 4.34\times10^{-2}\text{ cm}^3.$$

Taking the cube root:

$$D_b = 0.35\text{ cm}.$$

Note that unity brackets are squared or cubed when appropriate, e.g. when converting ft^3 to cm^3. This is permissible since the value of the unity bracket is 1, and 1^2 or 1^3 is still 1.

2.3 Force and Weight

According to Newton's law, the force exerted on a body in motion is proportional to its mass multiplied by the acceleration. As listed in Table 2.2, the dimensions of force are LMT^{-2}; the *natural units* of force in the SI system are kg m s^{-2}. Analogously, g cm s^{-2} and lb ft s^{-2} are the natural units of force in the metric and British systems, respectively.

Force occurs frequently in engineering calculations, and *derived units* are used more commonly than natural units. In SI, the derived unit is the *newton*, abbreviated as N:

$$1\text{ N} = 1\text{ kg m s}^{-2}. \tag{2.15}$$

In the British or imperial system, the derived unit for force is defined as (1 lb mass) × (gravitational acceleration at sea level and 45° latitude). The derived force-unit in this case is called the *pound-force*, and is denoted lb_f:

$$1\text{ lb}_f = 32.174\text{ lb}_m\text{ ft s}^{-2} \tag{2.16}$$

as gravitational acceleration at sea level and 45° latitude is 32.174 ft s^{-2}. Note that pound-mass, represented usually as lb, has been shown here using the abbreviation, lb_m, to distinguish it from lb_f. Use of the pound in the imperial system for reporting both mass and force can be a source of confusion and requires care.

In order to convert force from a defined unit to a natural unit, a special dimensionless unity-bracket called g_c is used. The form of g_c depends on the units being converted. From Eqs (2.15) and (2.16):

$$g_c = 1 = \left|\frac{1\text{ N}}{1\text{ kg m s}^{-2}}\right| = \left|\frac{1\text{ lb}_f}{32.174\text{ lb}_m\text{ft s}^{-2}}\right|. \tag{2.17}$$

Application of g_c is illustrated in Example 2.2.

Example 2.2 Use of g_c

Calculate the kinetic energy of 250 lb_m liquid flowing through a pipe at 35 ft s^{-1}. Express your answer in units of ft lb_f.

Solution:

Kinetic energy is given by the equation:

$$\text{kinetic energy} = E_k = {}^1\!/_2 Mv^2$$

where M is mass and v is velocity. Using the values given:

$$E_k = {}^1\!/_2(250\text{ lb}_m)\left(35\frac{\text{ft}}{\text{s}}\right)^2 = 1.531\times10^5\,\frac{\text{lb}_m\text{ft}^2}{\text{s}^2}.$$

Multiplying by g_c from Eq. (2.17) gives:

$$E_k = 1.531\times10^5\,\frac{\text{lb}_m\text{ft}^2}{\text{s}^2}\cdot\left|\frac{1\text{ lb}_f}{32.174\text{ lb}_m\text{ft s}^{-2}}\right|.$$

Calculating and cancelling units gives the answer:

$$E_k = 4760\text{ ft lb}_f.$$

Weight is the force with which a body is attracted by gravity to the centre of the earth. It changes according to the value of the gravitational acceleration *g*, which varies by about 0.5% over the earth's surface. In SI units *g* is approximately 9.8 m s^{-2}; in imperial units *g* is about 32.2 ft s^{-2}. Using Newton's law and depending on the exact value of *g*, the weight of a mass of 1 kg is about 9.8 newtons; the weight of a mass of 1 lb is about 1 lb_f. Note that although the value of *g* changes with position on the earth's surface (or in the universe), the value of g_c within a given system of units does not. g_c is a factor for converting units, not a physical variable.

2.4 Measurement Conventions

Familiarity with common physical variables and methods for expressing their magnitude is necessary for engineering analysis of bioprocesses. This section covers some useful definitions and engineering conventions that will be applied throughout the text.

2.4.1 Density

Density is a substantial variable defined as mass per unit volume. Its dimensions are $L^{-3}M$, and the usual symbol is ρ. Units for density are, for example, g cm^{-3}, kg m^{-3} and lb ft^{-3}. If the density of acetone is 0.792 g cm^{-3}, the mass of 150 cm^3 acetone can be calculated as follows:

$$150\ \mathrm{cm}^3\left(\frac{0.792\ \mathrm{g}}{\mathrm{cm}^3}\right) = 119\ \mathrm{g}.$$

Densities of solids and liquids vary slightly with temperature. The density of water at 4°C is 1.0000 g cm^{-3}, or 62.4 lb ft^{-3}. The density of solutions is a function of both concentration and temperature. Gas densities are highly dependent on temperature and pressure.

2.4.2 Specific Gravity

Specific gravity, also known as 'relative density', is a dimensionless variable. It is the ratio of two densities, that of the substance in question and that of a specified reference material. For liquids and solids, the reference material is usually water. For gases, air is commonly used as reference, but other reference gases may also be specified.

As mentioned above, liquid densities vary somewhat with temperature. Accordingly, when reporting specific gravity the temperatures of the substance and its reference material are specified. If the specific gravity of ethanol is given as $0.789^{20°C}_{4°C}$, this means that the specific gravity is 0.789 for ethanol at 20°C referenced against water at 4°C. Since the density of water at 4°C is almost exactly 1.0000 g cm^{-3}, we can say immediately that the density of ethanol at 20°C is 0.789 g cm^{-3}.

2.4.3 Specific Volume

Specific volume is the inverse of density. The dimensions of specific volume are L^3M^{-1}.

2.4.4 Mole

In the SI system, a mole is 'the amount of substance of a system which contains as many elementary entities as there are atoms in 0.012 kg of carbon-12' [3]. This means that a mole in the SI system is about 6.02×10^{23} molecules, and is denoted by the term *gram-mole* or *gmol*. One thousand gmol is called a *kilogram-mole* or *kgmol*. In the American engineering system, the basic mole unit is the *pound-mole* or *lbmol*, which is $6.02 \times 10^{23} \times 453.6$ molecules. The gmol, kgmol and lbmol therefore represent three different quantities. When molar quantities are specified simply as 'moles', gmol is usually meant.

The number of moles in a given mass of material is calculated as follows:

$$\text{gram moles} = \frac{\text{mass in grams}}{\text{molar mass in grams}} \tag{2.18}$$

$$\text{lb moles} = \frac{\text{mass in lb}}{\text{molar mass in lb}}. \tag{2.19}$$

Molar mass is the mass of one mole of substance, and has dimensions MN^{-1}. Molar mass is routinely referred to as *molecular weight*, although the molecular weight of a compound is a dimensionless quantity calculated as the sum of the atomic weights of the elements constituting a molecule of that compound. The *atomic weight* of an element is its mass relative to carbon-12 having a mass of exactly 12; atomic weight is also dimensionless. The terms 'molecular weight' and 'atomic weight' are frequently used by engineers and chemists instead of the more correct terms, 'relative molecular mass' and 'relative atomic mass'.

2.4.5 Chemical Composition

Process streams usually consist of mixtures of components or solutions of one or more solutes. The following terms are used to define the composition of mixtures and solutions.

The *mole fraction* of component A in a mixture is defined as:

$$\text{mole fraction A} = \frac{\text{number of moles of A}}{\text{total number of moles}} \; . \tag{2.20}$$

Mole percent is mole fraction × 100. In the absence of chemical reactions and loss of material from the system, the composition of a mixture expressed in mole fraction or mole percent does not vary with temperature.

The *mass fraction* of component A in a mixture is defined as:

$$\text{mass fraction A} = \frac{\text{mass of A}}{\text{total mass}} \; . \tag{2.21}$$

Mass percent is mass fraction × 100; mass fraction and mass percent are also called *weight fraction* and *weight percent*, respectively. Another common expression for composition is weight-for-weight percent (%w/w); although not so well defined, this is usually considered to be the same as weight percent. For example, a solution of sucrose in water with a concentration of 40% w/w contains 40 g sucrose per 100 g solution, 40 tonnes sucrose per 100 tonnes solution, 40 lb sucrose per 100 lb solution, and so on. In the absence of chemical reactions and loss of material from the system, mass and weight percent do not change with temperature.

Because the composition of liquids and solids is usually reported using mass percent, this can be assumed even if not specified. For example, if an aqueous mixture is reported to contain 5% NaOH and 3% $MgSO_4$, it is conventional to assume that there are 5 g NaOH and 3 g $MgSO_4$ in every 100 g solution. Of course, mole or volume percent may be used for liquid and solid mixtures; however this should be stated explicitly, e.g. 10 vol% or 50 mole%.

The *volume fraction* of component A in a mixture is:

$$\text{volume fraction A} = \frac{\text{volume of A}}{\text{total volume}} \; . \tag{2.22}$$

Volume percent is volume fraction × 100. Although not as clearly defined as volume percent, volume-for-volume percent (%v/v) is usually interpreted in the same way as volume percent; for example, an aqueous sulphuric acid mixture containing 30 cm^3 acid in 100 cm^3 solution is referred to as a 30% (v/v) solution. Weight-for-volume percent (%w/v) is also commonly used; a codeine concentration of 0.15% w/v generally means 0.15 g codeine per 100 ml solution.

Compositions of gases are commonly given in volume percent; if percentage figures are given without specification, volume percent is assumed. According to the *International Critical Tables* [4], the composition of air is 20.99% oxygen, 78.03% nitrogen, 0.94% argon and 0.03% carbon dioxide; small amounts of hydrogen, helium, neon, krypton and xenon make up the remaining 0.01%. For most purposes, all inerts are lumped together with nitrogen; the composition of air is taken as approximately 21% oxygen and 79% nitrogen. This means that any sample of air will contain about 21% oxygen *by volume*. At low pressure, gas volume is directly proportional to number of moles; therefore, the composition of air stated above can be interpreted as 21 *mole%* oxygen. Since temperature changes at low pressure produce the same relative change in partial volumes of constituent gases as in the total volume, volumetric composition of gas mixtures is not altered by variation in temperature. Temperature changes affect the component gases equally, so the overall composition is unchanged.

There are many other choices for expressing the concentration of a component in solutions and mixtures:

(i) Moles per unit volume, e.g. gmol l^{-1}, lbmol ft^{-3}.
(ii) Mass per unit volume, e.g. kg m^{-3}, g l^{-1}, lb ft^{-3}.
(iii) Parts per million, ppm. This is used for very dilute solutions. Usually, ppm is a mass fraction for solids and liquids and a mole fraction for gases. For example, an aqueous solution of 20 ppm manganese contains 20 g manganese per 10^6 g solution. A sulphur dioxide concentration of 80 ppm in air means 80 gmol SO_2 per 10^6 gmol gas mixture. At low pressures this is equivalent to 80 litres SO_2 per 10^6 litres gas mixture.
(iv) Molarity, gmol l^{-1}.
(v) Molality, gmol per 1000 g solvent.
(vi) Normality, mole equivalents l^{-1}. A normal solution contains one equivalent gram-weight of solute per litre of solution. For an acid or base, an equivalent gram-weight is the weight of solute in grams that will produce or react with one gmol hydrogen ions. Accordingly, a 1 N solution of HCl is the same as a 1 M solution; on the other hand, a 1 N H_2SO_4 or 1 N $Ca(OH)_2$ solution is 0.5 M.
(vii) Formality, formula gram-weight l^{-1}. If the molecular weight of a solute is not clearly defined, formality may be used to express concentration. A formal solution contains one formula gram-weight of solute per litre of solution. If the formula gram-weight and molecular gram-weight are the same, molarity and formality are the same.

In several industries, concentration is expressed in an indirect way using specific gravity. For a given solute and solvent, the density and specific gravity of solutions are directly dependent on concentration of solute. Specific gravity is conveniently measured using a hydrometer which may be calibrated using special scales. The *Baumé scale*, originally developed in France to measure levels of salt in brine, is in common use. One

Baumé scale is used for liquids lighter than water; another is used for liquids heavier than water. For liquids heavier than water such as sugar solutions:

$$\text{degrees Baumé } (^\circ\text{Bé}) = 145 - \frac{145}{G} \tag{2.23}$$

where G is specific gravity. Unfortunately, the reference temperature for the Baumé and other gravity scales is not standardised world-wide. If the Baumé hydrometer is calibrated at 60°F (15.6°C), G in Eq (2.23) would be the specific gravity at 60°F relative to water at 60°F; however another common reference temperature is 20°C (68°F). The Baumé scale is used widely in the wine and food industries as a measure of sugar concentration. For example, readings of °Bé from grape juice help determine when grapes should be harvested for wine making. The Baumé scale gives only an approximate indication of sugar levels; there is always some contribution to specific gravity from soluble compounds other than sugar.

Degrees Brix (°Brix), or *degrees Balling*, is another hydrometer scale used extensively in the sugar industry. Brix scales calibrated at 15.6°C and 20°C are in common use. With the 20°C scale, each degree Brix indicates 1 gram of sucrose per 100 g liquid.

2.4.6 Temperature

Temperature is a measure of the thermal energy of a body at thermal equilibrium. It is commonly measured in degrees *Celsius* (centigrade) or *Fahrenheit*. In science, the Celsius scale is most common; 0°C is taken as the ice point of water and 100°C the normal boiling point of water. The Fahrenheit scale has everyday use in the USA; 32°F represents the ice point and 212°F the normal boiling point of water. Both Fahrenheit and Celsius scales are *relative temperature scales*, i.e. their zero points have been arbitrarily assigned.

Sometimes it is necessary to use *absolute temperatures*. Absolute-temperature scales have as their zero point the lowest temperature believed possible. Absolute temperature is used in application of the ideal gas law and many other laws of thermodynamics. A scale for absolute temperature with degree units the same as on the Celsius scale is known as the *Kelvin* scale; the absolute-temperature scale using Fahrenheit degree-units is the *Rankine* scale. Units on the Kelvin scale used to be termed 'degrees Kelvin' and abbreviated °K. It is modern practice, however, to name the unit simply 'kelvin'; the SI symbol for kelvin is K. Units on the Rankine scale are denoted °R. 0°R = 0 K = −459.67°F = −273.15°C. Comparison of the four temperature scales is shown in Figure 2.1. One unit on the Kelvin–Celsius scale corresponds to a temperature difference of 1.8 times a single unit on the Rankine–Fahrenheit scale; the range of 180 Rankine–Fahrenheit degrees between the freezing and boiling points of water corresponds to 100 degrees on the Kelvin–Celsius scale.

Equations for converting temperature units are as follows; T represents the temperature reading:

$$T(\text{K}) = T(^\circ\text{C}) + 273.15 \tag{2.24}$$

$$T(^\circ\text{R}) = T(^\circ\text{F}) + 459.67 \tag{2.25}$$

$$T(^\circ\text{R}) = 1.8\ T(\text{K}) \tag{2.26}$$

$$T(^\circ\text{F}) = 1.8\ T(^\circ\text{C}) + 32. \tag{2.27}$$

2.4.7 Pressure

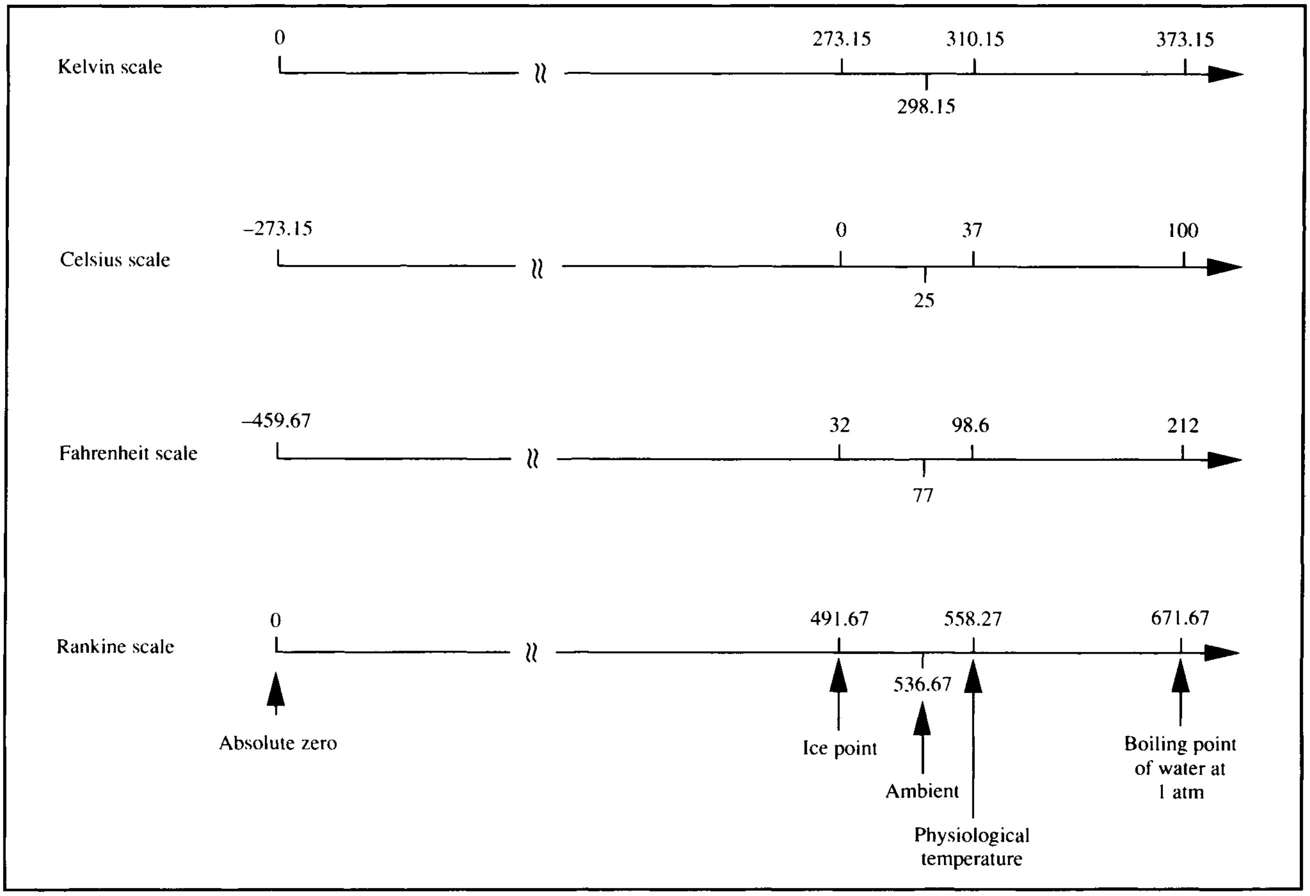
Pressure is defined as force per unit area, and has dimensions $L^{-1}MT^{-2}$. Units of pressure are numerous, including pounds per square inch (psi), millimetres of mercury (mmHg), standard atmospheres (atm), bar, newtons per square metre (N m^{-2}), and many others. The SI pressure unit, N m^{-2}, is called a pascal (Pa). Like temperature, pressure may be expressed using absolute or relative scales.

Absolute pressure is pressure relative to a complete vacuum. Because this reference pressure is independent of location, temperature and weather, absolute pressure is a precise and invariant quantity. However, absolute pressure is not commonly measured. Most pressure-measuring devices sense the difference in pressure between the sample and the surrounding atmosphere at the time of measurement. Measurements using these instruments give *relative pressure*, also known as *gauge pressure*. Absolute pressure can be calculated from gauge pressure as follows:

$$\text{absolute pressure} = \text{gauge pressure} + \text{atmospheric pressure}. \tag{2.28}$$

As you know from listening to weather reports, atmospheric pressure varies with time and place and is measured using a *barometer*. Atmospheric pressure or *barometric pressure* should not be confused with the standard unit of pressure called the standard atmosphere (atm), defined as 1.013×10^5 N m^{-2}, 14.70 psi, or 760 mmHg at 0°C. Sometimes the units for pressure include information about whether the pressure is absolute or relative. Pounds per square inch is abbreviated *psia* for absolute pressure or *psig* for gauge pressure. *Atma* denotes standard atmospheres of absolute pressure.

Figure 2.1 Comparison of temperature scales.



Vacuum pressure is another pressure term, used to indicate pressure below barometric pressure. A gauge pressure of −5 psig, or 5 psi below atmospheric, is the same as a vacuum of 5 psi. A perfect vacuum corresponds to an absolute pressure of zero.

2.5 Standard Conditions and Ideal Gases

A *standard state* of temperature and pressure has been defined and is used when specifying properties of gases, particularly molar volumes. Standard conditions are needed because the volume of a gas depends not only on the quantity present but also on the temperature and pressure. The most widely-adopted standard state is 0°C and 1 atm.

Relationships between gas volume, pressure and temperature were formulated in the 18th and 19th centuries. These correlations were developed under conditions of temperature and pressure so that the average distance between gas molecules was great enough to counteract the effect of intramolecular forces, and the volume of the molecules themselves could be neglected. Under these conditions, a gas became known as an *ideal gas*. This term now in common use refers to a gas which obeys certain simple physical laws, such as those of Boyle, Charles and Dalton. Molar volumes for an ideal gas at standard conditions are:

$$1 \text{ gmol} = 22.4 \text{ litres} \tag{2.29}$$

$$1 \text{ kgmol} = 22.4 \text{ m}^3 \tag{2.30}$$

$$1 \text{ lbmol} = 359 \text{ ft}^3. \tag{2.31}$$

No real gas is an ideal gas at all temperatures and pressures. However, light gases such as hydrogen, oxygen and air deviate negligibly from ideal behaviour over a wide range of conditions. On the other hand, heavier gases such as sulphur dioxide and hydrocarbons can deviate considerably from ideal, particularly at high pressures. Vapours near the boiling point also deviate markedly from ideal. Nevertheless, for many applications in bioprocess engineering, gases can be considered ideal without much loss of accuracy.

Eqs (2.29)–(2.31) can be verified using the *ideal gas law*:

$$pV = nRT \tag{2.32}$$

where p is absolute pressure, V is volume, n is moles, T is absolute temperature and R is the *ideal gas constant.* Eq. (2.32) can be applied using various combinations of units for the physical variables, as long as the correct value and units of R are employed. Table 2.5 gives a list of R values in different systems of units.

Table 2.5 Values of the ideal gas constant, R

(From R.E. Balzhiser, M.R. Samuels and J.D. Eliassen, 1972, Chemical Engineering Thermodynamics, *Prentice-Hall, New Jersey)*

Energy unit	*Temperature unit*	*Mole unit*	R
cal	K	gmol	1.9872
J	K	gmol	8.3144
cm^3 atm	K	gmol	82.057
l atm	K	gmol	0.082057
m^3 atm	K	gmol	0.000082057
l mmHg	K	gmol	62.361
l bar	K	gmol	0.083144
$kg_f m^{-2}$ l	K	gmol	847.9
$kg_f cm^{-2}$ l	K	gmol	0.08479
mmHg ft^3	K	lbmol	998.9
atm ft^3	K	lbmol	1.314
Btu	°R	lbmol	1.9869
psi ft^3	°R	lbmol	10.731
lb_f ft	°R	lbmol	1545
atm ft^3	°R	lbmol	0.7302
in.Hg ft^3	°R	lbmol	21.85
hp h	°R	lbmol	0.0007805
kW h	°R	lbmol	0.0005819
mmHg ft^3	°R	lbmol	555

Example 2.3 Ideal gas law

Gas leaving a fermenter at close to 1 atm pressure and 25°C has the following composition: 78.2% nitrogen, 19.2% oxygen, 2.6% carbon dioxide. Calculate:

(a) the mass composition of the fermenter off-gas; and
(b) the mass of CO_2 in each cubic metre of gas leaving the fermenter.

Solution:

Molecular weights: nitrogen = 28
oxygen = 32
carbon dioxide = 44.

(a) Because the gas is at low pressure, percentages given for composition can be considered mole percentages. Therefore, using the molecular weights, 100 gmol off-gas contains:

$$78.2\text{ gmol }N_2 \cdot \left| \frac{28\text{ g }N_2}{1\text{ gmol }N_2} \right| = 2189.6\text{ g }N_2$$

$$19.2\text{ gmol }O_2 \cdot \left| \frac{32\text{ g }O_2}{1\text{ gmol }O_2} \right| = 614.4\text{ g }O_2$$

$$2.6\text{ gmol }CO_2 \cdot \left| \frac{44\text{ g }CO_2}{1\text{ gmol }CO_2} \right| = 114.4\text{ g }CO_2.$$

Therefore, the total mass is (2189.6 + 614.4 + 114.4) g = 2918.4 g. The mass composition can be calculated as follows:

$$\text{Mass percent }N_2 = \frac{2189.6\text{ g}}{2918.4\text{ g}} \times 100 = 75.0\%$$

$$\text{Mass percent }O_2 = \frac{614.4\text{ g}}{2918.4\text{ g}} \times 100 = 21.1\%$$

$$\text{Mass percent }CO_2 = \frac{114.4\text{ g}}{2918.4\text{ g}} \times 100 = 3.9\%.$$

Therefore, the composition of the gas is 75.0 mass% N_2, 21.1 mass% O_2 and 3.9 mass% CO_2.

(b) As the gas composition is given in volume percent, in each cubic metre of gas there must be 0.026 m^3 CO_2. The relationship between moles of gas and volume at 1 atm and 25°C is determined using Eq. (2.32) and Table 2.5:

$$(1\text{ atm})\,(0.026\text{ m}^3) = n\,(0.000082057\ \frac{\text{m}^3\text{ atm}}{\text{gmol K}})\ (298.15\text{ K}).$$

Calculating the moles of CO_2 present:

$n = 1.06$ gmol.

Converting to mass of CO_2:

$$1.06\text{ gmol} \cdot \left| \frac{44\text{ g}}{1\text{ gmol}} \right| = 46.8\text{ g}.$$

Therefore, each cubic metre of fermenter off-gas contains 46.8 g CO_2.

2.6 Physical and Chemical Property Data

Information about the properties of materials is often required in engineering calculations. Because measurement of physical and chemical properties is time-consuming and expensive, handbooks containing this information are a tremendous resource. You may already be familiar with some handbooks of physical and chemical data, including:

(i) *International Critical Tables* [4]
(ii) *Handbook of Chemistry and Physics* [5]; and
(iii) *Handbook of Chemistry* [6].

To these can be added:

(iv) *Chemical Engineers' Handbook* [7];

and, for information about biological materials,

(v) *Biochemical Engineering and Biotechnology Handbook* [8].

A selection of physical and chemical property data is included in Appendix B.

2.7 Stoichiometry

In chemical or biochemical reactions, atoms and molecules rearrange to form new groups. Mass and molar relationships between the reactants consumed and products formed can be determined using stoichiometric calculations. This information is deduced from correctly-written reaction equations and relevant atomic weights.

As an example, consider the principal reaction in alcohol fermentation: conversion of glucose to ethanol and carbon dioxide:

$$C_6H_{12}O_6 \rightarrow 2\ C_2H_6O + 2\ CO_2. \tag{2.33}$$

This reaction equation states that one molecule of glucose breaks down to give two molecules of ethanol and two molecules of carbon dioxide. Applying molecular weights, the equation shows that reaction of 180 g glucose produces 92 g ethanol and 88 g carbon dioxide. During chemical or biochemical reactions, the following two quantities are conserved:

(i) *total mass*, i.e. total mass of reactants = total mass of products; and
(ii) *number of atoms of each element*, e.g. the number of C, H and O atoms in the reactants = the number of C, H and O atoms, respectively, in the products.

Note that there is no corresponding law for conservation of moles; moles of reactants ≠ moles of products.

Example 2.4 Stoichiometry of amino acid synthesis

The overall reaction for microbial conversion of glucose to L-glutamic acid is:

$$\underset{\text{(glucose)}}{C_6H_{12}O_6} + NH_3 + {}^3/_2 O_2 \rightarrow \underset{\text{(glutamic acid)}}{C_5H_9NO_4} + CO_2 + 3\ H_2O.$$

What mass of oxygen is required to produce 15 g glutamic acid?

Solution:

Molecular weights: oxygen = 32
glutamic acid = 147

In the following equation, g glutamic acid is converted to gmol using a unity bracket for molecular weight, the stoichiometric equation is applied to convert gmol glutamic acid to gmol oxygen, and finally gmol oxygen are converted to g for the final answer:

$$15\text{ g glumatic acid}\cdot\left|\frac{1\text{ gmol glutamic acid}}{147\text{ g glutamic acid}}\right|\cdot\left|\frac{{}^3/_2\text{ gmol }O_2}{1\text{ gmol glutamic acid}}\right|\cdot\left|\frac{32\text{ g }O_2}{1\text{ gmol }O_2}\right| = 4.9\text{ g }O_2.$$

Therefore, 4.9 g oxygen is required. More oxygen will be needed if microbial growth also occurs.

By themselves, equations such as (2.33) suggest that all the reactants are converted into the products specified in the equation, and that the reaction proceeds to completion. This is often not the case for industrial reactions. Because the stoichiometry may not be known precisely, or in order to manipulate the reaction beneficially, reactants are not usually supplied in the exact proportions indicated by the reaction equation. Excess quantities of some reactants may be provided; this excess material is found in the product mixture once the reaction is stopped. In addition, reactants are often consumed in side reactions to make products not described by the principal reaction equation; these side-products also form part of the final reaction mixture. In these circumstances, additional information is needed before the amounts of product formed or reactants consumed can be calculated. Terms used to describe partial and branched reactions are outlined below.

(i) The *limiting reactant* is the reactant present in the smallest *stoichiometric* amount. While other reactants may be present in smaller absolute quantities, at the time when the last molecule of the limiting reactant is consumed, residual amounts of all reactants except the limiting reactant will be present in the reaction mixture. As an illustration, for the glutamic acid reaction of Example 2.4, if 100 g glucose, 17 g NH_3 and 48 g O_2 are provided for conversion, glucose will be the limiting reactant even though a greater mass of it is available compared with the other substrates.

(ii) An *excess reactant* is a reactant present in an amount in excess of that required to combine with all of the limiting reactant. It follows that an excess reactant is one remaining in the reaction mixture once all the limiting reactant is consumed. The *percentage excess* is calculated using the amount of excess material relative to the quantity required for complete consumption of the limiting reactant:

$$\%\ \text{excess} = \frac{\left(\begin{array}{c}\text{moles present} - \text{moles required to react}\\ \text{completely with the limiting reactant}\end{array}\right)}{\left(\begin{array}{c}\text{moles required to react}\\ \text{completely with the limiting reactant}\end{array}\right)} \times 100 \tag{2.34}$$

or

$$\%\ \text{excess} = \frac{\left(\begin{array}{c}\text{mass present} - \text{mass required to react}\\ \text{completely with the limiting reactant}\end{array}\right)}{\left(\begin{array}{c}\text{mass required to react}\\ \text{completely with the limiting reactant}\end{array}\right)} \times 100. \tag{2.35}$$

The *required* amount of a reactant is the stoichiometric quantity needed for complete conversion of the limiting reactant. In the above glutamic acid example, the required amount of NH_3 for complete conversion of 100 g glucose is 9.4 g; therefore if 17 g NH_3 are provided the percent excess NH_3 is 80%. Even if only part of the reaction actually occurs, required and excess quantities are based on the entire amount of the limiting reactant.

Other reaction terms are not as well defined with multiple definitions in common use:

(iii) *Conversion* is the fraction or percentage of a reactant converted into products.

(iv) *Degree of completion* is usually the fraction or percentage of the limiting reactant converted into products.

(v) *Selectivity* is the amount of a particular product formed as a fraction of the amount that would have been formed if all the feed material had been converted to that product.

(vi) *Yield* is the ratio of mass or moles of product formed to the mass or moles of reactant consumed. If more than one product or reactant is involved in the reaction, the particular compounds referred to must be stated, e.g. the yield of glutamic acid from glucose was 0.6 g g^{-1}. Because of the complexity of metabolism and the frequent occurrence of side reactions, yield is an important term in bioprocess analysis. Application of the yield concept for cell and enzyme reactions is described in more detail in Chapter 11.

Example 2.5 Incomplete reaction and yield

Depending on culture conditions, glucose can be catabolised by yeast to produce ethanol and carbon dioxide, or can be diverted into other biosynthetic reactions. An inoculum of yeast is added to a solution containing 10 g l^{-1} glucose. After some time only 1 g l^{-1} glucose remains while the concentration of ethanol is 3.2 g l^{-1}. Determine:

(a) the fractional conversion of glucose to ethanol; and
(b) the yield of ethanol from glucose.

Solution:

(a) To find the fractional conversion of glucose to ethanol, we must first determine exactly how much glucose was directed into ethanol biosynthesis. Using a basis of 1 litre and Eq. (2.33) for ethanol fermentation, we can calculate the mass of glucose required for synthesis of 3.2 g ethanol:

$$3.2\ \text{g ethanol} \cdot \left|\frac{1\ \text{gmol ethanol}}{46\ \text{g ethanol}}\right| \cdot \left|\frac{1\ \text{gmol glucose}}{2\ \text{gmol ethanol}}\right| \cdot \left|\frac{180\ \text{g glucose}}{1\ \text{gmol glucose}}\right| = 6.3\ \text{g glucose}.$$

Therefore, based on the total amount of glucose provided per litre (10 g), the fractional conversion of glucose to ethanol was 0.63. Based on the amount of glucose actually consumed per litre (9 g), the fractional conversion to ethanol was 0.70.

(b) Yield of ethanol from glucose is based on the total mass of glucose consumed. Since 9 g glucose was consumed per litre to provide 3.2 g l^{-1} ethanol, the yield of ethanol from glucose was 0.36 g g^{-1}. We can also conclude that, per litre, 2.7 g glucose was consumed but not used for ethanol synthesis.

2.8 Summary of Chapter 2

Having studied the contents of Chapter 2, you should:

(i) understand dimensionality and be able to convert units with ease;
(ii) understand the terms *mole, molecular weight, density, specific gravity, temperature* and *pressure*, know various ways of expressing *concentration* of solutions and mixtures, and be able to work simple problems involving these concepts;
(iii) be able to apply the ideal gas law;
(iv) know where to find physical and chemical property data in the literature; and
(v) understand reaction terms such as *limiting reactant, excess reactant, conversion, degree of completion, selectivity* and *yield*, and be able to apply stoichiometric principles to reaction problems.

Problems

2.1 Unit conversion

(a) Convert 1.5×10^{-6} centipoise to kg s^{-1} cm^{-1}.
(b) Convert 0.122 horsepower (British) to British thermal units per minute (Btu min^{-1}).
(c) Convert 670 mmHg ft^3 to metric horsepower h.
(d) Convert 345 Btu lb^{-1} to kcal g^{-1}.

2.2 Unit conversion

Using Eq. (2.1) for the Reynolds number, calculate *Re* for the following two sets of data:

Parameter	*Case 1*	*Case 2*
D	2 mm	1 in.
u	3 cm s^{-1}	1 m s^{-1}
ρ	25 lb ft^{-3}	12.5 kg m^{-3}
μ	10^{-6} cP	0.14×10^{-4} lb_m s^{-1} ft^{-1}

2.3 Dimensionless groups and property data

The rate at which oxygen is transported from gas phase to liquid phase is a very important parameter in fermenter design. A well-known correlation for transfer of gas is:

$$Sh = 0.31\, Gr^{1/3}\, Sc^{1/3}$$

where *Sh* is the Sherwood number, *Gr* is the Grashof number and *Sc* is the Schmidt number. These dimensionless numbers are defined as follows:

$$Sh = \frac{k_L D_b}{\mathscr{D}}$$

$$Gr = \frac{D_b^3\, \rho_G (\rho_L - \rho_G) g}{\mu_L^2}$$

$$Sc = \frac{\mu_L}{\rho_L \mathscr{D}}$$

where k_L is mass-transfer coefficient, D_b is bubble diameter, $\mathscr{D}$ is diffusivity of gas in the liquid, ρ_G is density of gas, ρ_L is density of liquid, μ_L is viscosity of liquid, and g is gravitational acceleration = 32.17 ft s^{-2}.

A gas sparger in a fermenter operated at 28°C and 1 atm produces bubbles of about 2 mm diameter. Calculate the value of the mass transfer coefficient, k_L. Collect property data from, e.g. *Chemical Engineers' Handbook*, and assume that the culture broth has similar properties to water. (Do you think this is a reasonable assumption?) Report the literature source for any property data used. State explicitly any other assumptions you make.

2.4 Mass and weight

The density of water is 62.4 lb_m ft^{-3}. What is the weight of 10 ft^3 of water:

(a) at sea level and 45° latitude?; and
(b) somewhere above the earth's surface where g = 9.76 m s^{-2}?

2.5 Dimensionless numbers

The Colburn equation for heat transfer is:

$$\left(\frac{h}{C_p G}\right)\left(\frac{C_p \mu}{k}\right)^{2/3} = \frac{0.023}{\left(\frac{DG}{\mu}\right)^{0.2}}$$

where C_p is heat capacity, Btu lb^{-1} $°\text{F}^{-1}$; μ is viscosity, lb h^{-1} ft^{-1}; k is thermal conductivity, Btu h^{-1} ft^{-2} $(°\text{F ft}^{-1})^{-1}$; D is pipe diameter, ft; and G is mass velocity per unit area, lb h^{-1} ft^{-2}.

The Colburn equation is dimensionally consistent. What are the units and dimensions of the heat-transfer coefficient, h?

2.6 Dimensional homogeneity and g_c

Two students have reported different versions of the dimensionless power number N_P used to relate fluid properties to the power required for stirring:

(i) $N_P = \dfrac{Pg}{\rho N_i^3 D_i^5}$; and

(ii) $N_P = \dfrac{Pg_c}{\rho N_i^3 D_i^5}$

where P is power, g is gravitational acceleration, ρ is fluid density, N_i is stirrer speed, D_i is stirrer diameter and g_c is the force unity bracket. Which equation is correct?

2.7 Molar units

If a bucket holds 20.0 lb NaOH, how many:

(a) lbmol NaOH;
(b) gmol NaOH; and
(c) kgmol NaOH

does it contain?

2.8 Density and specific gravity

(a) The specific gravity of nitric acid is $1.5129^{20°\text{C}}_{4°\text{C}}$.
 (i) What is its density at 20°C in kg m^{-3}?
 (ii) What is its molar specific volume?
(b) The volumetric flow rate of carbon tetrachloride (CCl_4) in a pipe is 50 cm^3 min^{-1}. The density of CCl_4 is 1.6 g cm^{-3}.
 (i) What is the mass flow rate of CCl_4?
 (ii) What is the molar flow rate of CCl_4?

2.9 Molecular weight

Calculate the average molecular weight of air.

2.10 Mole fraction

A solution contains 30 wt% water, 25 wt% ethanol, 15 wt% methanol, 12 wt% glycerol, 10 wt% acetic acid and 8 wt% benzaldehyde. What is the mole fraction of each component?

2.11 Temperature scales

What is −40°F in degrees centigrade? degrees Rankine? kelvin?

2.12 Pressure scales

(a) The pressure gauge on an autoclave reads 15 psi. What is the absolute pressure in the chamber in psi? in atm?
(b) A vacuum gauge reads 3 psi. What is the pressure?

2.13 Stoichiometry and incomplete reaction

For production of penicillin ($C_{16}H_{18}O_4N_2S$) using *Penicillium* mould, glucose ($C_6H_{12}O_6$) is used as substrate, and phenylacetic acid ($C_8H_8O_2$) is added as precursor. The stoichiometry for overall synthesis is:

$$1.67\, C_6H_{12}O_6 + 2\, NH_3 + 0.5\, O_2 + H_2SO_4 + C_8H_8O_2 \rightarrow C_{16}H_{18}O_4N_2S + 2\, CO_2 + 9\, H_2O.$$

(a) What is the maximum theoretical yield of penicillin from glucose?
(b) When results from a particular penicillin fermentation were analysed, it was found that 24% of the glucose had been used for growth, 70% for cell maintenance activities (such as membrane transport and macromolecule turnover), and only 6% for penicillin synthesis. Calculate the yield of penicillin from glucose under these conditions.
(c) Batch fermentation under the conditions described in (b) is carried out in a 100-litre tank. Initially, the tank is filled with nutrient medium containing 50 g l^{-1} glucose and 4 g l^{-1} phenylacetic acid. If the reaction is stopped when the glucose concentration is 5.5 g l^{-1}, determine:
 (i) which is the limiting substrate if NH_3, O_2 and H_2SO_4 are provided in excess;
 (ii) the total mass of glucose used for growth;

(iii) the amount of penicillin produced; and
(iv) the final concentration of phenylacetic acid.

2.14 Stoichiometry, yield and the ideal gas law

Stoichiometric equations are used to represent growth of microorganisms provided a 'molecular formula' for the cells is available. The molecular formula for biomass is obtained by measuring the amounts of C, N, H, O and other elements in cells. For a particular bacterial strain, the molecular formula was determined to be $C_{4.4}H_{7.3}O_{1.2}N_{0.86}$.

These bacteria are grown under aerobic conditions with hexadecane ($C_{16}H_{34}$) as substrate. The reaction equation describing growth is:

$$C_{16}H_{34} + 16.28\,O_2 + 1.42\,NH_3 \rightarrow 1.65\,C_{4.4}H_{7.3}O_{1.2}N_{0.86} + 8.74\,CO_2 + 13.11\,H_2O.$$

(a) Is the stoichiometric equation balanced?
(b) Assuming 100% conversion, what is the yield of cells from hexadecane in g g^{-1}?
(c) Assuming 100% conversion, what is the yield of cells from oxygen in g g^{-1}?
(d) You have been put in charge of a small fermenter for growing the bacteria and aim to produce 2.5 kg of cells for inoculation of a pilot-scale reactor.
   (i) What minimum amount of hexadecane substrate must be contained in your culture medium?
   (ii) What must be the minimum concentration of hexadecane in the medium if the fermenter working volume is 3 cubic metres?
   (iii) What minimum volume of air at 20°C and 1 atm pressure must be pumped into the fermenter during growth to produce the required amount of cells?

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Suggestions for Further Reading

Units and Dimensions (see also refs 1 and 3)

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3

Presentation and Analysis of Data

Quantitative information is fundamental to scientific and engineering analysis. Information about bioprocesses, such as the amount of substrate fed into the system, the operating conditions, and properties of the product stream, is obtained by measuring pertinent physical and chemical variables. In industry, data are collected for equipment design, process control, trouble-shooting and economic evaluations. In research, experimental data are used to develop new theories and test theoretical predictions. In either case, quantitative interpretation of data is absolutely essential for making rational decisions about the system under investigation. The ability to extract useful and accurate information from data is an important skill for any scientist. Professional presentation and communication of results is also required.

Techniques for data analysis must take into account the existence of error in measurements. Because there is always an element of uncertainty associated with measured data, interpretation calls for a great deal of judgement. This is especially the case when critical decisions in design or operation of processes depend on data evaluation. Although computers and calculators make data processing less tedious, the data analyst must possess enough perception to use these tools effectively.

This chapter discusses sources of error in data and methods of handling errors in calculations. Presentation and analysis of data using graphs and equations and presentation of process information using flow sheets are described.

3.1 Errors in Data and Calculations

Measurements are never perfect. Experimentally-determined quantities are always somewhat inaccurate due to measurement error; absolutely 'correct' values of physical quantities (time, length, concentration, temperature, etc.) cannot be found. The significance or reliability of conclusions drawn from data must take measurement error into consideration. Estimation of error and principles of error propagation in calculations are important elements of engineering analysis and help prevent misleading representation of data. General principles for estimation and expression of errors are discussed in the following sections.

3.1.1 Significant Figures

Data used in engineering calculations vary considerably in accuracy. Economic projections may estimate market demand for a new biotechnology product to within ±100%; on the other hand, some properties of materials are known to within ±0.0001% or less. The uncertainty associated with quantities should be reflected in the way they are written. The number of figures used to report a measured or calculated variable is an indirect indication of the precision to which that variable is known. It would be absurd, for example, to quote the estimated income from sales of a new product using ten decimal places. Nevertheless, the mistake of quoting too many figures is not uncommon; display of superfluous figures on calculators is very easy but should not be transferred to scientific reports.

A *significant figure* is any digit, 1–9, used to specify a number. Zero may also be a significant figure when it is not used merely to locate the position of the decimal point. For example, the numbers 6304, 0.004321, 43.55 and 8.063×10^{10} each contain four significant figures. For the number 1200, however, there is no way of knowing whether or not the two zeros are significant figures; a direct statement or an alternative way of expressing the number is needed. For example, 1.2×10^3 has two significant figures, while 1.200×10^3 has four.

A number is rounded to *n* significant figures using the following rules:

(i) If the number in the $(n+1)$th position is less than 5, discard all figures to the right of the nth place.
(ii) If the number in the $(n+1)$th position is greater than 5, discard all figures to the right of the nth place, and increase the nth digit by 1.
(iii) If the number in the $(n+1)$th position is exactly 5, discard all figures to the right of the nth place, and increase the nth digit by 1.

For example, when rounding off to four significant figures:

1.426348 becomes 1.426;
1.426748 becomes 1.427; and
1.4265 becomes 1.427.

The last rule is not universal but is engineering convention; most electronic calculators and computers round up halves. Generally, rounding off means that the value may be wrong by up to 5 units in the next number-column not reported. Thus, 10.77 kg means that the mass lies somewhere between 10.765 kg and 10.775 kg, whereas 10.7754 kg represents a mass between 10.77535 kg and 10.77545 kg. These rules apply only to quantities based on measured values; some numbers used in calculations refer to precisely known or counted quantities. For example, there is no error associated with the number $^1/_2$ in the equation for kinetic energy:

$$\text{kinetic energy} = E_k = {}^1/_2\, M v^2$$

where M is mass and v is velocity.

It is good practice during calculations to carry along one or two extra significant figures for combination during arithmetic operations; final rounding-off should be done only at the end. How many figures should we quote in the final answer? There are several rules-of-thumb for rounding off after calculations, so rigid adherence to all rules is not always possible. However as a guide, after multiplication or division, the number of significant figures in the result should equal the smallest number of significant figures of any of the quantities involved in the calculation. For example:

$$(6.681 \times 10^{-2})\,(5.4 \times 10^{9}) = 3.608 \times 10^{8} \rightarrow 3.6 \times 10^{8}$$

and

$$\frac{6.16}{0.054677} = 112.6616310 \rightarrow 113.$$

For addition and subtraction, look at the position of the last significant figure in each number relative to the decimal point. The position of the last significant figure in the result should be the same as that most to the left, as illustrated below:

$$24.335 + 3.90 + 0.00987 = 28.24487 \rightarrow 28.24$$

and

$$121.808 - 112.87634 = 8.93166 \rightarrow 8.932.$$

3.1.2 Absolute and Relative Uncertainty

Uncertainty associated with measurements can be stated more explicitly than is possible using the rules of significant figures. For a particular measurement, we should be able to give our best estimate of the parameter value, and the interval representing its range of uncertainty. For example, we might be able to say with confidence that the prevailing temperature lies between 23.7°C and 24.3°C. Another way of expressing this result is 24 ± 0.3°C. The value ± 0.3°C is known as the *uncertainty* or *error* of the measurement, and allows us to judge the quality of the measuring process. Since ± 0.3°C represents the actual temperature range by which the reading is uncertain, it is known as the *absolute error*. An alternative expression for 24 ± 0.3°C is 24°C ± 1.25%; in this case the *relative error* is ± 1.25%.

Because most values of uncertainty must be estimated rather than measured, there is a rule-of-thumb that magnitudes of errors should be given with only one or two significant figures. A flow rate may be expressed as 146 ± 13 gmol h^{-1}, even though this means that two figures, i.e. the '4' and the '6' in the result, are uncertain. The number of digits used to express the result should be compatible with the magnitude of its estimated error. For example, in the statement: 2.1437 ± 0.12 grams, the estimated uncertainty of 0.12 grams shows that the last two digits in the result are superfluous. Use of more than three significant figures in this case gives a false impression of accuracy.

There are rules for combining errors during mathematical operations. The uncertainty associated with calculated results is found from the errors associated with the raw data. For addition and subtraction the rule is: *add absolute errors*. The total of the absolute errors becomes the absolute error associated with the final answer. For example, the sum of 1.25 ± 0.13 and 0.973 ± 0.051 is:

$$(1.25 + 0.973) \pm (0.13 + 0.051) = 2.22 \pm 0.18 = 2.22 \pm 8.1\%.$$

Considerable loss of accuracy can occur after subtraction, especially when two large numbers are subtracted to give an answer of small numerical value. Because the absolute error after subtraction of two numbers always increases, the relative error associated with a small-number answer can be very great. For example, consider the difference between two numbers, each with small relative error: 12 736 ± 0.5% and 12 681 ± 0.5%. For subtraction, the absolute errors are added:

$$(12\,736 \pm 64) - (12\,681 \pm 63) = 55 \pm 127 = 55 \pm 230\%.$$

Even though it could be argued that the two errors might almost cancel each other (if one were +64 and the other were −63, for example), we can never be certain that this would occur. 230% represents the worst case or *maximum possible error.* For measured values, any small number obtained by subtraction of two large numbers must be examined carefully and with justifiable suspicion. Unless explicit errors are reported, the large uncertainty associated with results can go unnoticed.

For multiplication and division: *add relative errors.* The total of the relative errors becomes the relative error associated with the answer. For example, 164 ± 1 divided by 790 ± 20 is the same as 164 ± 0.61% divided by 790 ± 2.5%:

$$({}^{790}/_{164}) \pm (2.5 + 0.61)\% = 4.82 \pm 3.1\% = 4.82 \pm 0.15.$$

Propagation of errors in more complex expressions will not be discussed here; more information and rules for combining errors can be found in other references [1–3].

So far we have considered the error occurring in a single observation. However, as discussed below, better estimates of errors are obtained by taking repeated measurements. Because this approach is useful only for certain types of measurement error, let us consider the various sources of error in experimental data.

3.1.3 Types of Error

There are two broad classes of measurement error: *systematic* and *random.* A systematic error is one which affects all measurements of the same variable in the same way. If the cause of systematic error is identified, it can be accounted for using a correction factor. For example, errors caused by an imperfectly calibrated analytical balance may be identified using standard weights; measurements with the balance can then be corrected to compensate for the error. Systematic errors easily go undetected; performing the same measurement using different instruments, methods and observers is required to detect systematic error [4].

Random or accidental errors are due to unknown causes. Random errors are present in almost all data; they are revealed when repeated measurements of an unchanging quantity give a 'scatter' of different results. As outlined in the next section, scatter from repeated measurements is used in statistical analysis to quantify random error. The term *precision* refers to the *reliability* or *reproducibility* of data, and indicates the extent to which a measurement is free from random error. *Accuracy,* on the other hand, requires both random and systematic errors to be small. Repeated weighings using a poorly-calibrated balance can give results that are very precise (because each reading is similar); however the result would be inaccurate because of the incorrect calibration and systematic error.

During experiments, large, isolated, one-of-a-kind errors can also occur. This type of error is different from the systematic and random errors mentioned above and can be described as a 'blunder'. Accounting for blunders in experimental data requires knowledge of the experimental process and judgement about the likely accuracy of the measurement.

3.1.4 Statistical Analysis

Measurements containing random errors but free of systematic errors and blunders can be analysed using statistical procedures. Details are available in standard texts, e.g. [5]; only the most basic techniques for statistical treatment will be described here. From readings containing random error, we aim to find the best estimate of the variable measured, and to quantify the extent to which random error affects the data.

In the following analysis, errors are assumed to follow a normal or Gaussian distribution. Normally-distributed random errors in a single measurement are just as likely to be positive as negative; thus, if an infinite number of repeated measurements were made of the same variable, random error would completely cancel out from the arithmetic mean of these values. For less than an infinite number of observations, the arithmetic mean of repeated measurements is still regarded as the best estimate of the variable, provided each measurement is made with equal care and under identical conditions. Taking replicate measurements is therefore standard practice in science; whenever possible, several readings of each datum point should be obtained. For variable x measured n times, the *arithmetic mean* is calculated as follows:

$$\bar{x} = \text{mean value of } x = \frac{\sum^{n} x}{n} = \frac{x_1 + x_2 + x_3 + \ldots x_n}{n}. \tag{3.1}$$

As indicated, the symbol $\sum^{n}$ represents the sum of n values; $\sum^{n} x$ means the sum of n values of parameter x.

In addition to the mean, we need some measure of the precision of the measurements; this is obtained by considering the scatter of individual values about the mean. The deviation of an individual value from the mean is known as the *residual*; an example of a residual is $(x_1 - \bar{x})$ where x_1 is a measurement in a set of replicates. The most useful indicator of the magnitude of the residuals is the *standard deviation.* For a set of experimental data, standard deviation σ is calculated as follows:

$$\sigma = \sqrt{\frac{\sum^{n}(x-\bar{x})^2}{n-1}}$$
$$= \sqrt{\frac{(x_1-\bar{x})^2+(x_2-\bar{x})^2+(x_3-\bar{x})^2+\ldots(x_n-\bar{x})^2}{n-1}}. \quad (3.2)$$

Eq. (3.2) is the definition used by most modern statisticians and manufacturers of electronic calculators; σ as defined in Eq. (3.2) is sometimes called the *sample standard deviation.*

Therefore, to report the results of repeated measurements, we quote the mean as the best estimate of the variable, and the standard deviation as a measure of the confidence we place in the result. The units and dimensions of the mean and standard deviation are the same as those of x. For less than an infinite number of repeated measurements, the mean and standard deviation calculated using one set of observations will produce a different result from that determined using another set. It can be shown mathematically that values of the mean and standard deviation become more reliable as n increases; taking replicate measurements is therefore standard practice in science. A compromise is usually struck between the conflicting demands of precision and the time and expense of experimentation; sometimes it is impossible to make a large number of replicate measurements. Sample size should always be quoted when reporting the outcome of statistical analysis. When substantial improvement in the accuracy of the mean and standard deviation is required, this is generally more effectively achieved by improving the intrinsic accuracy of the measurement rather than by just taking a multitude of repeated readings.

Example 3.1 Mean and standard deviation

The final concentration of L-lysine produced by a regulatory mutant of *Brevibacterium lactofermentum* is measured 10 times. The results in g l^{-1} are: 47.3, 51.9, 52.2, 51.8, 49.2, 51.1, 52.4, 47.1, 49.1 and 46.3. How should the lysine concentration be reported?

Solution:
For this sample, $n = 10$. From Eq. (3.1):

$$\bar{x} = \frac{47.3+51.9+52.2+51.8+49.2+51.1+52.4+47.1+49.1+46.3}{10}$$
$$= 49.84 \text{ g l}^{-1}.$$

Substituting this result into Eq. (3.2) gives:

$$\sigma = \sqrt{\frac{49.24}{9}} = 2.34 \text{ g l}^{-1}.$$

Therefore, from 10 repeated measurements, the lysine concentration was 49.8 g l^{-1} with standard deviation 2.3 g l^{-1}.

Methods for combining standard deviations in calculations are discussed elsewhere [1, 2]. Remember that standard statistical analysis does not account for systematic error; parameters such as the mean and standard deviation are useful only if the error in measurements is random. *The effect of systematic error cannot be minimised using standard statistical analysis or by collecting repeated measurements.*

3.2 Presentation of Experimental Data

Experimental data are often collected to examine relationships between variables. The role of these variables in the experimental process is clearly defined. *Dependent variables* or *response variables* are uncontrolled during the experiment; dependent variables are measured as they respond to changes in one or more *independent variables* which are controlled or fixed. For example, if we wanted to determine how UV radiation affects the frequency of mutation in a culture, radiation dose would be the independent variable and number of mutants the dependent variable.

There are three general methods for presenting data:

(i) tables;
(ii) graphs; and
(iii) equations.

Each has its own strengths and weaknesses. Tables listing data have highest accuracy, but can easily become too long and the overall result or trend of the data cannot be readily visualised. Graphs or plots of data create immediate visual impact since relationships between variables are represented directly. Graphs also allow easy interpolation of data, which can be difficult with tables. By convention, independent variables are plotted along the *abscissa* (the X-axis), while one or more dependent variables are plotted along the *ordinate* (Y-axis). Plots show at a glance the general pattern of data, and can help identify whether there are anomalous points; it is good practice to plot raw experimental data as they are being measured. In addition, graphs can be used directly for quantitative data analysis.

Physical phenomena can be represented using equations or *mathematical models*; for example, balanced growth of microorganisms is described using the model:

$$x = x_0 \, e^{\mu t} \tag{3.3}$$

where x is the cell concentration at time t, x_0 is the initial cell concentration, and μ is the specific growth rate. Mathematical models can be either mechanistic or empirical. *Mechanistic models* are founded on theoretical assessment of the phenomenon being measured. An example is the Michaelis–Menten equation for enzyme reaction:

$$v = \frac{v_{max}\, s}{K_m + s} \tag{3.4}$$

where v is rate of reaction, v_{max} is maximum rate of reaction, K_m is the Michaelis constant, and s is substrate concentration. The Michaelis–Menten equation is based on a loose analysis of reactions supposed to occur during simple enzyme catalysis. On the other hand, *empirical models* are used when no theoretical hypothesis can be postulated. Empirical models may be the only feasible option for correlating data from complicated processes. As an example, the following correlation has been developed to relate the power required to stir aerated liquids to that required in non-aerated systems:

$$\frac{P_g}{P_0} = 0.10 \left(\frac{F_g}{N_i\, V} \right)^{-0.25} \left(\frac{N_i^2\, D_i^4}{g\, W_i\, V^{2/3}} \right)^{-0.20} \tag{3.5}$$

In Eq. (3.5) P_g is power consumption with sparging, P_0 is power consumption without sparging, F_g is volumetric gas flow rate, N_i is stirrer speed, V is liquid volume, D_i is impeller diameter, g is gravitational acceleration, and W_i is impeller blade width. There is no easy theoretical explanation for this relationship; the equation is based on many observations using different impellers, gas flow rates and rates of stirring. Equations such as Eq. (3.5) are a short, concise means for communicating the results of a large number of experiments. However, they are one step removed from the raw data and can be only an approximate representation of all the information collected.

3.3 Data Analysis

Once experimental data are collected, what we do with them depends on the information being sought. Data are generally collected for one or more of the following reasons:

(i) to visualise the general trend of influence of one variable on another;
(ii) to test the applicability of a particular model to a process;
(iii) to estimate the value of coefficients in process models; and
(iv) to develop new empirical models.

Analysis of data would be enormously simplified if each datum point did not contain error. For example, after an experiment in which the apparent viscosity of a mycelial broth is measured as a function of temperature, if all points on a plot of viscosity versus temperature lay perfectly along a line and there were no scatter, it would be very easy to determine unequivocally the relationship between the variables. In reality, however, procedures for data analysis must be closely linked with statistical mathematics to account for random errors in measurement. Despite their importance, detailed description of statistical analysis is beyond the scope of this book; there are entire texts devoted to the subject. Rather than presenting methods for data analysis as such, the following sections discuss some of the ideas behind interpretation of experiments. Once the general approach is understood, the actual procedures involved can be obtained from the references listed at the end of this chapter.

As we shall see, interpreting experimental data requires a great deal of judgement and sometimes involves difficult decisions. Nowadays most scientists and engineers have access to computers or calculators equipped with software for data processing. These facilities are very convenient and have removed much of the tedium associated with statistical analysis. There is a danger, however, that software packages are applied without appreciation of inherent assumptions in the analysis or its mathematical limitations. Thus, the user cannot know how valuable or otherwise are the generated results.

As already mentioned in Section 3.1.4, standard statistical methods consider only random error, not systematic error. In practical terms, this means that most procedures for data processing are unsuitable when errors are due to poor instrument calibration, repetition of the same mistakes in measurement, or preconceived ideas about the expected result. All effort must be made to eliminate these types of error before treating the data. As also noted in Section 3.1.4, the reliability of results from statistical analysis improves if many readings are taken. No amount of sophisticated mathematical or other type of manipulation can make up for sparse, inaccurate data.

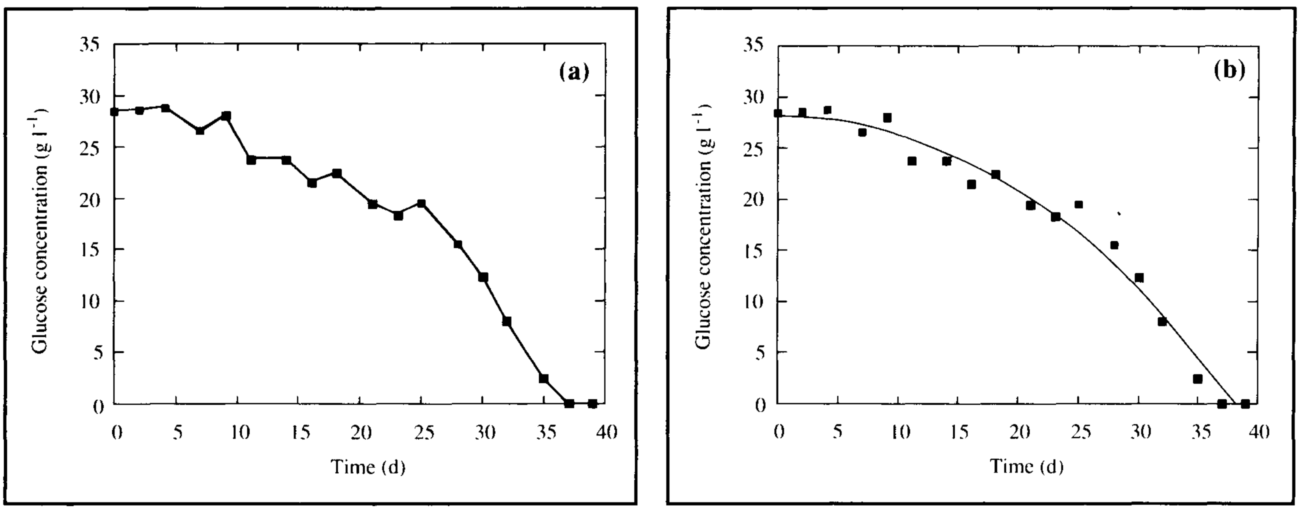
3.3.1 Trends

Consider the data plotted in Figure 3.1 representing consumption of glucose during batch culture of plant cells. If there were serious doubt about the trend of the data we could present the plot as a scatter of individual points without any lines drawn through them. Sometimes data are simply connected using line segments as shown in Figure 3.1(a); the problem with this representation is that it suggests that the ups and downs of glucose concentration are real. If, as with these data, we are assured there is a progressive downward trend in sugar concentration despite the occasional apparent increase, we should *smooth* the data by drawing a curve through the points as shown in Figure 3.1(b). Smoothing moderates the effects of experimental error. By drawing a particular curve we are indicating that, although the scatter of points is considerable, we believe the actual behaviour of the system is smooth and continuous, and that all of the data without experimental error would lie on that line. Usually there is great flexibility as to where the smoothing curve is placed, and several questions arise. To which points should the curve pass closest? Should all the data points be included, or are some points clearly in error? It soon becomes apparent that many equally-acceptable curves can be drawn through the data.

Various techniques are available for smoothing. A smooth line can be drawn freehand or with French or flexible curves and other drafting equipment; this is called *hand smoothing*. Procedures for minimising bias during hand smoothing can be applied; some examples are discussed further in Chapter 11. The danger involved in smoothing manually: that we tend to smooth the expected response into the data, is well recognised. Another method is to use a computer software package; this is called *machine smoothing*. Computer routines, by smoothing data according to pre-programmed mathematical or statistical principles, eliminate the subjective element but are still capable of introducing bias into the results. For example, abrupt changes in the trend of data are generally not recognised by statistical analysis. The advantage of hand smoothing is that judgements about the significance of individual data points can be taken into account.

Choice of curve is critical if smoothed data are to be applied in subsequent analysis. The data of Figure 3.1 may be used to calculate the *rate* of glucose consumption as a function of time; procedures for this type of analysis are described further in Chapter 11. In rate analysis, different smoothing curves can lead to significantly different results. Because final

Figure 3.1 Glucose concentration during batch culture of plant cells; (a) data connected directly by line segments; (b) data represented by a smooth curve.



interpretation of the data depends on decisions made during smoothing, it is important to minimise any errors introduced. One obvious way of doing this is to take as many readings as possible; when smooth curves are drawn through too few points it is very difficult to justify the smoothing process.

3.3.2 Testing Mathematical Models

Most applications of data analysis involve correlating measured data with existing mathematical models. The model proposes some functional relationship between two or more variables; our primary objective is to compare the properties of the model with those of the experimental system.

As an example, consider Figure 3.2 which shows the results from experiments in which rates of heat production and oxygen consumption are measured for several microbial cultures [6]. Although there is considerable scatter in these data we could be led to believe that the relationship between rate of heat production and rate of oxygen consumption is linear, as indicated by the straight line in Figure 3.2. However there is an infinite number of ways to represent any set of data; how do we know that this linear relationship is the best? For instance we might consider whether the data could be fitted by the curve shown in Figure 3.3. This non-linear, oscillating model seems to follow the data reasonably well; should we conclude that there is a more complex non-linear relationship between heat production and oxygen consumption?

Ultimately, we cannot know if a particular relationship holds between variables. This is because we can only test a selection of possible relationships and determine which *of them* fits closest to the data. We can determine which model, linear or oscillating, is the *better* representation of the data in Figure 3.2, but we can never conclude that the relationship between the variables is *actually* linear or oscillating. This fundamental limitation of data analysis has important consequences and must be accommodated in our approach. We must start off with a hypothesis about how the parameters are related and use data to determine whether this hypothesis is supported. A basic tenet in the philosophy of science is that it is only possible to *disprove* hypotheses by showing that experimental data do not conform to the model. The idea that the primary business of science is to falsify theories, not verify

Figure 3.2 Correlation between rate of heat evolution and rate of oxygen consumption for a variety of microbial fermentations. (○) *Escherichia coli*, glucose medium; (◆) *Candida intermedia*, glucose medium; (△) *C. intermedia*, molasses medium; (▽) *Bacillus subtilis*, glucose medium; (■) *B. subtilis*, molasses medium; (◓) *B. subtilis*, soybean-meal medium; (◇) *Aspergillus niger*, glucose medium; (●) *Asp. niger*, molasses medium. (From C.L. Cooney, D.I.C. Wang and R.I. Mateles, Measurement of heat evolution and correlation with oxygen consumption during microbial growth, *Biotechnol. Bioeng.* **11**, 269–281; Copyright © 1968. Reprinted by permission of John Wiley and Sons, Inc.)

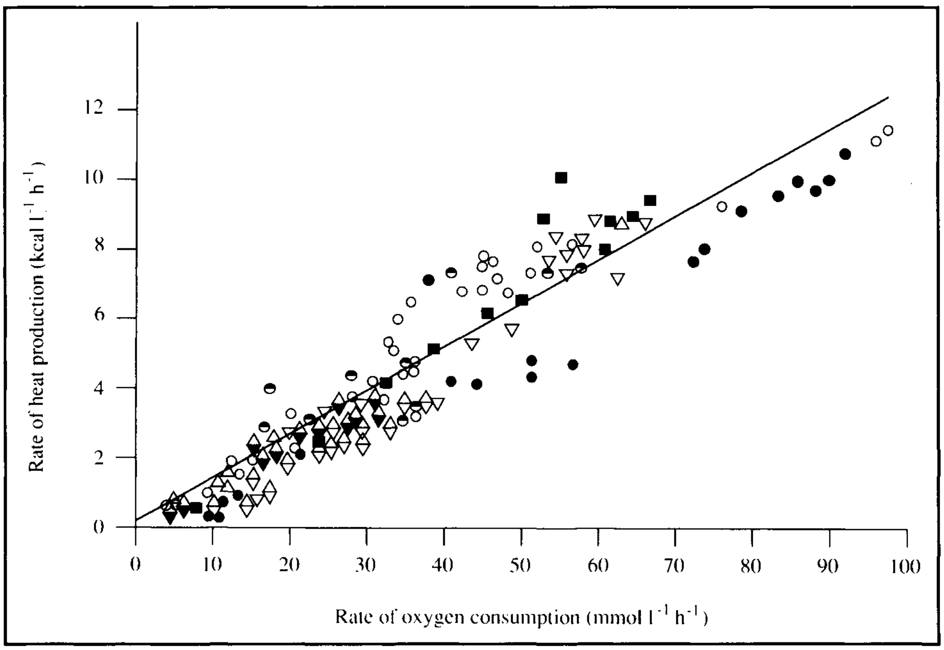
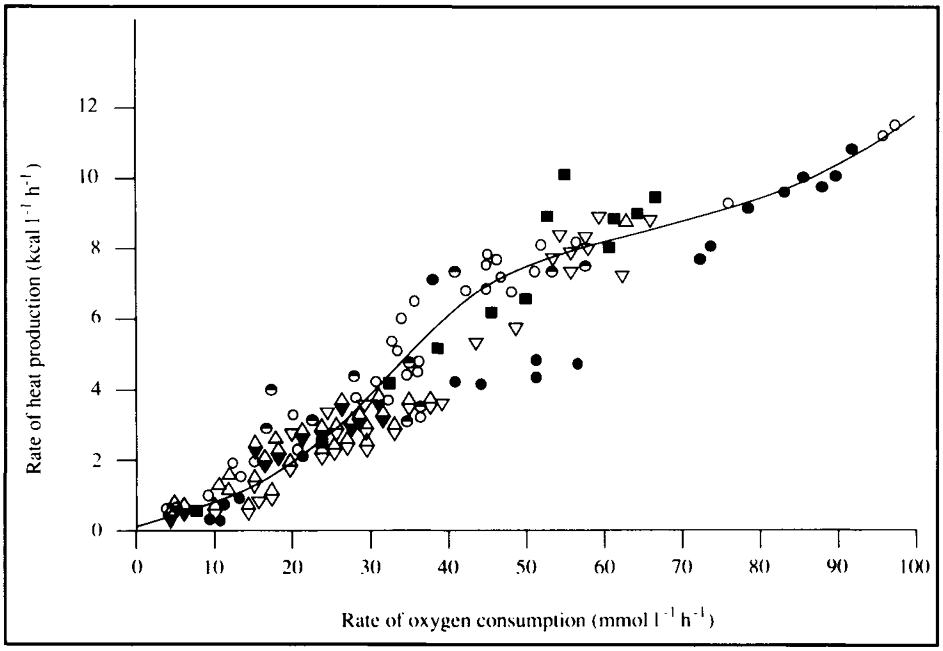


Figure 3.3 Alternative non-linear correlation for the data of Figure 3.2.



them, was developed this century by Austrian philosopher, Karl Popper. Popper's philosophical excursions into the meaning of scientific truth make extremely interesting reading, e.g. [7, 8]; his theories have direct application in analysis of measured data. We can never deduce with absolute certainty the physical relationships between variables using experiments. Language used to report the results of data analysis must reflect these limitations; particular models used to correlate data cannot be described as 'correct' or 'true' descriptions of the system, only 'satisfactory' or 'adequate' for our purposes and measurement precision.

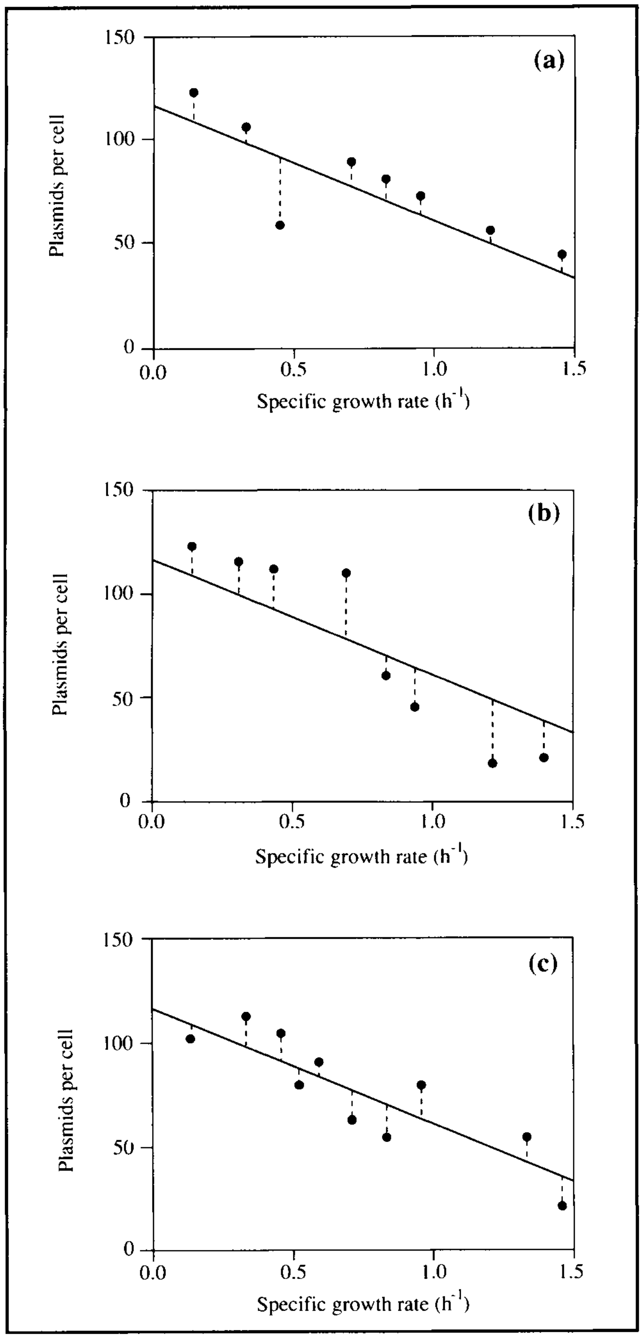
3.3.3 Goodness of Fit: Least-Squares Analysis

Determining how well data conform to a particular model requires numerical procedures. Generally, these techniques rely on measurement of the deviations or residuals of each datum point from the curve or line representing the model being tested. For example, residuals after correlating cell plasmid content with growth rate using a linear model are shown by the dashed lines in Figure 3.4. A curve or line producing small residuals is considered a good fit of the data.

A popular technique for locating the line or curve which minimises the residuals is *least-squares analysis*. This statistical procedure is based on minimising the *sum of squares of the residuals*. There are several variations of the procedure: Legendre's method minimises the sum-of-squares of residuals of the dependent variable; Gauss's and Laplace's methods minimise the sum of squares of weighted residuals where the weighting factors depend on the scatter of replicate data points. Each method gives different results; it should be remembered that the curve of 'best' fit is ultimately a matter of opinion. For example, by minimising the sum of squares of the residuals, least-squares analysis could produce a curve which does not pass close to particular data points known beforehand to be more accurate than the rest. Alternatively, we could choose to define the best fit as that which minimises the absolute values of the residuals, or the sum of the residuals raised to the fourth power. The decision to use the sum of squares is an arbitrary one; many alternative approaches are equally valid mathematically.

As well as minimising the residuals, other factors must be taken into account when correlating data. First, the curve used to fit the data should create approximately equal numbers of positive and negative residuals. As shown in Figure 3.4(a), when there are more positive than negative deviations from the points, even though the sum of the residuals is relatively small, the line representing the data cannot be considered a good fit. The fit is also poor when, as shown in Figure 3.4(b), all the positive residuals occur at low values of the independent variable while the negative residuals occur at high values. There should be no significant correlation of the residuals with either the dependent or independent variable. The best

Figure 3.4 Residuals in plasmid content after fitting a straight line to experimental data.



straight-line fit is shown in Figure 3.4(c); the residuals are relatively small, well distributed in both positive and negative directions, and there is no relationship between the residuals and either variable.

Some data sets contain one or more points which deviate substantially from predicted values, more than is expected from 'normal' random experimental error. These points known as 'outliers' have large residuals and, therefore, strongly influence regression methods using the sum-of-squares approach. It is usually inappropriate to eliminate outliers; they may be legitimate experimental results reflecting the true behaviour of the system and could be explained and fitted using an alternative model not yet considered. The best way to handle outliers is to analyse the data with and without the aberrant values to make sure their elimination does not influence discrimination between models. It must be emphasised that only one point at a time and only very rare data points, if any, should be eliminated from data sets.

Measuring individual residuals and applying least-squares analysis would be very useful in determining which of the two curves in Figures 3.2 and 3.3 fits the data more closely. However, as well as mathematical considerations, other factors can influence choice of model for experimental data. Consider again the data of Figures 3.2 and 3.3. Unless the fit obtained with the oscillatory model were very much improved compared with the linear model, we might prefer the straight-line correlation because it is simple, and because it conforms with what we know about microbial metabolism and the thermodynamics of respiration. It is difficult to find a credible theoretical justification for representing the relationship with an oscillating curve, so we could be persuaded to reject the non-linear model even though it fits the data reasonably well. Choosing between models on the basis of supposed mechanism requires a great deal of judgement. Since we cannot know for sure what the relationship is between the two parameters, choosing between models on the basis of supposed mechanism brings in an element of bias. This type of presumptive judgement is the reason why it is so difficult to overturn established scientific theories; even if data are available to support a new hypothesis there is a tendency to reject it because it does not agree with accepted theory. Nevertheless, if we wanted to fly in the face of convention and argue that an oscillatory relationship between rates of heat evolution and oxygen consumption is more reasonable than a straight-line relationship, we would undoubtedly have to support our claim with more evidence than the data shown in Figure 3.3.

3.3.4 Linear and Non-Linear Models

A straight line is represented by the equation:

$$y = Ax + B. \tag{3.6}$$

B is the *intercept* of the straight line on the ordinate; A is the *slope*. A and B are also called the *coefficients*, *parameters* or *adjustable parameters* of Eq. (3.6). Once a straight line is drawn, A is found by taking any two points (x_1, y_1) and (x_2, y_2) on the line, and calculating:

$$A = \frac{(y_2 - y_1)}{(x_2 - x_1)}. \tag{3.7}$$

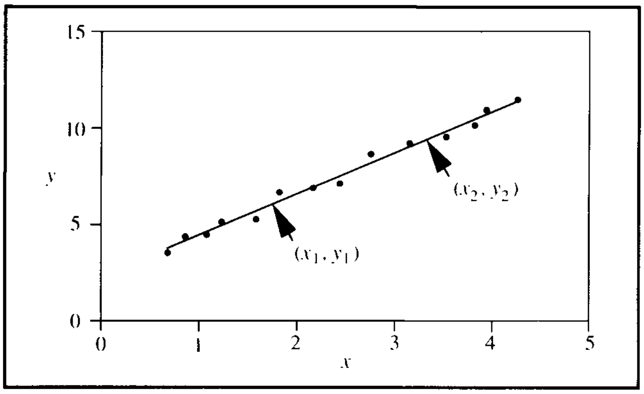
As indicated in Figure 3.5, (x_1, y_1) and (x_2, y_2) are points *on the line* through the data; they are not measured data values. Once A is known, B is calculated as:

$$B = y_1 - A\,x_1 \quad \text{or} \quad B = y_2 - A\,x_2. \tag{3.8}$$

Suppose we measure n pairs of values of two variables x and y, and a plot of the dependent variable y versus the independent variable x suggests a straight-line relationship. In testing correlation of the data with Eq. (3.6), changing the values of A and B will affect how well the model fits the data. Values of A and B giving the best straight line are determined by *linear regression* or *linear least-squares analysis*. This procedure is one of the most frequently used in data analysis; linear-regression routines are part of many computer packages and are available on hand-held calculators. Linear regression methods fit data by finding the straight line which minimises the sum of squares of the residuals. Details of the method can be found in statistics texts [5, 9, 10].

Because linear regression is so accessible, it can be readily applied without proper regard for its appropriateness or the assumptions incorporated in its method. Unless the following points are considered before using regression analysis, biased estimates of parameter values will be obtained.

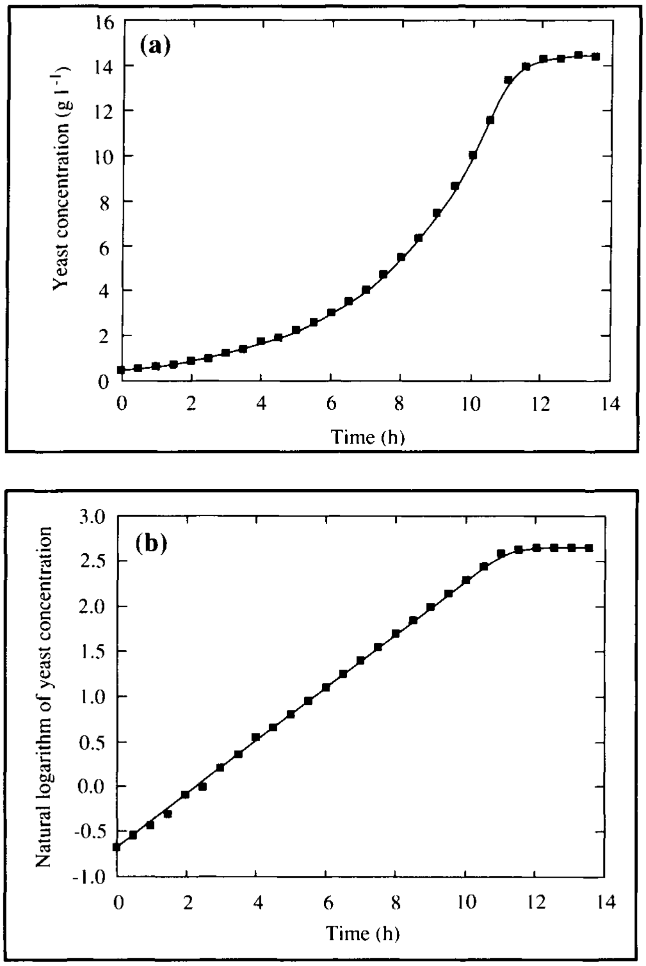
Figure 3.5 Straight-line correlation for calculation of model parameters.



(i) Least-squares analysis applies only to data containing random error.
(ii) The variables x and y must be independent.
(iii) Simple linear-regression methods are restricted to the special case of all uncertainty being associated with one variable. If the analysis uses a regression of y on x, then y should be the variable involving the largest errors. More complicated techniques are required to deal with errors in x and y simultaneously.
(iv) Simple linear-regression methods assume that each data point has equal significance. Modified procedures must be used if some points are considered more or less important than others, or if the line must pass through some specified point, e.g. the origin. It is also assumed that each point is equally precise, i.e. the standard deviation or random error associated with individual readings is the same. In experiments, the degree of fluctuation in the response variable often changes within the range of interest; for example, measurements may be more or less affected by instrument noise at the high or low end of the scale, so that data collected at the beginning of an experiment will have different errors compared with those measured at the end. Under these conditions, simple least-squares analysis is not appropriate.
(v) As already mentioned with respect to Figures 3.4(a) and 3.4(b), positive and negative residuals should be approximately evenly distributed, and residuals should be independent of both x and y variables.

Correlating data with straight lines is a relatively easy form of data analysis. When experimental data deviate markedly from a straight line, correlation using non-linear models is required. It is usually more difficult to decide which model to test and obtain parameter values when data do not follow linear relationships. As an example, consider growth of *Saccharomyces cerevisiae* yeast, which is expected to follow the non-linear model of Eq. (3.3). We could attempt to check whether measured cell-concentration data are consistent with Eq. (3.3) by plotting the values on linear graph paper as shown in Figure 3.6(a). The data appear to exhibit an exponential response typical of simple growth kinetics but it is not clear that an exponential model is appropriate. It is also difficult to ascertain some of the finer points of culture behaviour; for instance, whether the initial points represent a lag phase or whether exponential growth commenced immediately. Furthermore, the value of μ for this culture is not readily discernible from Figure 3.6(a).

Figure 3.6 Growth curve for *Saccharomyces cerevisiae*
(**a**) data plotted directly on linear graph-paper;
(**b**) linearisation of growth data by plotting logarthims of cell concentration versus time.



A convenient graphical approach to this problem is to transform the model equation into a linear form. Following the rules for logarithms outlined in Appendix D, taking the natural logarithm of both sides of Eq (3.3) gives:

$$\ln x = \ln x_0 + \mu t. \tag{3.9}$$

Eq. (3.9) indicates a linear relationship between $\ln x$ and t, with intercept $\ln x_0$ and slope μ. Accordingly, if Eq. (3.3) is a good model of yeast growth, a plot of the natural logarithm of cell concentration versus time should, during the growth phase, yield a straight line. Results of this linear transformation are shown in Figure 3.6(b). All points before stationary phase appear to lie on a straight line suggesting absence of a lag phase; the value of μ is readily calculated from the slope of the line. Graphical linearisation has the advantage that gross deviations from the model are immediately evident upon visual inspection. Other non-linear relationships and suggested methods for yielding straight-line plots are given in Table 3.1.

Once data have been transformed to produce straight lines, it is tempting to apply linear least-squares analysis to determine the model parameters. We could enter the values of time and logarithms of cell concentration into a computer or calculator programmed for linear regression. This analysis would give us the straight line through the data which minimises the sum-of-squares of the residuals. Most users of linear regression choose this technique because they believe it will automatically give them an objective and unbiased analysis of their data. However, *application of linear least-squares analysis to linearised data can result in biased estimates of model parameters.* The reason is related to the assumption in least-squares analysis that each datum point has equal random error associated with it.

When data are linearised, the error structure is changed so that distribution of errors becomes biased [11, 12]. Although standard deviations for each raw datum point may be approximately constant over the range of measurement, when logarithms are calculated, the error associated with each datum point becomes dependent on its magnitude. This also happens when data are inverted, as in some of the transformations suggested in Table 3.1. Small errors in y lead to enormous errors in $^1/_y$ when y is small; for large values of y the same errors are barely noticeable in $^1/_y$. This effect is shown in Figure 3.7; the error bars represent a constant error in y of $\pm$ 0.05 y'. When the magnitude of errors after transformation is dependent on the value of the variable, simple least-squares analysis should not be used.

In such cases, modifications must be made to the analysis. One alternative is to apply *weighted least-squares techniques.* The usual way of doing this is to take replicate measurements of the variable, transform the data, calculate the standard deviations for the transformed variable, and then weight the values by $^1/_{\sigma^2}$. Correctly weighted linear regression often gives satisfactory parameter values for non-linear models; details of the procedures can be found elsewhere [2, 9].

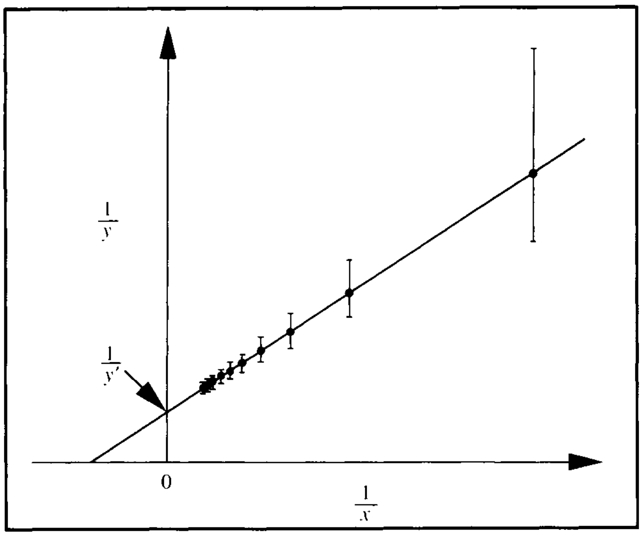
Techniques of *non-linear regression* usually give better results than weighted linear regression. In non-linear regression, equations such as those in Table 3.1 are fitted directly to the data. However, determining an optimal set of parameters by non-linear regression can be difficult and reliability of the results more difficult to interpret. The most common non-linear

Table 3.1 Methods for plotting data as straight lines

$y = A x^n$	Plot y vs x on logarithmic coordinates.
$y = A + B x^2$	Plot y vs x^2.
$y = A + B x^n$	First obtain A as the intercept on a plot of y vs x; then plot $(y - A)$ vs x on logarithmic coordinates.
$y = B^x$	Plot y vs x on semi-logarithmic coordinates.
$y = A + (B/x)$	Plot y vs $1/x$.
$y = \dfrac{1}{Ax + B}$	Plot $1/y$ vs x.
$y = \dfrac{x}{(A + Bx)}$	Plot x/y vs x, or $1/y$ vs. $1/x$.
$y = 1 + (A x^2 + B)^{1/2}$	Plot $(y - 1)^2$ vs x^2.
$y = A + Bx + Cx^2$	Plot $\dfrac{(y - y_n)}{(x - x_n)}$ vs x, where (y_n, x_n) are the coordinates of any point on a smooth curve through the experimental points.
$y = \dfrac{x}{(A + Bx)} + C$	Plot $\dfrac{(x - x_n)}{(y - y_n)}$ vs x, where (y_n, x_n) are the coordinates of any point on a smooth curve through the experimental points.

methods, such as the Gauss–Newton procedure widely available as computer software, are based on gradient, search or linearisation algorithms and use iterative solution techniques. More information about non-linear approaches to data analysis is available in other books [5, 9, 10, 13].

Figure 3.7 Error bars for $1/y$ vary in magnitude even though the error in y is the same for each datum point.



3.4 Graph Paper With Logarithmic Coordinates

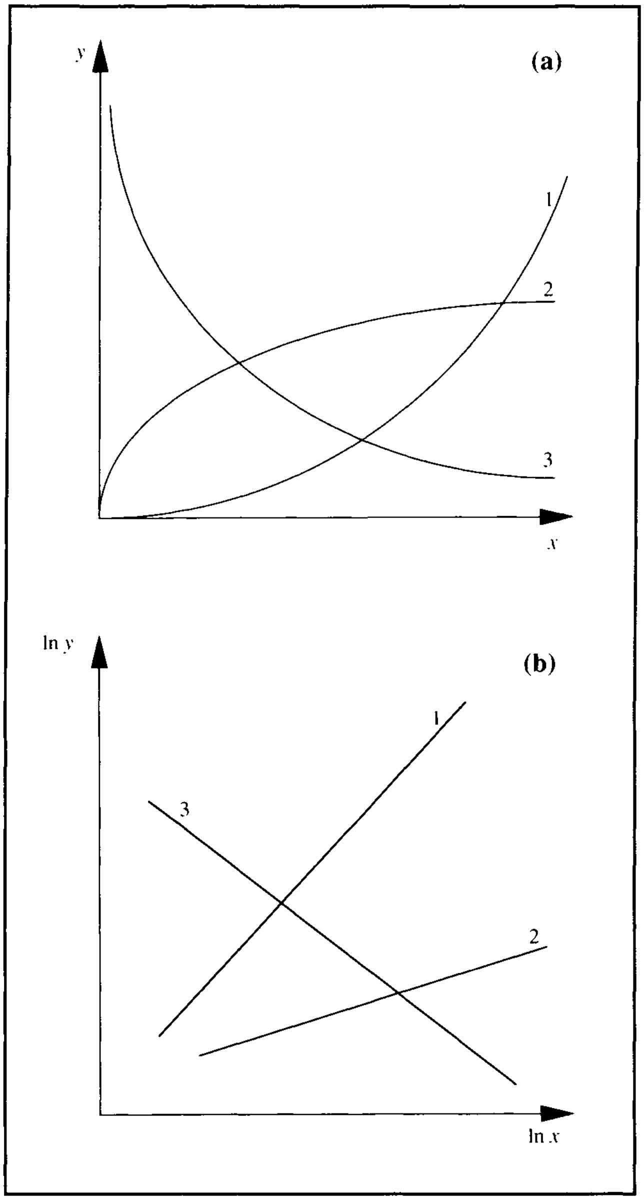
Two frequently-occurring non-linear functions are the *power law*, $y = B\,x^A$, and the *exponential function*, $y = B\,e^{Ax}$. These relationships are often presented using graph paper with logarithmic coordinates. Before proceeding, it may be necessary for you to review the mathematical rules for logarithms outlined in Appendix D.

3.4.1 Log–Log Plots

When plotted on a linear scale, some data have the form of either 1, 2 or 3 in Figure 3.8(a), all of which are not straight lines. Note also that none of these curves intersects either axis except at the origin. If straight-line representation is required, we must transform the data by calculating logarithms; plots of log or ln x versus log or ln y yield straight lines as shown in Figure 3.8(b). The best straight line through the data can be estimated using suitable non-linear regression analysis as discussed in Section 3.3.4.

When there are many data points, calculating the logarithms of x and y can be very time-consuming. An alternative is to use log–log graph paper. The raw data, not logarithms, are plotted directly on log–log paper; the resulting graph is as if logarithms were calculated to base e. Graph paper with both axes scaled logarithmically is shown in Figure 3.9; each axis in this example covers two logarithmic cycles. On log–log plots

Figure 3.8 Equivalent curves on linear and log–log graph paper. (See text.)

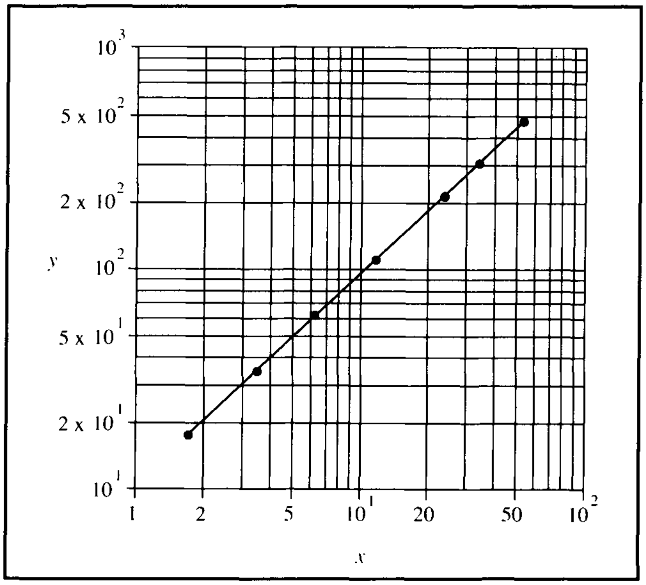


the origin (0,0) can never be represented; this is because ln 0 (or $\log_{10}0$) is not defined.

A straight line on log–log graph paper corresponds to the equation:

$$y = Bx^A \tag{3.10}$$

Figure 3.9 Log–log plot.



or

$$\ln y = \ln B + A \ln x. \tag{3.11}$$

Inspection of Eq. (3.10) shows that, if A is positive, $y = 0$ when $x = 0$. Therefore, a positive value of A corresponds to either curve 1 or 2 passing through the origin of Figure 3.8(a). If A is negative, when $x = 0$, y is infinite; therefore, negative A corresponds to curve 3 in Figure 3.8(a) which is asymptotic to both linear axes.

A and B are obtained from a straight line on log–log paper as follows. A can be calculated in two ways:

(i) If the log–log graph paper is drawn so that the ordinate and abscissa scales are the same, i.e. the distance measured with a ruler for a 10-fold change in the y variable is the same as for a 10-fold change in the x variable, A is the actual slope of the line. A is obtained by taking two points (x_1, y_1) and (x_2, y_2) on the line, and measuring the distances between y_2 and y_1 and x_2 and x_1 with a ruler:

$$A = \frac{(\text{distance between } y_2 \text{ and } y_1)}{(\text{distance between } x_2 \text{ and } x_1)}. \tag{3.12}$$

(ii) Alternatively, A is obtained by reading from the axes the coordinates of two points on the line, (x_1, y_1) and (x_2, y_2), and making the calculation:

$$A = \frac{(\ln y_2 - \ln y_1)}{(\ln x_2 - \ln x_1)} = \frac{\ln (y_2/y_1)}{\ln (x_2/x_1)} \,. \tag{3.13}$$

Note that all points x_1, y_1, x_2 and y_2 used in these calculations are points *on the line* through the data; they are not measured data values.

Once A is known, B is calculated from Eq. (3.11) as follows:

$$\ln B = \ln y_1 - A \ln x_1 \quad \text{or} \quad \ln B = \ln y_2 - A \ln x_2 \tag{3.14}$$

where $B = e^{(\ln B)}$. B can also be determined as the value of y when $x = 1$.

3.4.2 Semi-Log Plots

When plotted on linear-scale graph paper, some data show exponential rise or decay as illustrated in Figure 3.10(a). Curves 1 and 2 can be transformed into straight lines if log y or ln y is plotted against x, as shown in Figure 3.10(b).

An alternative to calculating logarithms is a semi-log plot, also known as a *linear-log plot*. As shown in Figure 3.11 raw data, not logarithms, are plotted directly on semi-log paper; the resulting graph is as if all logarithms were calculated to base e. Zero cannot be represented on the log-scale axis of semi-log plots. In Figure 3.11, values of the dependent variable were fitted within one logarithmic cycle from 10 to 100; semi-log paper with multiple cycles is also available.

A straight line on semi-log paper corresponds to the equation:

$$y = B e^{Ax} \tag{3.15}$$

or

$$\ln y = \ln B + A x. \tag{3.16}$$

Values of A and B are obtained from the straight line as follows. If two points (x_1, y_1) and (x_2, y_2), are located on the line, A is given by:

$$A = \frac{(\ln y_2 - \ln y_1)}{(x_2 - x_1)} = \frac{\ln(y_2/y_1)}{(x_2 - x_1)} \,. \tag{3.17}$$

B is the value of y at $x = 0$, i.e. B is the intercept of the line at the ordinate. Alternatively, once A is known, B is determined as follows:

$$\ln B = \ln y_1 - A x_1 \quad \text{or} \quad \ln B = \ln y_2 - A x_2. \tag{3.18}$$

B is calculated as $e^{(\ln B)}$.

Figure 3.10 Equivalent curves on linear and semi-log graph paper. (See text.)

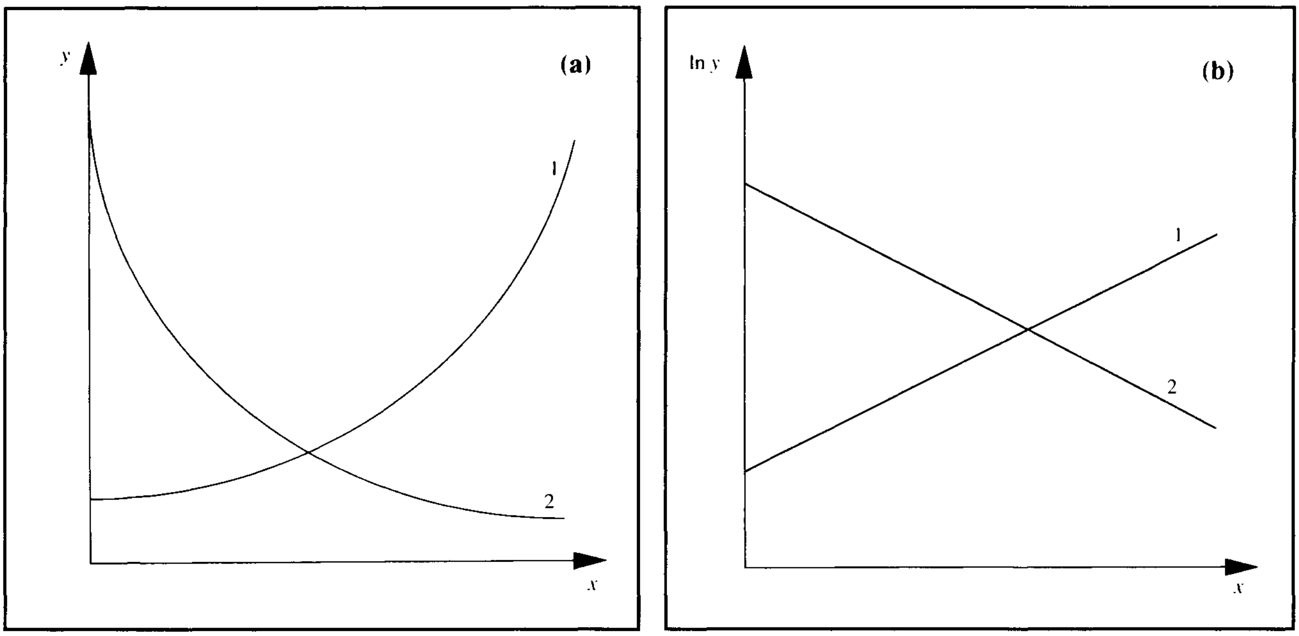
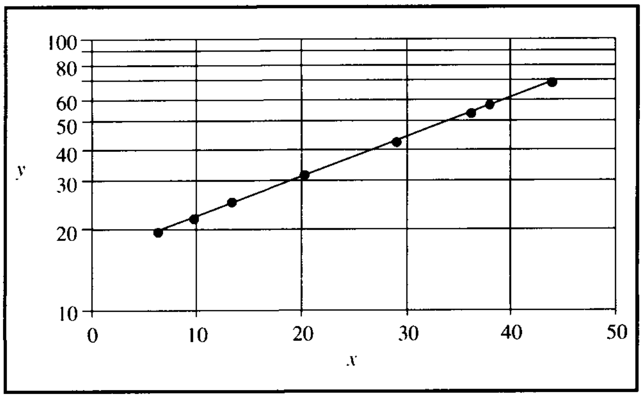


Figure 3.11 Semi-log plot.



Example 3.2 Cell growth data

Data for cell concentration x versus time t are plotted on semi-log graph paper. Points ($t_1 = 0.5$ h, $x_1 = 3.5$ g l^{-1}) and ($t_2 = 15$ h, $x_2 = 10.6$ g l^{-1}) fall on a straight line passing through the data.

(a) Determine the equation relating x and t.
(b) What is the value of the specific growth rate for this culture?

Solution:

(a) A straight line on semi-log graph paper means that x and t are correlated with the equation $x = B\,e^{At}$. A and B are calculated from Eqs (3.17) and (3.18):

$$A = \frac{\ln 10.6 - \ln 3.5}{15 - 0.5} = 0.076$$

and

$$\ln B = \ln 10.6 - (0.076)\,(15) = 1.215$$

or

$$B = 3.37.$$

Therefore, the equation for cell concentration as a function of time is:

$$x = 3.37 e^{0.076\,t}.$$

This result should be checked, for example, by substituting $t_1 = 0.5$:

$$B\,e^{At_1} = 3.37 e^{(0.076)(0.5)} = 3.5 = x_1.$$

(b) After comparing the empirical equation with Eq. (3.3), the specific growth rate μ is 0.076 h^{-1}.

3.5 General Procedures For Plotting Data

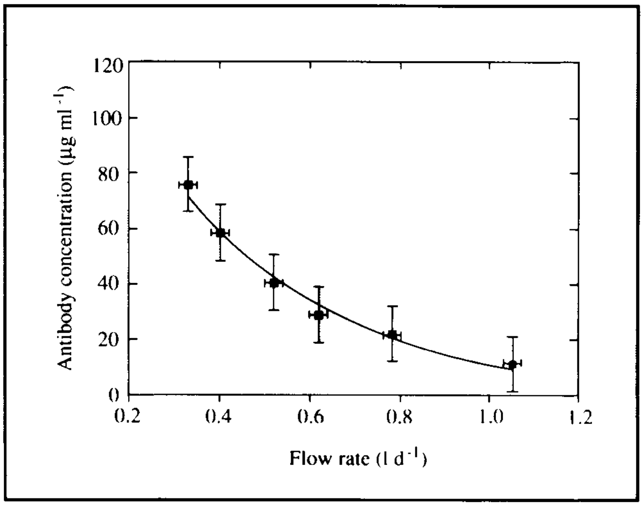
Axes on plots must be labelled for the graph to carry any meaning. The units associated with all physical variables should be stated explicitly. If more than one curve is plotted on a single graph, each curve must be identified with a label.

It is good practice to indicate the precision of data on graphs using *error bars*. As an example, Table 3.2 lists values of monoclonal-antibody concentration as a function of medium flow rate during continuous culture of hybridoma cells in a stirred fermenter. The flow rate was measured to within ± 0.02 litres per day. Measurement of antibody concentration was more difficult and somewhat imprecise; these values are estimated to involve errors of ± 10 μg ml^{-1}. Errors associated with the data are indicated in Figure 3.12 using error bars to show the possible range of each variable.

Table 3.2 Antibody concentration during continuous culture of hybridoma cells

Flow rate (l d^{-1})	*Antibody concentration* (μg ml^{-1})
0.33	75.9
0.40	58.4
0.52	40.5
0.62	28.9
0.78	22.0
1.05	11.5

Figure 3.12 Error bars for antibody concentration and medium flow rate measured during continuous culture of hybridoma cells.



3.6 Process Flow Diagrams

This chapter is concerned with ways of presenting and analysing data. Because of the complexity of large-scale manufacturing processes, communicating information about these systems requires special methods. *Flow diagrams* or *flow sheets* are simplified pictorial representations of processes and are used to present relevant process information and data. Flow sheets vary in complexity from simple block diagrams to highly complex schematic drawings showing main and auxiliary process equipment such as pipes, valves, pumps and by-pass loops.

Figure 3.13 is a simplified process flow diagram showing the major operations for production of the antibiotic, bacitracin. This qualitative flow sheet indicates the flow of materials, the sequence of process operations, and the principal equipment in use. When flow diagrams are applied in calculations, the operating conditions, masses and concentrations of material handled by the process are also specified. An example is Figure 3.14, which represents recovery operations for 2,3-butanediol produced commercially by fermentation of whole wheat mash. The quantities and compositions of streams undergoing processes such as distillation, evaporation, screening and drying are shown to allow calculation of product yields and energy costs.

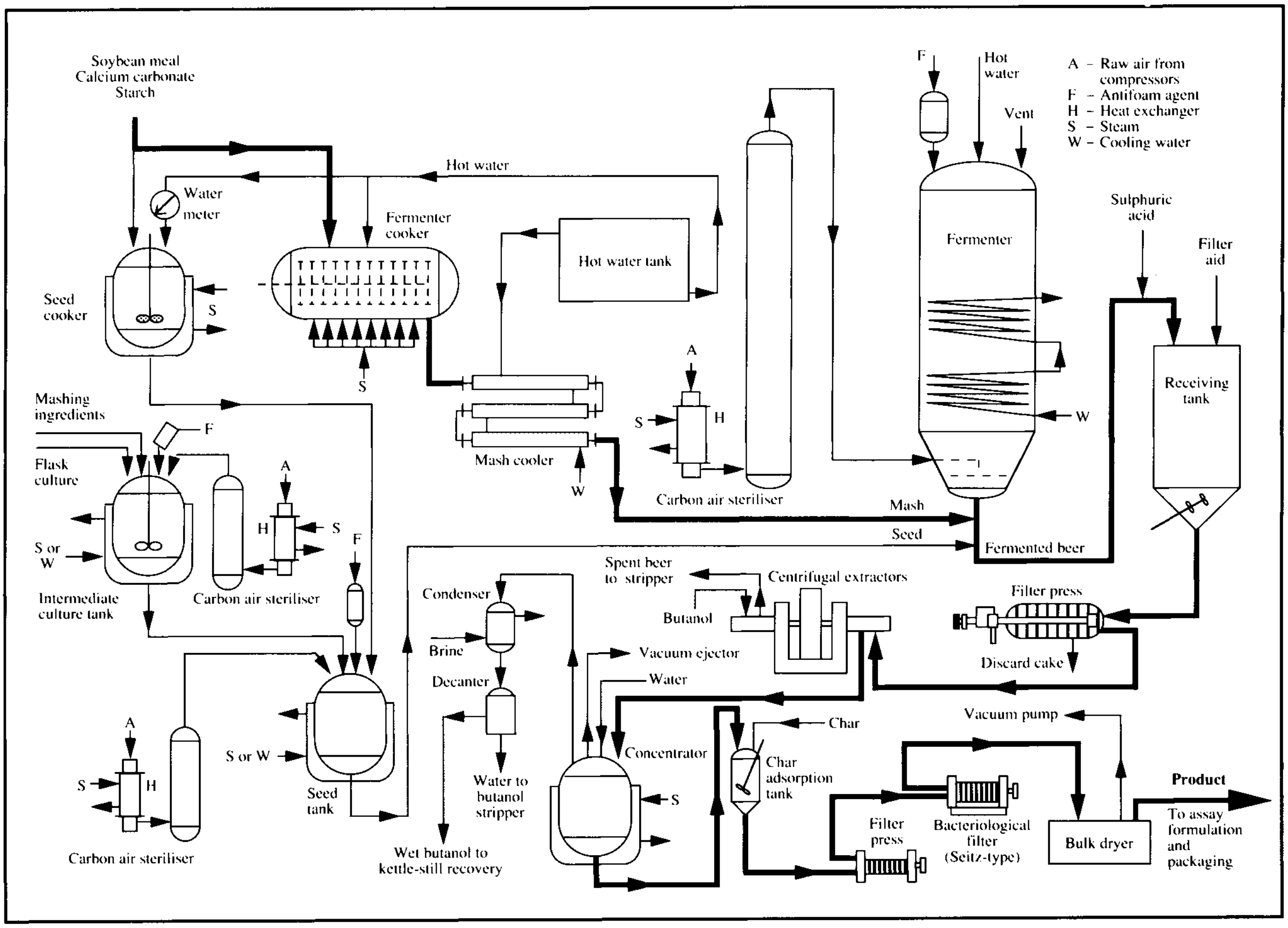
Detailed engineering flowsheets such as Figure 3.15 are useful for plant construction work and trouble-shooting because they show all piping, valves, drains, pumps and safety equipment. Standard symbols are adopted to convey the information as concisely as possible. Figure 3.15 represents a pilot-scale fermenter with separate vessels for antifoam, acid and alkali. All air, medium inlet and harvest lines are shown, as are the steam and condensate-drainage lines for *in situ* steam sterilisation of the entire apparatus.

In addition to those illustrated here, other specialised types of flow diagram are used to specify instrumentation for process control networks in large-scale processing plants, and for utilities such as steam, water, fuel and air supplies. We will not be applying complicated or detailed diagrams such as Figure 3.15 in our analysis of bioprocessing; their use is beyond the scope of this book. However, simplified versions of flow diagrams are extremely useful, especially for material- and energy-balance calculations; we will be applying block diagram flow sheets in Chapters 4–6 for this purpose. You should become familiar with flow diagrams for showing data and other process information.

3.7 Summary of Chapter 3

This chapter covers a range of topics related to data presentation and analysis. After studying Chapter 3 you should:

Figure 3.13 Process flowsheet showing the major operations for production of bacitracin. (From G.C. Inskeep, R.E. Bennett, J.F. Dudley and M.W. Shepard, 1951, Bacitracin: product of biochemical engineering, *Ind. Eng. Chem.* **43**, 1488–1498.)



(i) understand use of *significant figures*;

(ii) know the types of *error* which affect accuracy of experimental data and which errors can be accounted for with statistical techniques;

(iii) be able to combine errors in simple calculations;

(iv) be able to report the results of replicate measurements in terms of the *mean* and *standard deviation*;

(v) understand the fundamental limitations associated with use of experimental data for testing mathematical models;

(vi) be familiar with *least-squares analysis* and its assumptions;

(vii) be able to analyse linear plots and determine model parameters;

(viii) understand how simple non-linear models can be linearised to obtain straight-line plots;

(ix) be able to use *log* and *semi-log graph paper* with ease; and

(x) be familiar with simple *process flow diagrams.*

Problems

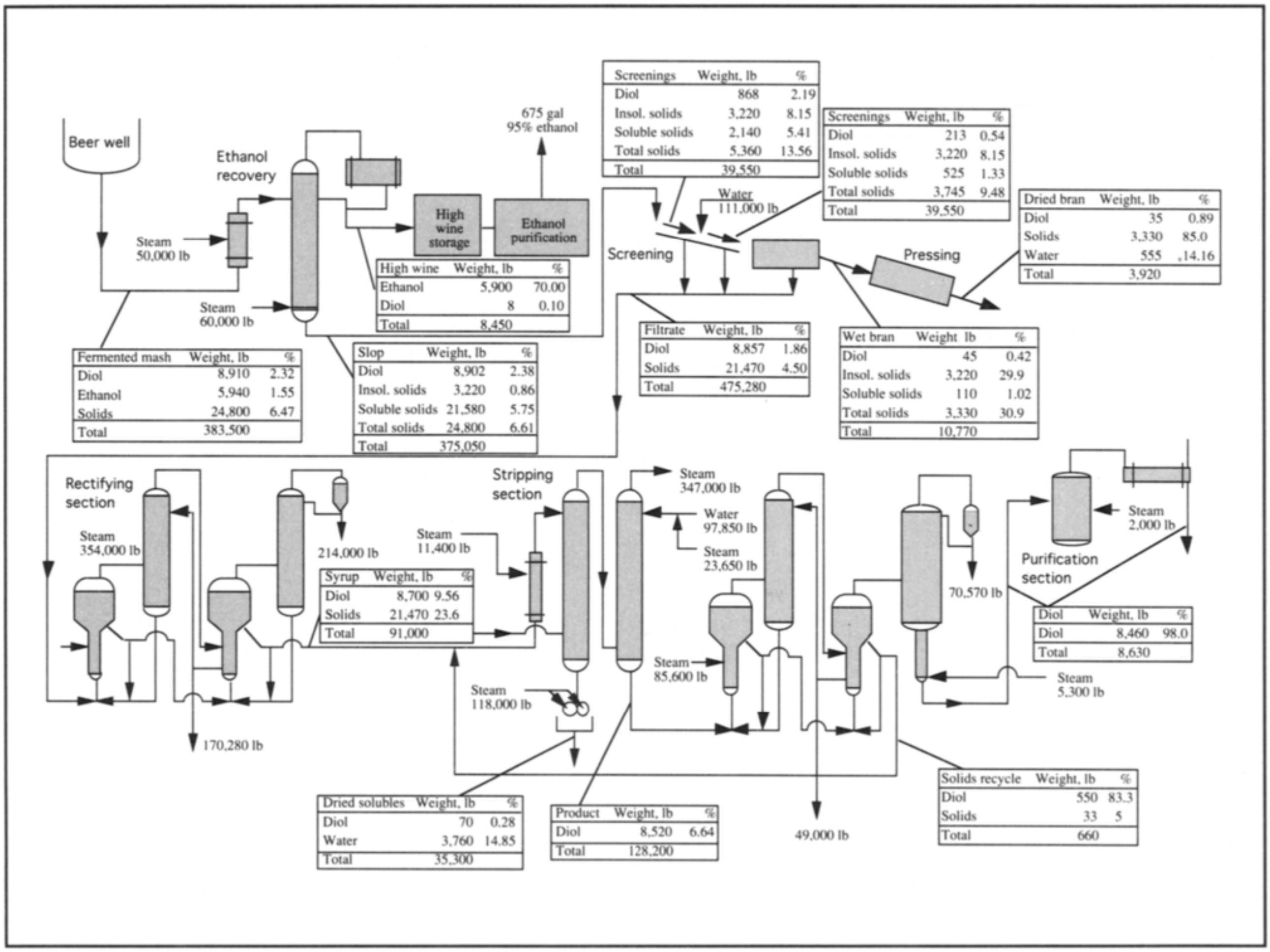
3.1 Combination of errors

The oxygen mass-transfer coefficient in fermentation vessels is determined from experimental measurements using the formula:

$$k_L a = \frac{OTR}{C^*_{AL} - C_{AL}}$$

where $k_L a$ is the mass-transfer coefficient, OTR is the oxygen transfer rate, C^*_{AL} is the solubility of oxygen in the fermentation

Figure 3.14 Quantitative flowsheet for the downstream processing of 2,3-butanediol based on fermentation of 1000 bushels wheat per day by *Aerobacillus polymyxa.* (From J.A. Wheat, J.D. Leslie, R.V. Tomkins, H.E. Mitton, D.S. Scott and G.A. Ledingham, 1948, Production and properties of 2,3-butanediol, XXVIII: Pilot plant recovery of *levo*-2,3-butanediol from whole wheat mashes fermented by *Aerobacillus polymyxa, Can. J. Res.* **26F**, 469–496.)



Fermented mash	Weight, lb	%
Diol	8,910	2.32
Ethanol	5,940	1.55
Solids	24,800	6.47
Total	383,500	

High wine	Weight, lb	%
Ethanol	5,900	70.00
Diol	8	0.10
Total	8,450	

Slop	Weight, lb	%
Diol	8,902	2.38
Insol. solids	3,220	0.86
Soluble solids	21,580	5.75
Total solids	24,800	6.61
Total	375,050	

Screenings	Weight, lb	%
Diol	868	2.19
Insol. solids	3,220	8.15
Soluble solids	2,140	5.41
Total solids	5,360	13.56
Total	39,550	

Screenings	Weight, lb	%
Diol	213	0.54
Insol. solids	3,220	8.15
Soluble solids	525	1.33
Total solids	3,745	9.48
Total	39,550	

Dried bran	Weight, lb	%
Diol	35	0.89
Solids	3,330	85.0
Water	555	14.16
Total	3,920	

Filtrate	Weight, lb	%
Diol	8,857	1.86
Solids	21,470	4.50
Total	475,280	

Wet bran	Weight lb	%
Diol	45	0.42
Insol. solids	3,220	29.9
Soluble solids	110	1.02
Total solids	3,330	30.9
Total	10,770	

Syrup	Weight, lb	%
Diol	8,700	9.56
Solids	21,470	23.6
Total	91,000	

Diol	Weight, lb	%
Diol	8,460	98.0
Total	8,630	

Solids recycle	Weight, lb	%
Diol	550	83.3
Solids	33	5
Total	660	

Dried solubles	Weight, lb	%
Diol	70	0.28
Water	3,760	14.85
Total	35,300	

Product	Weight, lb	%
Diol	8,520	6.64
Total	128,200	

broth, and C_{AL} is the dissolved-oxygen concentration. C^*_{AL} is estimated to be 0.25 mol m^{-3} with an uncertainty of ± 4%; C_{AL} is measured as 0.183 mol m^{-3} ± 4%. If the *OTR* is 0.011 mol m^{-3} s^{-1} ± 5%, what is the uncertainty associated with $k_L a$?

3.2 Mean and standard deviation

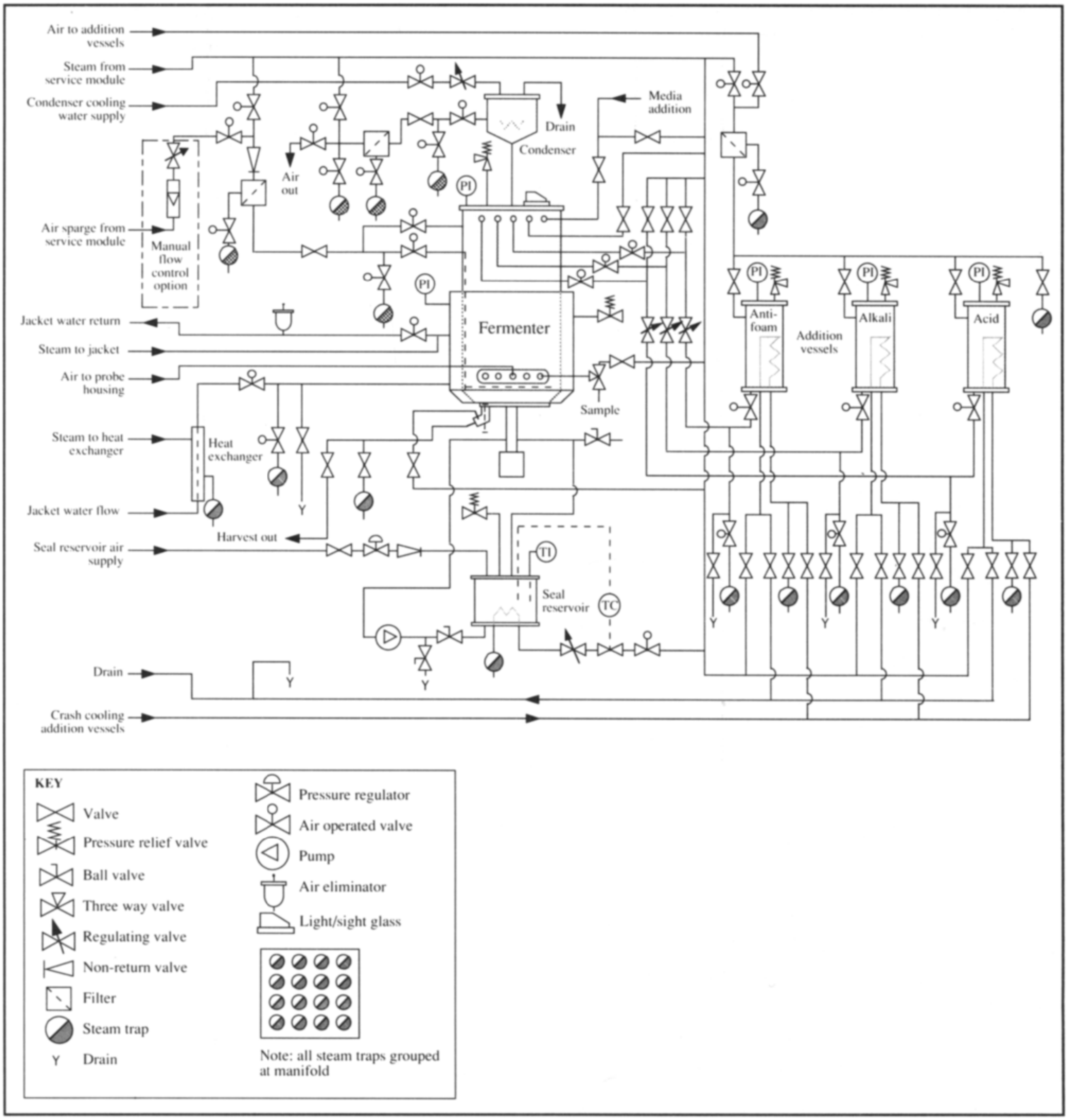
The pH for maximum activity of β-amylase enzyme is measured four times. The results are: 5.15, 5.45, 5.50 and 5.35.

(a) What is the best estimate of optimal pH?
(b) How accurate is this value?
(c) If the experiment were stopped after only the first two measurements were taken, what would be the result and its accuracy?
(d) If an additional four measurements were made with the same results as those above, how would this change the outcome of the experiment?

3.3 Linear and non-linear models

Determine the equation for y as a function of x using the following information. Reference to coordinate point (x, y) means that x is the abscissa value and y is the ordinate value.

Figure 3.15 Detailed equipment diagram for pilot-plant fermentation system. (Reproduced with permission from LH Engineering Ltd, a member of the Inceltech Group of companies. Copyright 1983.)



(a) A plot of y versus x on linear graph paper gives a straight line passing through the points (1, 10) and (8, 0.5).
(b) A plot of y versus $x^{1/2}$ on linear graph paper gives a straight line passing through points (3.2, 14.5) and (8.9, 38.5).
(c) A plot of $^1/_y$ versus x^2 on linear graph paper gives a straight line passing through points (5, 6) and (1, 3).
(d) A plot of y versus x on log-log paper gives a straight line passing through (0.5, 25) and (550, 2600).
(e) A plot of y versus x on semi-log paper gives a straight line passing through (1.5, 2.5) and (10, 0.036).

3.4 Linear curve fitting

Sucrose concentration in a fermentation broth is measured using HPLC. Chromatogram peak areas are measured for five standard sucrose solutions to calibrate the instrument. Measurements are performed in triplicate with results as follows:

Sucrose concentration ($g\,l^{-1}$)	*Peak area*
6.0	55.55, 57.01, 57.95
12.0	110.66, 114.76, 113.05
18.0	168.90, 169.44, 173.55
24.0	233.66, 233.89, 230.67
30.0	300.45, 304.56, 301.11

(a) Determine the mean peak areas for each sucrose concentration, and the standard deviations.
(b) Plot the data. Plot the standard deviations as error bars.
(c) Find an equation for sucrose concentration as a function of peak area.
(d) A sample containing sucrose gives a peak area of 209.86. What is the sucrose concentration?

3.5 Non-linear model: calculation of parameters

The mutation rate of *E. coli* increases with temperature. The following data were obtained by measuring the frequency of mutation of *his*$^-$ cells to produce *his*$^+$ colonies:

Temperature (°C)	*Relative mutation frequency*, α
15	4.4×10^{-15}
20	2.0×10^{-14}
25	8.6×10^{-14}
30	3.5×10^{-13}
35	1.4×10^{-12}

Mutation frequency is expected to obey an Arrhenius-type equation:

$$\alpha = \alpha_0 \, e^{-E/RT}$$

where α_0 is the mutation rate parameter, E is activation energy, R is the universal gas constant and T is absolute temperature.

(a) Test the model using an appropriate plot on log graph paper.
(b) What is the activation energy for the mutation reaction?
(c) What is the value of α_0?

3.6 Linear regression: distribution of residuals

Medium conductivity is sometimes used to monitor cell growth during batch culture. Experiments are carried out to relate decrease in conductivity to increase in plant-cell biomass during culture of *Catharanthus roseus* in an airlift fermenter. The results are tabulated below.

Decrease in medium conductivity ($mS\,cm^{-1}$)	*Increase in biomass concentration* ($g\,l^{-1}$)
0	0
0.12	2.4
0.31	2.0
0.41	2.8
0.82	4.5
1.03	5.1
1.40	5.8
1.91	6.0
2.11	6.2
2.42	6.2
2.44	6.2
2.74	6.6
2.91	6.0
3.53	7.0
4.39	9.8
5.21	14.0
5.24	12.6
5.55	14.6

(a) Plot the points on linear coordinates and obtain an equation for the 'best' straight line through the data using linear least-squares analysis.
(b) Plot the residuals in biomass increase versus conductivity change after comparing the model equation with the

actual data. What do you conclude about the goodness of fit for the straight line?

3.7 Discriminating between rival models

In bioreactors where the liquid contents are mixed by sparging air into the vessel, the liquid velocity is dependent directly on the gas velocity. The following results were obtained from experiments with 0.15 M NaCl solution.

Gas superficial velocity, u_G ($m\ s^{-1}$)	*Liquid superficial velocity*, u_L ($m\ s^{-1}$)
0.02	0.060
0.03	0.066
0.04	0.071
0.05	0.084
0.06	0.085
0.07	0.086
0.08	0.091
0.09	0.095
0.095	0.095

(a) How well are these data fitted with a linear model? Determine the equation for the 'best' straight line relating gas and liquid velocities.

(b) It has been reported in the literature that fluid velocities in air-driven reactors are related using the power equation:

$$u_L = \alpha\, u_G^{\,v}$$

where α and v are constants. Is this model an appropriate description of the experimental data?

(c) Which equation, linear or non-linear, is the better model for the reactor system?

3.8 Non-linear model: calculation of parameters

When nutrient medium is autoclaved, the number of viable microorganisms decreases with time spent in the autoclave. An experiment is conducted to measure the number of viable cells N in a bottle of glucose solution after various sterilisation times t. Triplicate measurements are taken of cell number; the mean and standard deviation of each measurement are listed in the following table.

t (min)	*Mean* N	*Standard deviation of* N
5	3.6×10^3	0.20×10^3
10	6.3×10^2	0.40×10^2
15	1.07×10^2	0.09×10^2
20	1.8×10^1	0.12×10^1
30	< 1	–

From what is known about thermal death kinetics for microorganisms, it is expected that the relationship between N and t is of the form:

$$N = N_0\, e^{-k_d t}$$

where k_d is the specific death constant and N_0 is the number of viable cells present before autoclaving begins.

(a) Plot the results on suitable graph paper to obtain a straight line through the data.

(b) Plot the standard deviations on the graph as error bars.

(c) What are the values of k_d and N_0?

(d) What are the units and dimensions of k_d and N_0?

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Suggestions for Further Reading

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Part 2
Material and Energy Balances

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4

Material Balances

One of the simplest concepts in process engineering is the material or mass balance. Because mass in biological systems is conserved at all times, the law of conservation of mass provides the theoretical framework for material balances.

In steady-state material balances, masses entering a process are summed up and compared with the total mass leaving the system; the term 'balance' implies that masses entering and leaving should be equal. Essentially, material balances are accounting procedures: total mass entering must be accounted for at the end of the process, even if it undergoes heating, mixing, drying, fermentation or any other operation (except nuclear reaction) within the system. Usually it is not feasible to measure the masses and compositions of all streams entering and leaving a system; unknown quantities can be calculated using mass-balance principles. Mass-balance problems have a constant theme: given the masses of some input and output streams, calculate the masses of others.

Mass balances provide a very powerful tool in engineering analysis. Many complex situations are simplified by looking at the movement of mass and equating what comes out to what goes in. Questions such as: what is the concentration of carbon dioxide in the fermenter off-gas? what fraction of the substrate consumed is not converted into products? how much reactant is needed to produce *x* grams of product? how much oxygen must be provided for this fermentation to proceed? can be answered using mass balances. This chapter explains how the law of conservation of mass is applied to atoms, molecular species and total mass, and sets up formal techniques for solving material-balance problems with and without reaction. Aspects of metabolic stoichiometry are also discussed for calculation of nutrient and oxygen requirements during fermentation processes.

4.1 Thermodynamic Preliminaries

Thermodynamics is a fundamental branch of science dealing with the properties of matter. Thermodynamic principles are useful in setting up material balances; some terms borrowed from thermodynamics are defined below.

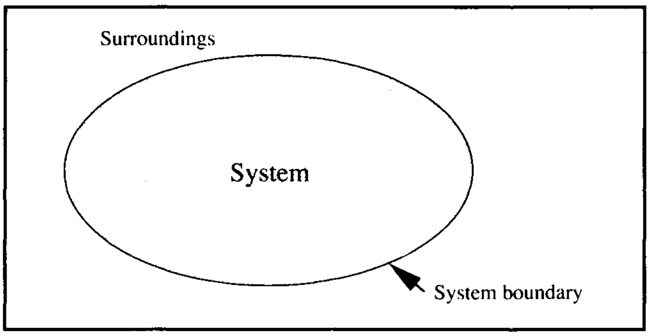
4.1.1 System and Process

In thermodynamics, a *system* consists of any matter identified for investigation. As indicated in Figure 4.1, the system is set apart from the *surroundings*, which are the remainder of the *universe*, by a *system boundary*. The system boundary may be real and tangible, such as the walls of a beaker or fermenter, or imaginary. If the boundary does not allow mass to pass from system to surroundings and vice versa, the system is a *closed* system with constant mass. Conversely, a system able to exchange mass with its surroundings is an *open* system.

A *process* causes changes in the system or surroundings. Several terms are commonly used to describe processes.

(i) A *batch process* operates in a closed system. All materials are added to the system at the start of the process; the system is then closed and products removed only when the process is complete.
(ii) A *semi-batch process* allows either input or output of mass, but not both.

Figure 4.1 Thermodynamic system.



(iii) A *fed-batch process* allows input of material to the system but not output.
(iv) A *continuous process* allows matter to flow in and out of the system. If rates of mass input and output are equal, continuous processes can be operated indefinitely.

4.1.2 Steady State and Equilibrium

If all properties of a system, such as temperature, pressure, concentration, volume, mass, etc. do not vary with time, the process is said to be at *steady state.* Thus, if we monitor any variable of a steady-state system, its value will be unchanging with time.

According to this definition of steady state, batch, fed-batch and semi-batch processes cannot operate under steady-state conditions. Mass of the system is either increasing or decreasing with time during fed-batch and semi-batch processes; in batch processes, even though the total mass is constant, changes occurring inside the system cause the system properties to vary with time. Such processes are called *transient* or *unsteady-state* processes. On the other hand, continuous processes may be either steady state or transient. It is usual to run continuous processes as close to steady state as possible; however, unsteady-state conditions will exist during start-up and for some time after any change in operating conditions.

Steady state is an important and useful concept in engineering analysis. However, it is often confused with another thermodynamic term, *equilibrium.* A system at equilibrium is one in which all opposing forces are exactly counter-balanced so that the properties of the system do not change with time. From experience we know that systems tend to approach an equilibrium condition when they are isolated from their surroundings. At equilibrium there is no net change in either the system or the universe. Equilibrium implies that there is no net driving force for change; the energy of the system is at a minimum and, in rough terms, the system is 'static', 'unmoving' or 'inert'. For example, when liquid and vapour are in equilibrium in a closed vessel, although there may be constant exchange of molecules between the phases, there is no net change in either the system or the surroundings.

To convert raw materials into useful products there must be an overall change in the universe. Because systems at equilibrium produce no net change, equilibrium is of little value in processing operations. The best strategy is to avoid equilibrium by continuously disturbing the system in such a way that raw material will always be undergoing transformation into the desired product. In continuous processes at steady state, mass is constantly exchanged with the surroundings; this disturbance drives the system away from equilibrium so that a net change in both the system and the universe can occur. Large-scale equilibrium does not often occur in engineering systems; steady states are more common.

4.2 Law of Conservation of Mass

Mass is conserved in ordinary chemical and physical processes. Consider the system of Figure 4.2 operating as a continuous process with input and output streams containing glucose. The mass flow rate of glucose into the system is $\hat{M}_i$ kg h^{-1}; the mass flow rate out is $\hat{M}_o$ kg h^{-1}. If $\hat{M}_i$ and $\hat{M}_o$ are different, there are four possible explanations:

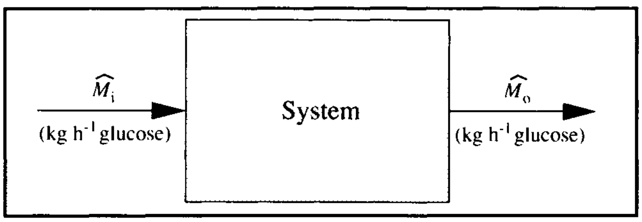
(i) measurements of $\hat{M}_i$ and $\hat{M}_o$ are wrong;
(ii) the system has a leak allowing glucose to enter or escape undetected;
(iii) glucose is consumed or generated by chemical reaction within the system; or
(iv) glucose accumulates within the system.

If we assume that the measurements are correct and there are no leaks, the difference between $\hat{M}_i$ and $\hat{M}_o$ must be due to consumption or generation by reaction, and/or accumulation. A mass balance for the system can be written in a general way to account for these possibilities:

$$\left\{\begin{matrix}\text{mass in}\\\text{through}\\\text{system}\\\text{boundaries}\end{matrix}\right\}-\left\{\begin{matrix}\text{mass out}\\\text{through}\\\text{system}\\\text{boundaries}\end{matrix}\right\}+\left\{\begin{matrix}\text{mass}\\\text{generated}\\\text{within}\\\text{system}\end{matrix}\right\}-\left\{\begin{matrix}\text{mass}\\\text{consumed}\\\text{within}\\\text{system}\end{matrix}\right\}=\left\{\begin{matrix}\text{mass}\\\text{accumulated}\\\text{within}\\\text{system}\end{matrix}\right\}. \tag{4.1}$$

The accumulation term in the above equation can be either positive or negative; negative accumulation represents depletion of pre-existing reserves. Eq. (4.1) is known as the *general mass-balance equation.* The mass referred to in the equation can be total mass, mass of a particular molecular or atomic species, or biomass. Use of Eq. (4.1) is illustrated in Example 4.1.

Figure 4.2 Flow sheet for a mass balance on glucose.



Example 4.1 General mass-balance equation

A continuous process is set up for treatment of wastewater. Each day, 10^5 kg cellulose and 10^3 kg bacteria enter in the feed stream, while 10^4 kg cellulose and 1.5×10^4 kg bacteria leave in the effluent. The rate of cellulose digestion by the bacteria is 7×10^4 kg d^{-1}. The rate of bacterial growth is 2×10^4 kg d^{-1}; the rate of cell death by lysis is 5×10^2 kg d^{-1}. Write balances for cellulose and bacteria in the system.

Solution:
Cellulose is not generated by the process, only consumed. Using a basis of 1 day, the cellulose balance in kg from Eq. (4.1) is:

$$(10^5 - 10^4 + 0 - 7 \times 10^4) = \text{accumulation}.$$

Therefore, 2×10^4 kg cellulose accumulates in the system each day.
Performing the same balance for bacteria:

$$(10^3 - 1.5 \times 10^4 + 2 \times 10^4 - 5 \times 10^2) = \text{accumulation}.$$

Therefore, 5.5×10^3 kg bacterial cells accumulate in the system each day.

4.2.1 Types of Material Balance

The general mass-balance equation (4.1) can be applied with equal ease to two different types of mass-balance problem, depending on the data provided. For continuous processes it is usual to collect information about the process referring to a particular instant in time. Amounts of mass entering and leaving the system are specified using flow rates, e.g. molasses enters the system at a rate of 50 lb h^{-1}; at the same instant in time, fermentation broth leaves at a rate of 20 lb h^{-1}. These two quantities can be used directly in Eq. (4.1) as the input and output terms. A mass balance based on rates is called a *differential balance.*

An alternative approach is required for batch and semi-batch processes. Information about these systems is usually collected over a period of time rather than at a particular instant. For example: 100 kg substrate is added to the reactor; after 3 days' incubation, 45 kg product is recovered. Each term of the mass-balance equation in this case is a quantity of mass, not a rate. This type of balance is called an *integral balance.*

In this chapter, we will be using differential balances for continuous systems operating at steady state, and integral balances for batch and semi-batch systems between initial and final states. Calculation procedures for the two types of material balance are very similar.

4.2.2 Simplification of the General Mass-Balance Equation

Eq. (4.1) can be simplified in certain situations. If a continuous process is at steady state, the accumulation term on the right-hand side must be zero. This follows from the definition of steady state: because all properties of the system, including its mass, must be unchanging with time, a system at steady state cannot accumulate mass. Under these conditions, Eq. (4.1) becomes:

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}. \tag{4.2}$$

Eq. (4.2) is called the *general steady-state mass-balance equation.* Eq. (4.2) also applies over the entire duration of batch and fed-batch processes; 'mass out' in this case is the total mass harvested from the system so that at the end of the process there is no accumulation.

If reaction does not occur in the system, or if the mass balance is applied to a substance that is neither a reactant nor product of reaction, the generation and consumption terms in Eqs (4.1) and (4.2) are zero. Because total mass can be neither created nor destroyed except in nuclear reaction, generation and consumption terms must also be zero in balances applied to total mass. Similarly, generation and consumption of atomic species such as C, N, O, etc. cannot occur in normal chemical reaction. Therefore at steady state, for balances on total mass or atomic species or when reaction does not occur, Eq. (4.2) can be further simplified to:

$$\text{mass in} = \text{mass out}. \tag{4.3}$$

Table 4.1 summarises the types of material balance for which direct application of Eq. (4.3) is valid. Because total number of moles does not balance in systems with reaction, we will carry out all material balances using mass.

Table 4.1 Application of the simplified mass balance Eq. (4.3)

Material	*At steady state, does mass in = mass out?*	
	Without reaction	*With reaction*
Total mass	yes	yes
Total number of moles	yes	no
Mass of a molecular species	yes	no
Number of moles of a molecular species	yes	no
Mass of an atomic species	yes	yes
Number of moles of an atomic species	yes	yes

4.3 Procedure For Material-Balance Calculations

The first step in material-balance calculations is to understand the problem. Certain information is available about a process; the task is to calculate unknown quantities. Because it is sometimes difficult to sort through all the details provided, it is best to use standard procedures to translate process information into a form that can be used in calculations.

Material balances should be carried out in an organised manner; this makes the solution easy to follow, check, or use by others. In this chapter, a formalised series of steps is followed for each mass-balance problem. For easier problems these procedures may seem long-winded and unnecessary; however a standard method is helpful when you are first learning mass-balance techniques. The same procedures are used in the next chapter as a basis for energy balances.

These points are essential.

(i) *Draw a clear process flow diagram showing all relevant information.* A simple box diagram showing all streams entering or leaving the system allows information about a process to be organised and summarised in a convenient way. All given quantitative information should be shown on the diagram. Note that the variables of interest in material balances are masses, mass flow rates and mass compositions; if information about particular streams is given using volume or molar quantities, mass flow rates and compositions should be calculated before labelling the flow sheet.

(ii) *Select a set of units and state it clearly.* Calculations are easier when all quantities are expressed using consistent units. Units must also be indicated for all variables shown on process diagrams.

(iii) *Select a basis for the calculation and state it clearly.* In approaching mass-balance problems it is helpful to focus on a specific quantity of material entering or leaving the system. For continuous processes at steady state we usually base the calculation on the amount of material entering or leaving the system within a specified period of time. For batch or semi-batch processes, it is convenient to use either the total amount of material fed to the system or the amount withdrawn at the end. Selection of a basis for calculation makes it easier to visualise the problem; the way this works will become apparent in the worked examples of the next section.

(iv) *State all assumptions applied to the problem.* To solve problems in this and the following chapters, you will need to apply some 'engineering' judgement. Real-life situations are complex, and there will be times when one or more assumptions are required before you can proceed with calculations. To give you experience with this, problems posed in this text may not give you all the necessary information. The details omitted can be assumed, provided your assumptions are reasonable. Engineers make assumptions all the time; knowing when an assumption is permissible and what constitutes a reasonable assumption is one of the marks of a skilled engineer. When you make assumptions about a problem it is vitally important that you state them exactly. Other scientists looking through your calculations need to know the conditions under which your results are applicable; they will also want to decide whether your assumptions are acceptable or whether they can be improved.

In this chapter, differential mass balances on continuous processes are performed with the understanding that the system is at steady state; we can assume that mass flow rates and compositions do not change with time and the accumulation term of Eq. (4.1) is zero. If steady state does not prevail in continuous processes, information about the rate of accumulation would be required for solution of mass-balance problems. This is discussed further in Chapter 6.

Another assumption we must make in mass-balance problems is that the system under investigation does not leak. In totalling up all the masses entering and leaving the system, we must be sure that all streams are taken into account. When analysing real systems it is always a good idea to check for leaks before carrying out mass balances.

(v) *Identify which components of the system, if any, are involved in reaction.* This is necessary for determining which mass-balance equation (4.2) or (4.3), is appropriate. The

Example 4.2 Setting up a flow sheet

Humid air enriched with oxygen is prepared for a gluconic acid fermentation. The air is prepared in a special humidifying chamber. $1.5\ \mathrm{l\ h^{-1}}$ liquid water enters the chamber at the same time as dry air and $15\ \mathrm{gmol\ min^{-1}}$ dry oxygen gas. All the water is evaporated. The outflowing gas is found to contain 1% (w/w) water. Draw and label the flow sheet for this process.

Solution:

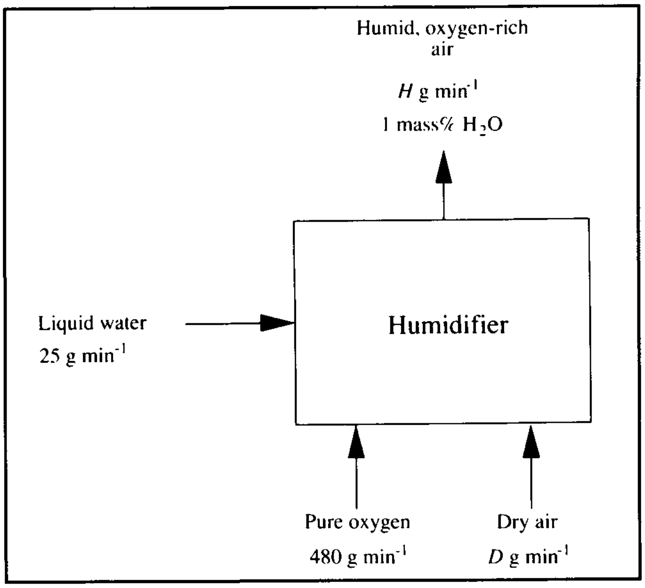
Let us choose units of g and min for this process; the information provided is first converted to mass flow rates in these units. The density of water is taken to be $10^3\ \mathrm{g\ l^{-1}}$; therefore:

$$1.5\ \mathrm{l\ h^{-1}} = \frac{1.5\ \mathrm{l}}{\mathrm{h}} \left(\frac{10^3\ \mathrm{g}}{\mathrm{l}}\right) \left|\frac{1\ \mathrm{h}}{60\ \mathrm{min}}\right| = 25\ \mathrm{g\ min^{-1}}.$$

As the molecular weight of O_2 is 32:

$$15\ \mathrm{gmol\ min^{-1}} = \frac{15\ \mathrm{gmol}}{\mathrm{min}} \cdot \left|\frac{32\ \mathrm{g}}{1\ \mathrm{gmol}}\right| = 480\ \mathrm{g\ min^{-1}}.$$

Figure 4E2.1 Flowsheet for oxygen enrichment and humidification of air.



Unknown flow rates are represented with symbols. As shown in Figure 4E2.1, the flow rate of dry air is denoted $D\ \mathrm{g\ min^{-1}}$ and the flow rate of humid, oxygen-rich air is $H\ \mathrm{g\ min^{-1}}$. The water content in the humid air is shown as 1 mass%.

simpler Eq. (4.3) can be applied to molecular species which are neither reactants nor products of reaction.

4.4 Material-Balance Worked Examples

Procedures for setting out mass-balance calculations are outlined in this section. Although not the only way to attack these problems, the method shown will assist your problem-solving efforts by formalising the mathematical approach. Mass-balance calculations are divided into four steps: *assemble*, *analyse*, *calculate* and *finalise*. Differential and integral mass balances with and without reaction are illustrated below.

Example 4.3 Continuous filtration

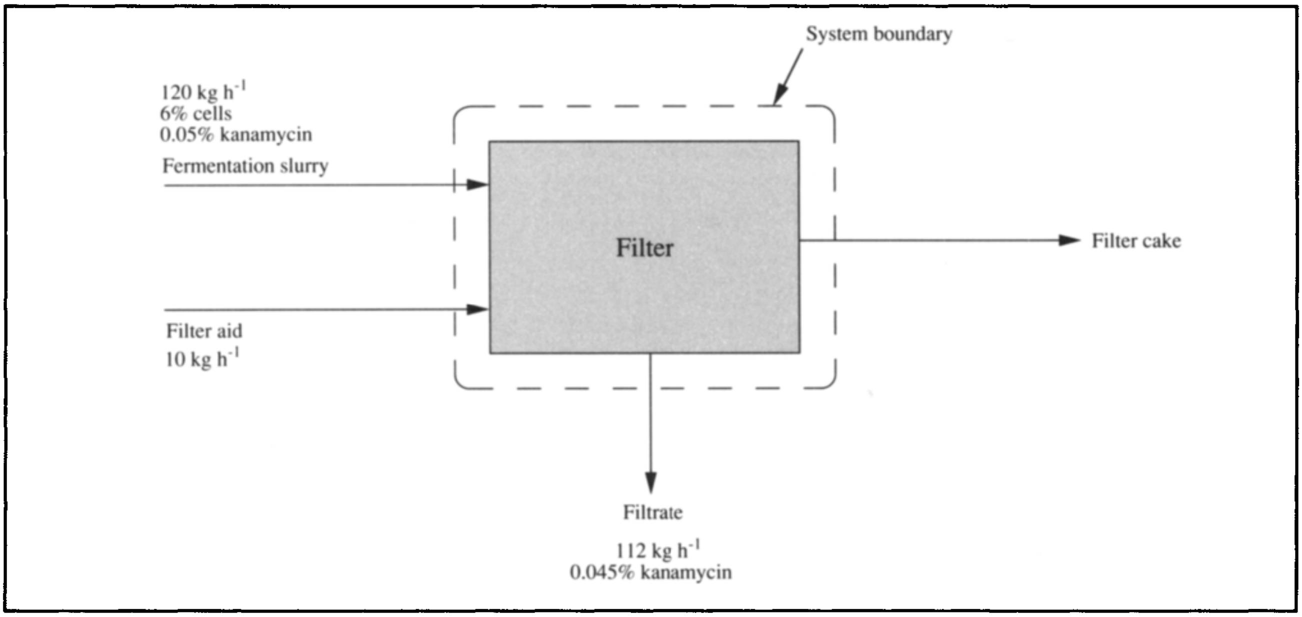
A fermentation slurry containing *Streptomyces kanamyceticus* cells is filtered using a continuous rotary vacuum filter. 120 kg h^{-1} slurry is fed to the filter; 1 kg slurry contains 60 g cell solids. To improve filtration rates, particles of diatomaceous-earth filter aid are added at a rate of 10 kg h^{-1}. The concentration of kanamycin in the slurry is 0.05% by weight. Liquid filtrate is collected at a rate of 112 kg h^{-1}; the concentration of kanamycin in the filtrate is 0.045% (w/w). Filter cake containing cells and filter aid is continuously removed from the filter cloth.

(a) What percentage liquid is the filter cake?
(b) If the concentration of kanamycin in the filter-cake liquid is the same as in the filtrate, how much kanamycin is absorbed per kg filter aid?

Solution:

1. *Assemble*
   (i) *Draw the flowsheet showing all data with units.*
   This is shown in Figure 4E3.1.

Figure 4E3.1 Flowsheet for continuous filtration.



   (ii) *Define the system boundary by drawing on the flowsheet.*
   The system boundary is shown in Figure 4E3.1.
2. *Analyse*
   (i) *State any assumptions.*
   — process is operating at steady state
   — system does not leak
   — filtrate contains no solids
   — cells do not absorb or release kanamycin during filtration
   — filter aid is dry
   — the liquid phase of the slurry, excluding kanamycin, can be considered water
   (ii) *Collect and state any extra data needed.*
   No extra data are required.

(iii) *Select and state a basis.*
The calculation is based on 120 kg slurry entering the filter, or 1 hour.
(iv) *List the compounds, if any, which are involved in reaction.*
No compounds are involved in reaction.
(v) *Write down the appropriate general mass-balance equation.*
The system is at steady state and no reaction occurs; therefore Eq. (4.3) is appropriate:

mass in = mass out.

3. *Calculate*
   (i) *Set up a calculation table showing all components of all streams passing across system boundaries. State the units used for the table. Enter all known quantities.*

As shown in Figure 4E3.1, four streams cross the system boundaries: fermentation slurry, filter aid, filtrate and filter cake. The components of these streams: cells, kanamycin, filter aid and water are represented in Table 4E3.1. The table is divided into two major sections: In and Out. Masses entering or leaving the system each hour are shown in the table; the units used are kg. Because filtrate and filter cake flow out of the system, there are no entries for these streams on the left hand-side of the table. Conversely, there are no entries for fermentation slurry or filter aid on the Out side of the table. The total mass of each stream is given in the last column on each side of the table. The total amount of each component flowing in and out of the system is shown in the last row. With all known quantities entered, several masses remain unknown; these quantities are indicated by question marks.

Table 4E3.1 Mass-balance table (kg)

Stream	*In*					*Out*				
	Cells	*Kanamycin*	*Filter aid*	*Water*	*Total*	*Cells*	*Kanamycin*	*Filter aid*	*Water*	*Total*
Fermentation slurry	7.2	0.06	0	?	120	–	–	–	–	–
Filter aid	0	0	10	0	10	–	–	–	–	–
Filtrate	–	–	–	–	–	0	0.05	0	?	112
Filter cake	–	–	–	–	–	?	?	?	?	?
Total	?	?	?	?	?	?	?	?	?	?

(ii) *Calculate unknown quantities, apply the mass-balance equation.*
To complete Table 4E3.1, let us consider each row and column separately. In the row representing fermentation slurry, the total mass of the stream is known as 120 kg and the masses of each component except water are known. The entry for water can therefore be determined as the difference between 120 kg and the sum of the known components: (120 – 7.2 – 0.06 – 0) kg = 112.74 kg. This mass for water has been entered in Table 4E3.2. The row for filter aid is already complete in Table 4E3.1; no cells or kanamycin are present in the diatomaceous earth, which we assume is dry. We can fill in the final row of the In side of the table; numbers in this row are obtained by adding the values in each vertical column. The total mass of cells input to the system in all streams is 7.2 kg, the total kanamycin entering is 0.06 kg, etc. The total mass of all components fed into the system is the sum of the last column of the left-hand side: (120 + 10) kg = 130 kg. On the Out side, we can complete the row for filtrate. We have assumed there are no solids such as cells or filter aid in the filtrate; therefore the mass of water in the filtrate is (112 – 0.05) kg = 111.95 kg. As yet, the entire composition and mass of the filter cake remain unknown.

Table 4E3.2 Completed mass-balance table (kg)

Stream	*In*					*Out*				
	Cells	*Kanamycin*	*Filter aid*	*Water*	*Total*	*Cells*	*Kanamycin*	*Filter aid*	*Water*	*Total*
Fermentation slurry	7.2	0.06	0	112.74	120	–	–	–	–	–
Filter aid	0	0	10	0	10	–	–	–	–	–
Filtrate	–	–	–	–	–	0	0.05	0	111.95	112
Filter cake	–	–	–	–	–	7.2	0.01	10	0.79	18
Total	7.2	0.06	10	112.74	130	7.2	0.06	10	112.74	130

To complete the table, we must consider the mass-balance equation relevant to this problem, Eq. (4.3). In the absence of reaction, this equation can be applied to total mass and to the masses of each component of the system.

Total mass balance
130 kg total mass in = total mass out.
∴ Total mass out = 130 kg.

Cell balance
7.2 kg cells in = cells out.
∴ Cells out = 7.2 kg.

Kanamycin balance
0.06 kg kanamycin in = kanamycin out.
∴ Kanamycin out = 0.06 kg.

Filter-aid balance
10 kg filter aid in = filter aid out.
∴ Filter aid out = 10 kg.

Water balance
112.74 kg water in = water out.
∴ Water out = 112.74 kg.

These results are entered in the last row of the Out side of Table 4E3.2; in the absence of reaction this row is always identical to the final row of In side. The component masses for filter cake can now be filled in as the difference between numbers in the final row and the masses of each component in the filtrate. Take time to look over Table 4E3.2; you should understand how all the numbers were obtained.

(iii) *Check that your results are reasonable and make sense.*

Mass-balance calculations must be checked. Make sure that all columns and rows of Table 4E3.2 add up to the totals shown.

4. *Finalise*

(i) *Answer the specific questions asked in the problem.*

The percentage liquid in the filter cake can be calculated from the results in Table 4E3.2. Dividing the mass of water in the filter cake by the total mass of this stream, the percentage liquid is:

$$\frac{0.79\ \text{kg}}{18\ \text{kg}} \times 100 = 4.39\%.$$

If the concentration of kanamycin in this liquid is only 0.045%, the mass of kanamycin is very close to:

$$\frac{0.045}{100} \times 0.79\text{ kg} = 3.6 \times 10^{-4}\text{ kg}.$$

However, we know from Table 4E3.2 that a total of 0.01 kg kanamycin is contained in the filter cake; therefore $(0.01 - 3.6 \times 10^{-4})$ kg = 0.0096 kg kanamycin is unaccounted for. Following our assumption that kanamycin is not adsorbed by the cells, 0.0096 kg kanamycin must be retained by the filter aid. 10 kg filter aid is present; therefore the kanamycin absorbed per kg filter aid is:

$$\frac{0.0096\text{ kg}}{10\text{ kg}} = 9.6 \times 10^{-4}\text{ kg kg}^{-1}.$$

(ii) *State the answers clearly and unambiguously, checking significant figures.*
   (a) The liquid content of the filter cake is 4.4%.
   (b) The amount of kanamycin absorbed by the filter aid is 9.6×10^{-4} kg kg^{-1}.

Note in Example 4.3 that the complete composition of the fermentation slurry was not provided. Cell and kanamycin concentrations were given; however the slurry most probably contained a variety of other components such as residual carbohydrate, minerals, vitamins, amino acids and additional fermentation products. These components were ignored in the mass balance; the liquid phase of the slurry was considered to be water only. This assumption is reasonable as the concentration of dissolved substances in fermentation broths is usually very small; water in spent broth usually accounts for more than 90% of the liquid phase.

Note also in this problem that the masses of some of the components were different by several orders of magnitude, e.g. the mass of kanamycin in the filtrate was of the order 10^{-2} whereas the total mass of this stream was of the order 10^2. Calculation of the mass of water by difference therefore involved subtracting a very small number from a large one and carrying more significant figures than warranted. This is an unavoidable feature of most mass balances for biological processes, which are characterised by dilute solutions, low product concentrations and large amounts of water. However, although excess significant figures were carried in the mass-balance table, the final answers were reported with due regard to data accuracy.

The above example illustrates mass-balance procedures for a simple steady-state process without reaction. An integral mass balance for a batch system without reaction is illustrated in Example 4.4.

Example 4.4 Batch mixing

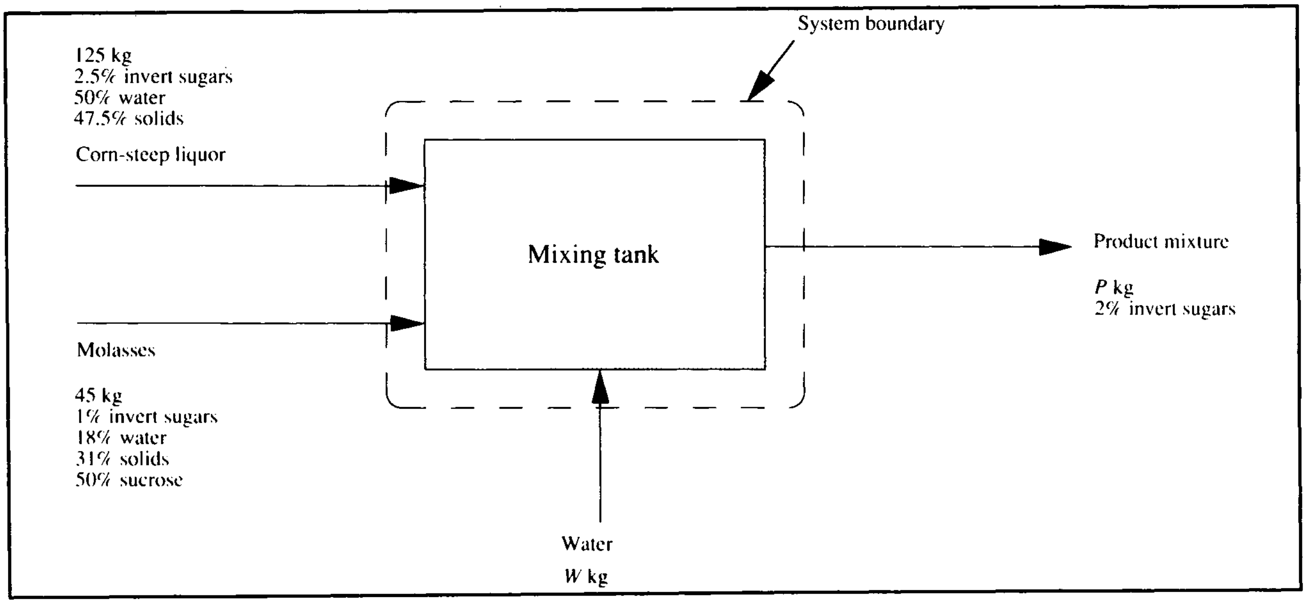
Corn-steep liquor contains 2.5% invert sugars and 50% water; the rest can be considered solids. Beet molasses containing 50% sucrose, 1% invert sugars, 18% water and the remainder solids, is mixed with corn-steep liquor in a mixing tank. Water is added to produce a diluted sugar mixture containing 2% (w/w) invert sugars. 125 kg corn-steep liquor and 45 kg molasses are fed into the tank.

(a) How much water is required?
(b) What is the concentration of sucrose in the final mixture?

Solution:

1. *Assemble*
   (i) *Flow sheet.*
   The flow sheet for this batch process is shown in Figure 4E4.1. Unlike in Figure 4E3.1 where the streams represented continuously-flowing inputs and outputs, the streams in Figure 4E4.1 represent masses added and removed at the beginning and end of the mixing process, respectively.

Figure 4E4.1 Flowsheet for batch mixing process.



(ii) *System boundary.*
The system boundary is indicated in Figure 4E4.1.

2. *Analyse*
(i) *Assumptions.*
— no leaks
— no inversion of sucrose to reducing sugars, or any other reaction
(ii) *Extra data.*
No extra data are required.
(iii) *Basis.*
125 kg corn-steep liquor.
(iv) *Compounds involved in reaction.*
No compounds are involved in reaction.
(v) *Mass-balance equation.*
The appropriate mass-balance equation is Eq. (4.3):

mass in = mass out.

3. *Calculate*
(i) *Calculation table.*
Table 4E4.1 shows all given quantities in kg. Rows and columns on each side of the table have been completed as much as possible from the information provided. Two unknown quantities are given symbols; the mass of water added is denoted *W*, the total mass of product mixture is denoted *P*.

Table 4E4.1 Mass-balance table (kg)

Stream	*In*					*Out*				
	Invert sugars	*Sucrose*	*Solids*	H_2O	*Total*	*Invert sugars*	*Sucrose*	*Solids*	H_2O	*Total*
Corn-steep liquor	3.125	0	59.375	62.5	125	–	–	–	–	–
Molasses	0.45	22.5	13.95	8.1	45	–	–	–	–	–
Water	0	0	0	W	W	–	–	–	–	–
Product mixture	–	–	–	–	–	0.02 P	?	?	?	P
Total	3.575	22.5	73.325	70.6 + W	170 + W	0.02 P	?	?	?	P

(ii) *Mass-balance calculations.*
Total mass balance

$(170 + W)$ kg total mass in $=$ P kg total mass out.
$\therefore 170 + W = P.$

$$(1)$$

Invert sugars balance

3.575 kg invert sugars in $= (0.02\ P)$ kg invert sugars out.
$\therefore 3.575 = 0.02\ P$
$P = 178.75$ kg.

Using this result in (1):

$W = 8.75$ kg.

$$(2)$$

Sucrose balance

22.5 kg sucrose in = sucrose out.
∴ Sucrose out = 22.5 kg.

Solids balance

73.325 kg solids in = solids out.
∴ Solids out = 73.325 kg.

H_2O balance

$(70.6 + W)$ kg in $=$ H_2O out.

Using the result from (2):

79.35 kg H_2O in = H_2O out.
$\therefore$ H_2O out = 79.35 kg.

These results allow the mass-balance table to be completed, as shown in Table 4E4.2.

Table 4E4.2 Completed mass-balance table (kg)

Stream	*In*					*Out*				
	Invert sugars	*Sucrose*	*Solids*	*H_2O*	*Total*	*Invert sugars*	*Sucrose*	*Solids*	*H_2O*	*Total*
Corn-steep liquor	3.125	0	59.375	62.5	125	–	–	–	–	–
Molasses	0.45	22.5	13.95	8.1	45	–	–	–	–	–
Water	0	0	0	8.75	8.75	–	–	–	–	–
Product mixture	–	–	–	–	–	3.575	22.5	73.325	79.35	178.75
Total	3.575	22.5	73.325	79.35	178.75	3.575	22.5	73.325	79.35	178.75

(iii) *Check the results.*
All columns and rows of Table 4E4.2 add up correctly.

4. *Finalise*
(i) *The specific questions.*
The water required is 8.75 kg. The sucrose concentration in the product mixture is:

$$\frac{22.5}{178.75} \times 100 = 12.6\%$$

(ii) *Answers.*
(a) 8.75 kg water is required.
(b) The product mixture contains 13% sucrose.

Material balances on reactive systems are slightly more complicated than Examples 4.3 and 4.4. To solve problems with reaction, stoichiometric relationships must be used in conjunction with mass-balance equations. These procedures are illustrated in Examples 4.5 and 4.6.

Example 4.5 Continuous acetic acid fermentation

Acetobacter aceti bacteria convert ethanol to acetic acid under aerobic conditions. A continuous fermentation process for vinegar production is proposed using non-viable *A. aceti* cells immobilised on the surface of gelatin beads. The production target is 2 kg h^{-1} acetic acid; however the maximum acetic acid concentration tolerated by the cells is 12%. Air is pumped into the fermenter at a rate of 200 gmol h^{-1}.

(a) What minimum amount of ethanol is required?
(b) What minimum amount of water must be used to dilute the ethanol to avoid acid inhibition?
(c) What is the composition of the fermenter off-gas?

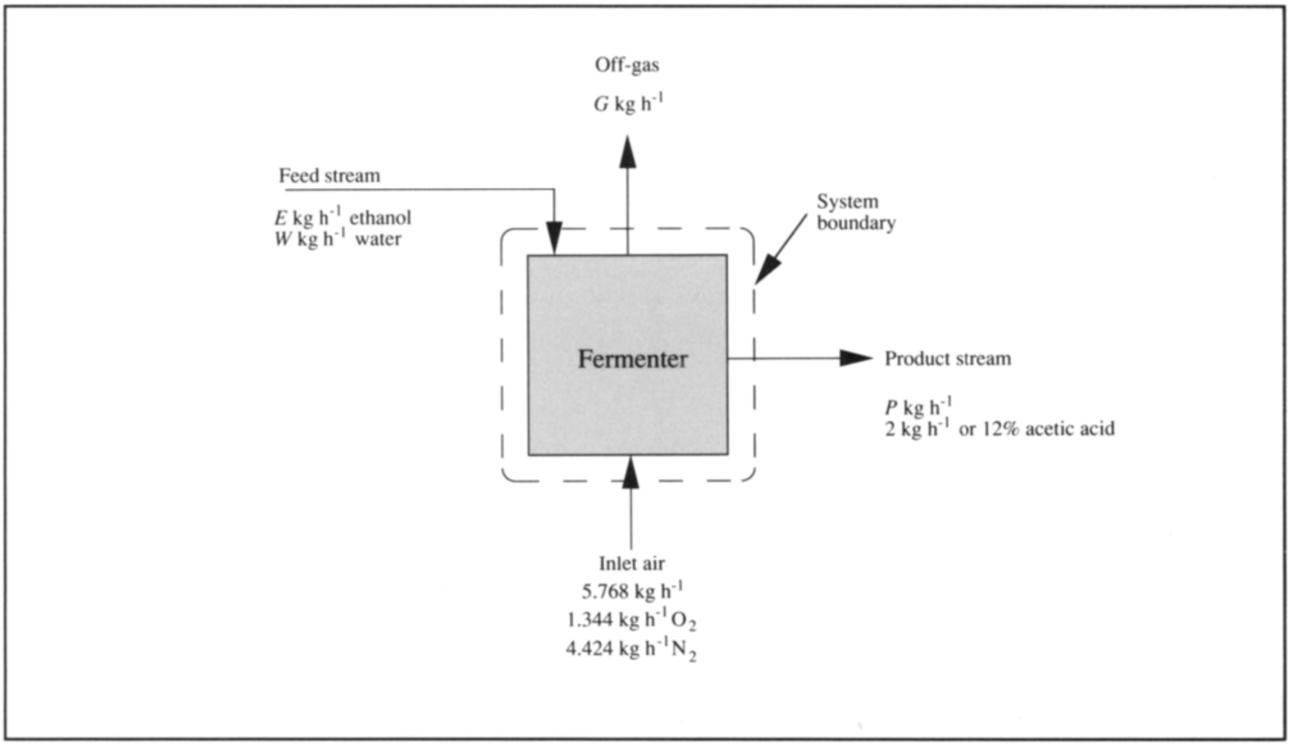
Solution:

1. *Assemble*

(i) *Flow sheet.*

The flow sheet for this process is shown in Figure 4E5.1.

Figure 4E5.1 Flow sheet for continuous acetic acid fermentation.



(ii) *System boundary.*

The system boundary is shown in Figure 4E5.1.

(iii) *Write down the reaction equation.*

In the absence of cell growth, maintenance or other metabolism of substrate, the reaction equation is:

$$\underset{\text{(ethanol)}}{C_2H_5OH} + O_2 \rightarrow \underset{\text{(acetic acid)}}{CH_3COOH} + H_2O.$$

2. *Analyse*

(i) *Assumptions.*

—steady state
—no leaks
—inlet air is dry
—gas volume% = mole%
—no evaporation of ethanol, H_2O or acetic acid
—complete conversion of ethanol
—ethanol is used by the cells for synthesis of acetic acid only; no side-reactions occur

—oxygen transfer is sufficiently rapid to meet the demands of the cells
—concentration of acetic acid in the product stream is 12%

(ii) *Extra data.*

Molecular weights: ethanol = 46
acetic acid = 60
$O_2 = 32$
$N_2 = 28$
$H_2O = 18$

Composition of air: 21% O_2, 79% N_2.

(iii) *Basis.*

The calculation is based on 2 kg acetic acid leaving the system, or 1 hour.

(iv) *Compounds involved in reaction.*

The compounds involved in reaction are ethanol, acetic acid, O_2 and H_2O. N_2 is not involved in reaction.

(v) *Mass-balance equations.*

For ethanol, acetic acid, O_2 and H_2O, the appropriate mass-balance equation is Eq. (4.2):

mass in + mass generated = mass out + mass consumed.

For total mass and N_2, the appropriate mass-balance equation is Eq. (4.3):

mass in = mass out.

3. *Calculate*

(i) *Calculation table.*

The mass-balance table with data provided is shown as Table 4E5.1; the units are kg. EtOH denotes ethanol; HAc is acetic acid. If 2 kg acetic acid represents 12 mass% of the product stream, the total mass of the product stream must be $^2/_{0.12}$ = 16.67 kg. If we assume complete conversion of ethanol, the only components of the product stream are acetic acid and water; therefore water must account for 88 mass% of the product stream = 14.67 kg. In order to represent what is known about the inlet air, some preliminary calculations are needed.

$$O_2 \text{ content} = (0.21)\,(200 \text{ gmol}\,.\left|\frac{32 \text{ g}}{\text{gmol}}\right| = 1344 \text{ g} = 1.344 \text{ kg}$$

$$N_2 \text{ content} = (0.79)\,(200 \text{ gmol})\,.\left|\frac{28 \text{ g}}{\text{gmol}}\right| = 4424 \text{ g} = 4.424 \text{ kg}.$$

Therefore, the total mass of air in = 5.768 kg. The masses of O_2 and N_2 can now be entered in the table, as shown.

Table 4E5.1 Mass-balance table (kg)

Stream	*In*						*Out*					
	EtOH	*HAc*	*H_2O*	*O_2*	*N_2*	*Total*	*EtOH*	*HAc*	*H_2O*	*O_2*	*N_2*	*Total*
Feed stream	*E*	0	*W*	0	0	*E* + *W*	–	–	–	–	–	–
Inlet air	0	0	0	1.344	4.424	5.768	–	–	–	–	–	–
Product stream	–	–	–	–	–	–	0	2	14.67	0	0	16.67
Off-gas	–	–	–	–	–	–	0	0	0	?	?	*G*
Total	*E*	0	*W*	1.344	4.424	5.768 + *E* + *W*	0	2	14.67	?	?	16.67 + *G*

E and W denote the unknown quantities of ethanol and water in the feed stream, respectively; G represents the total mass of off-gas. The question marks in the table show which other quantities must be calculated.

(ii) *Mass-balance and stoichiometry calculations.*
As N_2 is a tie component, its mass balance is straightforward.

N_2 balance

4.424 kg N_2 in = N_2 out.
∴ N_2 out = 4.424 kg.

To deduce the other unknowns, we must use stoichiometric analysis as well as mass balances.

HAc balance

0 kg HAc in + HAc generated = 2 kg HAc out + 0 kg HAc consumed.
∴ HAc generated = 2 kg.

$$2\text{ kg} = 2\text{ kg}\cdot\left|\frac{1\text{ kgmol}}{60\text{ kg}}\right| = 3.333\times10^{-2}\text{ kgmol}.$$

From reaction stoichiometry, we know that generation of 3.333×10^{-2} kgmol HAc requires 3.333×10^{-2} kgmol each of EtOH and O_2, and is accompanied by generation of 3.333×10^{-2} kgmol H_2O:

$$\therefore 3.333\times10^{-2}\text{ kgmol}\cdot\left|\frac{46\text{ kg}}{1\text{ kgmol}}\right| = 1.533\text{ kg EtOH is consumed}$$

$$3.333\times10^{-2}\text{ kgmol}\cdot\left|\frac{32\text{ kg}}{1\text{ kgmol}}\right| = 1.067\text{ kg } O_2 \text{ is consumed}$$

$$3.333\times10^{-2}\text{ kgmol}\cdot\left|\frac{18\text{ kg}}{1\text{ kgmol}}\right| = 0.600\text{ kg } H_2O \text{ is generated.}$$

We can use this information to complete the mass balances for EtOH, O_2 and H_2O.

EtOH balance

EtOH in + 0 kg EtOH generated = 0 kg EtOH out + 1.533 kg EtOH consumed.
∴ EtOH in = 1.533 kg = E.

O_2 balance

1.344 kg O_2 in + 0 kg O_2 generated = O_2 out + 1.067 kg O_2 consumed.
∴ O_2 out = 0.277 kg.

Therefore, summing the O_2 and N_2 components of the off-gas:

G = (0.277 + 4.424) kg = 4.701 kg.

H_2O balance

W kg H_2O in + 0.600 kg H_2O generated = 14.67 kg H_2O out + 0 kg H_2O consumed.
∴ W = 14.07 kg.

These results allow us to complete the mass-balance table, as shown in Table 4E5.2.

Table 4E5.2 Completed mass-balance table (kg)

Stream	*In*						*Out*					
	EtOH	*HAc*	H_2O	O_2	N_2	*Total*	*EtOH*	*HAc*	H_2O	O_2	N_2	*Total*
Feed stream	1.533	0	14.07	0	0	15.603	–	–	–	–	–	–
Inlet air	0	0	0	1.344	4.424	5.768	–	–	–	–	–	–
Product stream	–	–	–	–	–	–	0	2	14.67	0	0	16.67
Off-gas	–	–	–	–	–	–	0	0	0	0.277	4.424	4.701
Total	1.533	0	14.07	1.344	4.424	21.371	0	2	14.67	0.277	4.424	21.371

(iii) *Check the results.*
All rows and columns of Table 4E5.2 add up correctly.

4. *Finalise*
(i) *The specific questions.*
The ethanol required is 1.533 kg. The water required is 14.07 kg. The off-gas contains 0.277 kg O_2 and 4.424 kg N_2. Since gas compositions are normally expressed using volume or mole%, we must convert these values to moles:

$$O_2 \text{ content} = 0.277 \text{ kg} \cdot \left| \frac{1 \text{ kgmol}}{32 \text{ kg}} \right| = 8.656 \times 10^{-3} \text{ kgmol}$$

$$N_2 \text{ content} = 4.424 \text{ kg} \cdot \left| \frac{1 \text{ kgmol}}{28 \text{ kg}} \right| = 0.1580 \text{ kgmol.}$$

Therefore, the total molar quantity of off-gas is 0.1667 kgmol. The off-gas composition is:

$$\frac{8.656 \times 10^{-3} \text{ kgmol}}{0.1667 \text{ kgmol}} \times 100 = 5.19\% \, O_2$$

$$\frac{0.1580 \text{ kgmol}}{0.1667 \text{ kgmol}} \times 100 = 94.8\% \, N_2.$$

(ii) *Answers.*
Quantities are expressed in kg h^{-1} rather than kg to reflect the continuous nature of the process and the basis used for calculation.

(a) 1.5 kg h^{-1} ethanol is required.
(b) 14.1 kg h^{-1} water must be used to dilute the ethanol in the feed stream.
(c) The composition of the fermenter off-gas is 5.2% O_2 and 94.8% N_2.

There are several points to note about the problem and calculation of Example 4.5. First, cell growth and its requirement for substrate were not considered because the cells used in this process were non-viable. For fermentation with live cells, growth and other metabolic activity must be taken into account in the mass balance. This requires knowledge of growth stoichiometry, which is considered in Example 4.6 and discussed in more detail in Section 4.6. Use of non-growing immobilised cells in Example 4.5 meant that the cells were not components of any stream flowing in or out of the process, nor were they generated in reaction. Therefore, cell mass did not have to be included in the calculation.

Example 4.5 illustrates the importance of phase separations. Unreacted oxygen and nitrogen were assumed to leave the system as off-gas rather than as components of the liquid product stream. This assumption is reasonable due to the very poor solubility of oxygen and nitrogen in aqueous liquids; although the product stream most likely contains some dissolved gas, the quantities are relatively small. This assumption may need to be reviewed for gases with higher solubility, e.g. ammonia.

In the above problem, nitrogen did not react, nor were there more than one stream in and one stream out carrying nitrogen. A material which goes directly from one stream to another is called a *tie component*; the mass balance for a tie component is relatively simple. Tie components are useful because they can provide partial solutions to mass-balance problems making subsequent calculations easier. More than one tie component may be present in a particular process.

One of the listed assumptions in Example 4.5 is rapid oxygen transfer. Because cells use oxygen in dissolved form, oxygen must be transferred into the liquid phase from gas bubbles supplied to the fermenter. The speed of this process depends on the culture conditions and operation of the fermenter as described in more detail in Chapter 9. In mass-balance problems we assume that all oxygen required by the stoichiometric equation is immediately available to the cells.

Sometimes it is not possible to solve for unknown quantities in mass balances until near the end of the calculation. In such cases, symbols for various components rather than numerical values must be used in the balance equations. This is illustrated in the integral mass-balance of Example 4.6 which analyses batch culture of growing cells for production of xanthan gum.

Example 4.6 Xanthan gum production

Xanthan gum is produced using *Xanthomonas campestris* in batch culture. Laboratory experiments have shown that for each gram of glucose utilised by the bacteria, 0.23 g oxygen and 0.01 g ammonia are consumed, while 0.75 g gum, 0.09 g cells, 0.27 g gaseous CO_2 and 0.13 g H_2O are formed. Other components of the system such as phosphate can be neglected. Medium containing glucose and ammonia dissolved in 20 000 litres water is pumped into a stirred fermenter and inoculated with *X. campestris*. Air is sparged into the fermenter; the total amount of off-gas recovered during the entire batch culture is 1250 kg. Because of the high viscosity and difficulty in handling xanthan-gum solutions, the final gum concentration should not be allowed to exceed 3.5 wt%.

(a) How much glucose and ammonia are required?
(b) What percentage excess air is provided?

Solution:

1. *Assemble*

(i) *Flow sheet.*

The flow sheet for this process is shown in Figure 4E6.1.

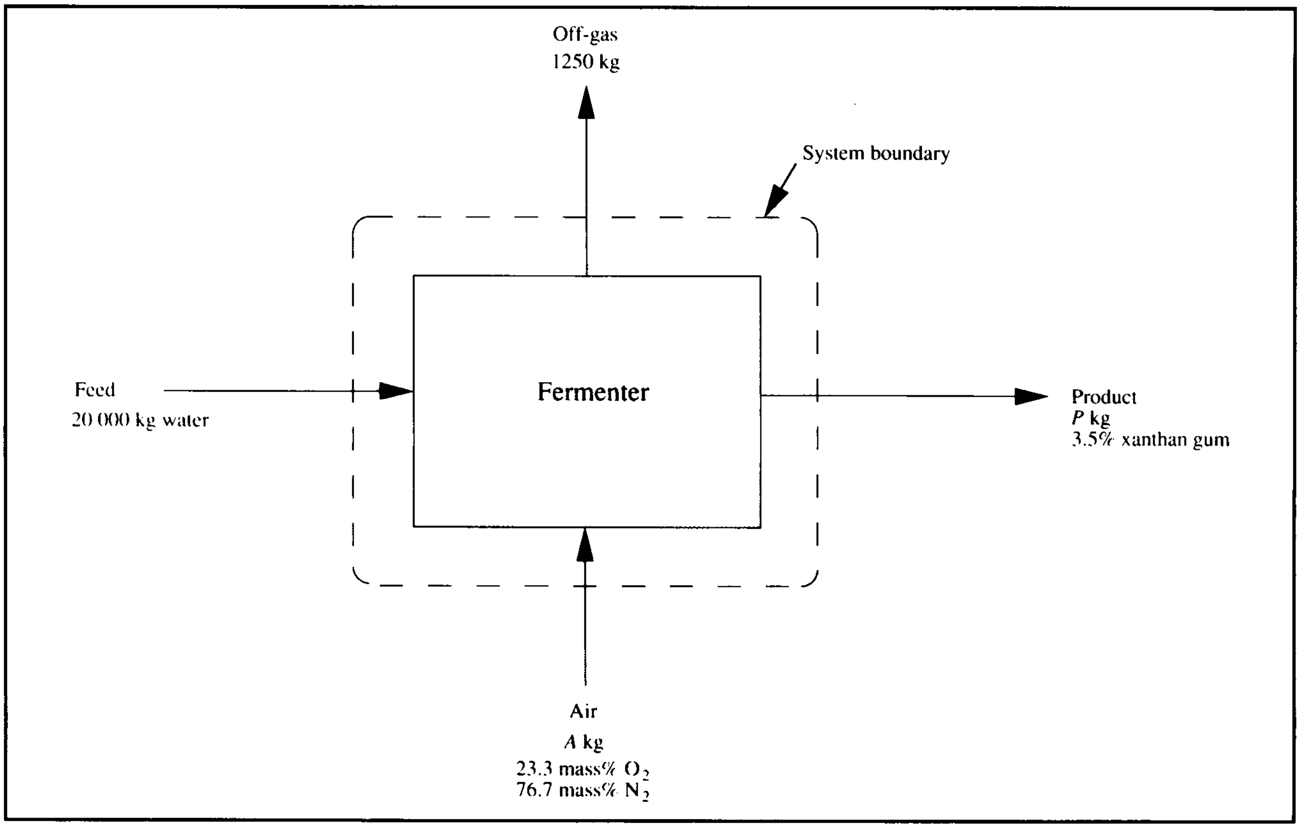
(ii) *System boundary.*

The system boundary is shown in Figure 4E6.1.

(iii) *Reaction equation.*

$$1\ \text{g glucose} + 0.23\ \text{g}\ O_2 + 0.01\ \text{g}\ NH_3 \rightarrow 0.75\ \text{g gum} + 0.09\ \text{g cells} + 0.27\ \text{g}\ CO_2 + 0.13\ \text{g}\ H_2O.$$

Figure 4E6.1 Flowsheet for xanthan gum fermentation.



2. *Analyse*

(i) *Assumptions.*

—no leaks

—inlet air and off-gas are dry

—conversion of glucose and NH_3 is 100% complete

—CO_2 leaves in the off-gas

(ii) *Extra data.*

Molecular weights: $O_2 = 32$
$N_2 = 28$

(iii) *Basis.*

1250 kg off-gas.

(iv) *Compounds involved in reaction.*

The compounds involved in reaction are glucose, O_2, NH_3, gum, cells, CO_2 and H_2O. N_2 is not involved in reaction.

(v) *Mass-balance equations.*

For glucose, O_2, NH_3, gum, cells, CO_2 and H_2O, the appropriate mass-balance equation is Eq. (4.2):

mass in + mass generated = mass out + mass consumed.

For total mass and N_2, the appropriate mass-balance equation is Eq. (4.3):

mass in = mass out.

3. *Calculate*

(i) *Calculation table.*

Some preliminary calculations are required to start the mass-balance table. First, using 1 kg l^{-1} as the density of water, 20 000 litres water is equivalent to 20 000 kg. Let A be the unknown mass of air added. At low pressure, air is composed of 21 mol% O_2 and 79 mol% N_2; we need to determine the composition of air as mass fractions. In 100 gmol air:

$$O_2 \text{ content} = 21 \text{ gmol} \cdot \left| \frac{32 \text{ g}}{1 \text{ gmol}} \right| = 672 \text{ g}.$$

$$N_2 \text{ content} = 79 \text{ gmol} \cdot \left| \frac{28 \text{ g}}{1 \text{ gmol}} \right| = 2212 \text{ g}.$$

If the total mass of air in 100 gmol is (2212 + 672) = 2884 g, the composition of air is:

$$\frac{672 \text{ g}}{2884 \text{ g}} \times 100 = 23.3 \text{ mass\% } O_2.$$

$$\frac{2212 \text{ g}}{2884 \text{ g}} \times 100 = 76.7 \text{ mass\% } N_2.$$

Therefore, the mass of O_2 in the inlet air is $0.233A$; the mass of N_2 is $0.767A$. Let F denote the total mass of feed medium added; let P denote the total mass of product. We will perform the calculation to produce the maximum-allowable gum concentration; therefore, the mass of gum in the product is $0.035P$. With the assumption of 100% conversion of glucose and NH_3, these compounds are not present in the product. Quantities known at the beginning of the problem are shown in Table 4E6.1.

Table 4E6.1 Mass-balance table (kg)

Stream	*In*									*Out*								
	Glucose	*O_2*	*N_2*	*CO_2*	*Gum*	*Cells*	*NH_3*	*H_2O*	*Total*	*Glucose*	*O_2*	*N_2*	*CO_2*	*Gum*	*Cells*	*NH_3*	*H_2O*	*Total*
Feed	?	0	0	0	0	0	?	20 000	F	–	–	–	–	–	–	–	–	–
Air	0	$0.233A$	$0.767A$	0	0	0	0	0	A	–	–	–	–	–	–	–	–	–
Off-gas	–	–	–	–	–	–	–	–	–	0	?	?	?	0	0	0	0	1250
Product	–	–	–	–	–	–	–	–	–	0	0	0	0	$0.035P$	?	0	0	P
Total	?	$0.233A$	$0.767A$	0	0	0	?	20 000	$F+A$	0	?	?	?	$0.035P$	?	0	?	$1250 + P$

(ii) *Mass-balance and stoichiometry calculations.*

Total mass balance

$(F+A)$ kg total mass in = $(1250 + P)$ kg total mass out.

$\therefore F+A = 1250 + P.$

(1)

Gum balance

0 kg gum in + gum generated = $(0.035P)$ kg gum out + 0 kg gum consumed.

$\therefore$ Gum generated = $(0.035P)$ kg.

From reaction stoichiometry, synthesis of $(0.035P)$ kg gum requires:

$$\frac{0.035P}{0.75}\ (1\ \text{kg}) = (0.0467P)\ \text{kg glucose}$$

$$\frac{0.035P}{0.75}\ (0.23\ \text{kg}) = (0.0107P)\ \text{kg}\ O_2$$

$$\frac{0.035P}{0.75}\ (0.01\ \text{kg}) = (0.00047P)\ \text{kg}\ NH_3$$

and produces:

$$\frac{0.035P}{0.75}\ (0.09\ \text{kg}) = (0.0042P)\ \text{kg cells}$$

$$\frac{0.035P}{0.75}\ (0.27\ \text{kg}) = (0.0126P)\ \text{kg}\ CO_2$$

$$\frac{0.035P}{0.75}\ (0.13\ \text{kg}) = (0.00607P)\ \text{kg}\ H_2O.$$

O_2 balance

$(0.233A)$ kg O_2 in + 0 kg O_2 generated = O_2 out + $(0.0107P)$ kg O_2 consumed.
$\therefore O_2$ out = $(0.233A - 0.0107P)$ kg. (2)

N_2 balance
N_2 is a tie component.

$(0.767A)$ kg N_2 in = N_2 out.
$\therefore N_2$ out = $(0.767A)$ kg. (3)

CO_2 balance

0 kg CO_2 in + $(0.0126P)$ kg CO_2 generated = CO_2 out + 0 kg CO_2 consumed.
$\therefore$ CO_2 out = $(0.0126P)$ kg. (4)

The total mass of gas out is 1250 kg. Therefore, adding the amounts of O_2, N_2 and CO_2 out from (2), (3) and (4):

$$\begin{aligned} 1250 &= (0.233A - 0.0107P) + (0.767A) + (0.0126P) \\ &= A + 0.0019P \end{aligned}$$
$$\therefore A = 1250 - 0.0019P. \quad (5)$$

Glucose balance

glucose in + 0 kg glucose generated = 0 kg glucose out + $(0.0467P)$ kg glucose consumed.
$\therefore$ Glucose in = $(0.0467\ P)$ kg. (6)

NH_3 balance

NH_3 in + 0 kg NH_3 generated = 0 kg NH_3 out + $(0.00047P)$ kg NH_3 consumed.
$\therefore$ NH_3 in = $(0.00047P)$ kg. (7)

We can now calculate the total mass of the feed, F:

F = glucose in + NH_3 in + water in.

From (6) and (7):

$$F = (0.0467P) \text{ kg} + (0.00047P) \text{ kg} + 20\,000 \text{ kg}$$
$$= (20\,000 + 0.04717P) \text{ kg}. \quad (8)$$

We can now use (8) and (5) in (1):

$$(20\,000 + 0.04717P) + (1250 - 0.0019P) = (1250 + P)$$
$$20\,000 = 0.95473\,P$$
$$\therefore P = 20\,948.3 \text{ kg}.$$
$\therefore$ Gum out = 733.2 kg.

Substituting this result in (5) and (8):

A = 1210.2 kg
F = 20 988.1 kg.

From Table 4E6.1:

O_2 in = 282.0 kg
N_2 in = 928.2 kg.

Using the results for P, A and F in (2), (3), (4), (6) and (7):

O_2 out = 57.8 kg
N_2 out = 928.2 kg
CO_2 out = 263.9 kg
Glucose in = 978.3 kg
NH_3 in = 9.8 kg.

Cell balance

0 kg cells in + $(0.0042P)$ kg cells generated = cells out + 0 kg cells consumed.
$\therefore$ Cells out = $(0.0042P)$ kg
Cells out = 88.0 kg.

H_2O balance

20 000 kg H_2O in + $(0.00607P)$ kg H_2O generated = H_2O out + 0 kg H_2O consumed.
$\therefore$ H_2O out = 20 000 + $(0.00607P)$ kg.
H_2O out = 20 127.2 kg.

These entries are included in Table 4E6.2.

Table 4E6.2 Completed mass-balance table (kg)

Stream	*In*									*Out*								
	Glucose	O_2	N_2	CO_2	*Gum*	*Cells*	NH_3	H_2O	*Total*	*Glucose*	O_2	N_2	CO_2	*Gum*	*Cells*	NH_3	H_2O	*Total*
Feed	978.3	0	0	0	0	0	9.8	20000	20988.1	–	–	–	–	–	–	–	–	–
Air	0	282.0	928.2	0	0	0	0	0	1210.2	–	–	–	–	–	–	–	–	–
Off-gas	–	–	–	–	–	–	–	–	–	0	57.8	928.2	263.9	0	0	0	0	1250
Product	–	–	–	–	–	–	–	–	–	0	0	0	0	733.2	88.0	0	20127.2	20948.3
Total	978.3	282.0	928.2	0	0	0	9.8	20000	22198.3	0	57.8	928.2	263.9	733.2	88.0	0	20127.2	22198.3

(iii) *Check the results.*

All the columns and rows of Table 4E6.2 add up correctly to within round-off error.

4. *Finalise*

(i) *The specific questions.*

From the completed mass-balance table, 978.3 kg glucose and 9.8 kg NH_3 are required. Calculation of percentage excess air is based on oxygen as oxygen is the reacting component of air. Percentage excess can be calculated using Eq. (2.35) in units of kg:

$$\%\ \text{excess air} = \frac{\left(\begin{array}{c}\text{kg } O_2 \text{ present} - \text{kg } O_2 \text{ required to react} \\ \text{completely with the limiting substrate}\end{array}\right)}{\left(\begin{array}{c}\text{kg } O_2 \text{ required to react completely} \\ \text{with the limiting substrate}\end{array}\right)} \times 100.$$

In this problem, both glucose and ammonia are limiting substrates. From stoichiometry and the mass-balance table, the mass of oxygen required to react completely with 978.3 kg glucose and 9.8 kg NH_3 is:

$$\frac{978.3 \text{ kg}}{1 \text{ kg}} (0.23 \text{ kg}) = 225.0 \text{ kg } O_2.$$

The mass provided is 282.0 kg; therefore:

$$\%\ \text{excess air} = \frac{282.0 - 225.0}{225.0} \times 100 = 25.3\%.$$

(ii) *Answers.*

(a) 980 kg glucose and 9.8 kg NH_3 are required.

(b) 25% excess air is provided.

4.5 Material Balances With Recycle, By-Pass and Purge Streams

So far, we have performed mass balances on simple single-unit processes. However, steady-state systems incorporating recycle, by-pass and purge streams are common in bioprocess industries; flow sheets illustrating these modes of operation are shown in Figure 4.3. Material-balance calculations for such systems can be somewhat more involved than those in Examples 4.3 to 4.6; several balances are required before all mass flows can be determined.

As an example, consider the system of Figure 4.4. Because cells are the catalysts in fermentation processes, it is often advantageous to recycle biomass from spent fermentation broth. Cell recycle requires a separation device, such as a centrifuge or gravity settling tank, to provide a concentrated recycle stream under aseptic conditions. The flow sheet for cell recycle is shown in Figure 4.5; as indicated, at least four

Figure 4.3 Flow sheet for processes with (a) recycle, (b) by-pass and (c) purge streams.

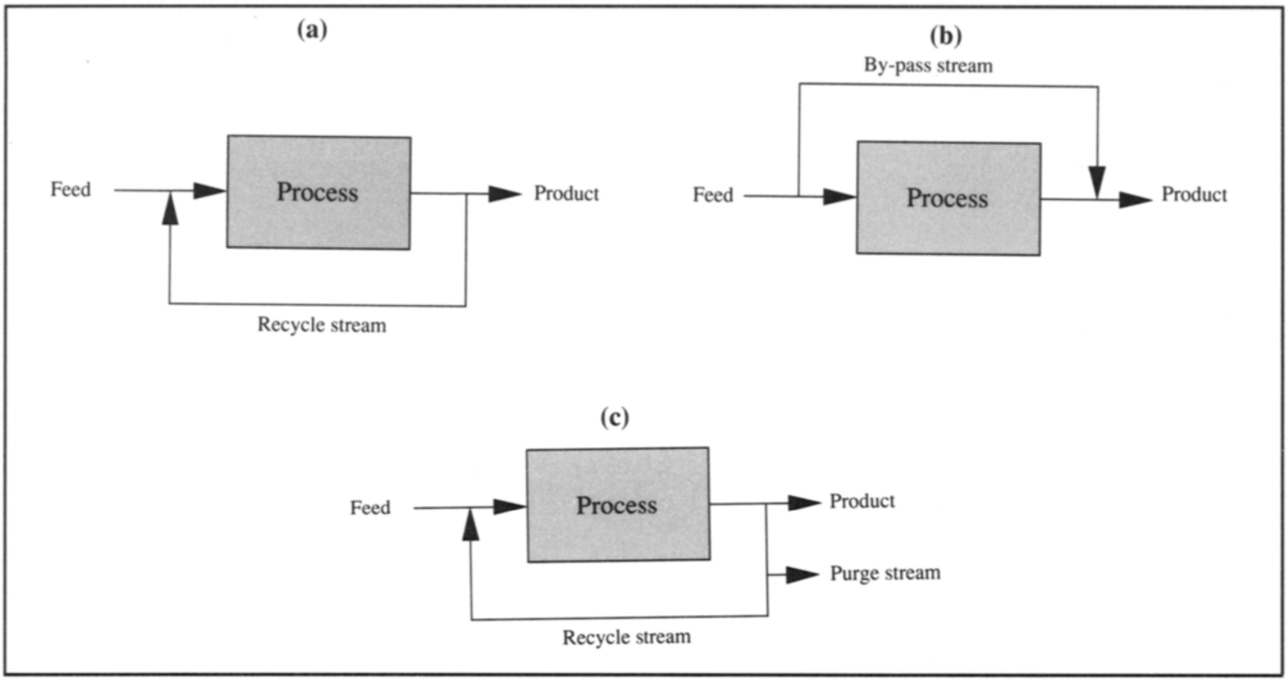
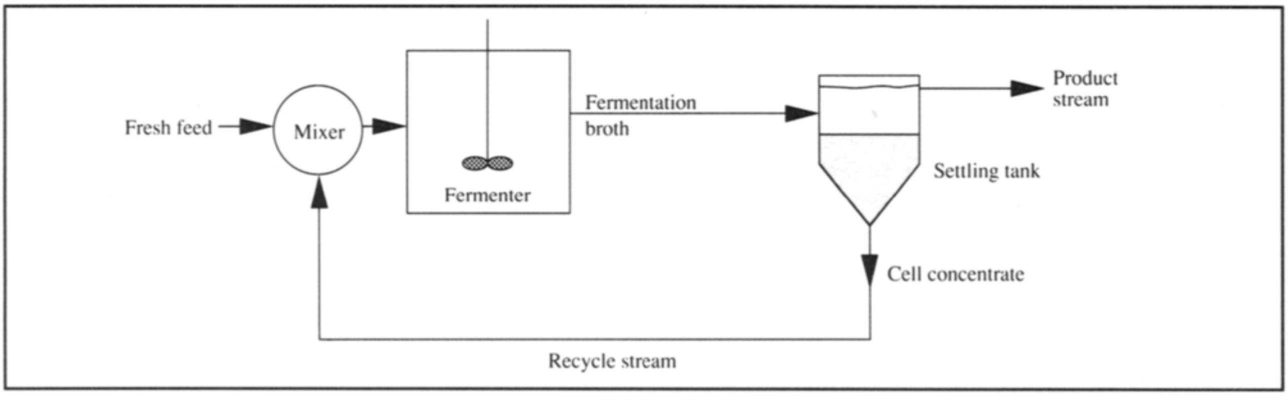


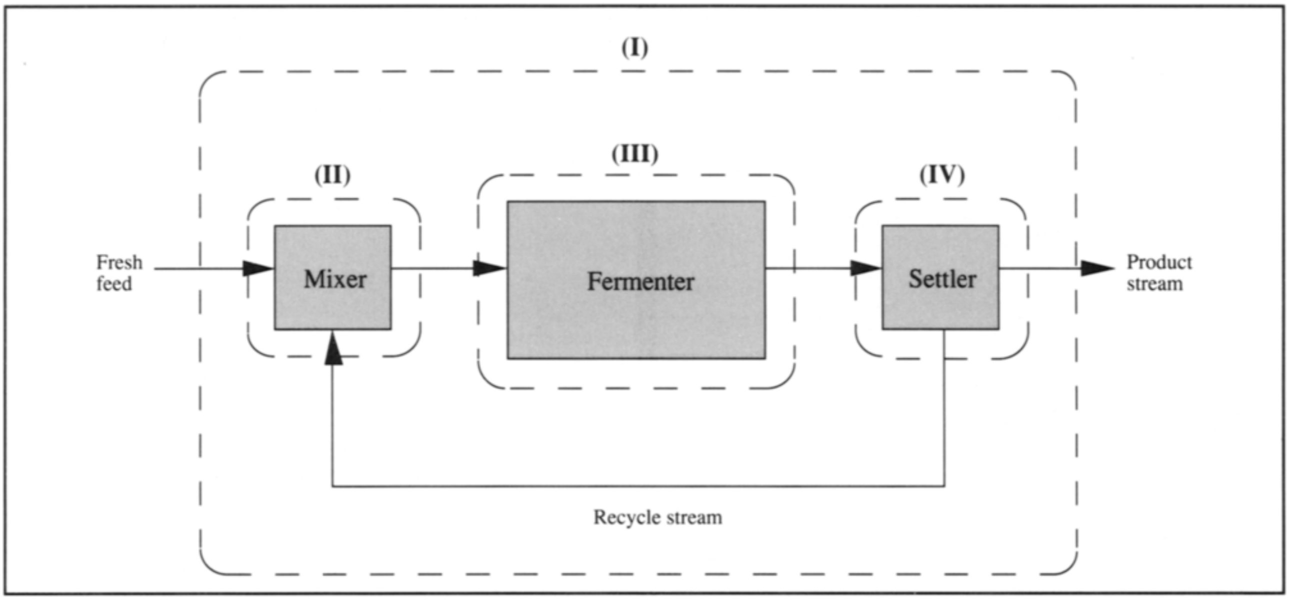
Figure 4.4 Fermenter with cell recycle.



different system boundaries can be defined. System I represents the overall recycle process; only the fresh feed and final product streams cross this system boundary. In addition, separate material balances can be performed over each process unit: the mixer, the fermenter and the settler. Other system boundaries could also be defined; for example, we could group the mixer and fermenter, or settler and fermenter, together. Material balances with recycle involve carrying out individual mass-balance calculations for each designated system. Depending on which quantities are known and what information is sought, analysis of more than one system may be required before the flow rates and compositions of all streams are known.

Mass balances with recycle, by-pass or purge usually involve longer calculations than for simple processes, but are not more difficult conceptually. Accordingly, we will not treat these types of process further. Examples of mass-balance procedures for multi-unit processes can be found in standard chemical-engineering texts, e.g. [1–3].

Figure 4.5 System boundaries for cell-recycle system.



4.6 Stoichiometry of Growth and Product Formation

So far in this chapter, the law of conservation of mass has been used to determine unknown quantities entering or leaving bioprocesses. For mass balances with reaction such as Examples 4.5 and 4.6, the stoichiometry of conversion must be known before the mass balance can be solved. When cell growth occurs, cells are a product of reaction and must be represented in the reaction equation. In this section we will discuss how reaction equations for growth and product synthesis are formulated. Metabolic stoichiometry has many applications in bioprocessing: as well as in mass and energy balances, it can be used to compare theoretical and actual product yields, check the consistency of experimental fermentation data, and formulate nutrient medium.

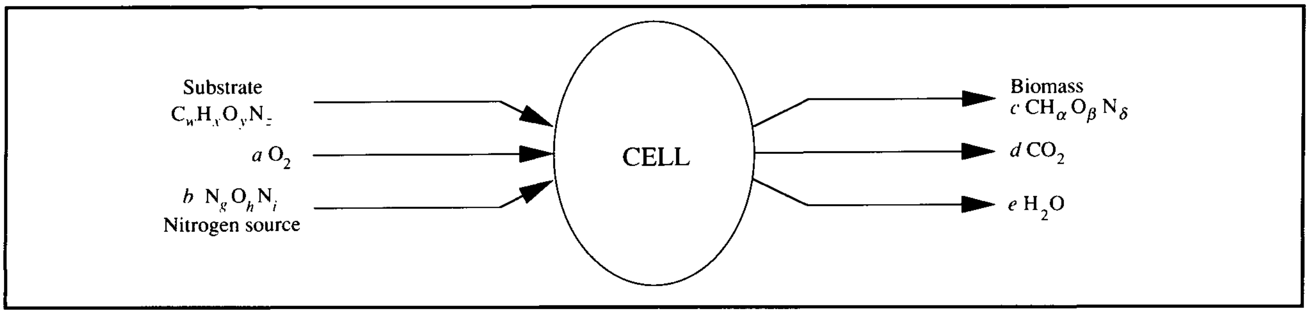
4.6.1 Growth Stoichiometry and Elemental Balances

Despite its complexity and the thousands of intracellular reactions involved, cell growth obeys the law of conservation of matter. All atoms of carbon, hydrogen, oxygen, nitrogen and other elements consumed during growth are incorporated into new cells or excreted as products. Confining our attention to those compounds taken up or produced in significant quantity, if the only extracellular products formed are CO_2 and H_2O, we can write the following equation for aerobic cell growth:

$$C_wH_xO_yN_z + a\,O_2 + b\,H_gO_hN_i \rightarrow c\,CH_\alpha O_\beta N_\delta + d\,CO_2 + e\,H_2O. \tag{4.4}$$

In Eq. (4.4), $C_wH_xO_yN_z$ is the chemical formula for the substrate (e.g. for glucose $w = 6$, $x = 12$, $y = 6$ and $z = 0$), $H_gO_hN_i$ is the chemical formula for the nitrogen source, and $CH_\alpha O_\beta N_\delta$ is the chemical 'formula' for dry biomass. a, b, c, d and e are stoichiometric coefficients. Eq. (4.4) is written on the basis of one mole of substrate; therefore a moles O_2 are consumed and d moles CO_2 are formed per mole substrate reacted, etc. As illustrated in Figure 4.6, the equation represents a macroscopic view of metabolism; it ignores the detailed structure of the system and considers only those components which have net interchange with the environment. Despite its simplicity, the macroscopic approach provides a powerful tool for thermodynamic analysis. Eq. (4.4) does not include a multitude of compounds such as ATP and NADH which are integral to metabolism and undergo exchange cycles in cells, but are not subject to net exchange with the environment. Compounds such as vitamins and minerals taken up during metabolism could be included; however, since these growth factors are generally consumed in small quantity we assume here that their contribution to the stoichiometry and energetics of

Figure 4.6 Conversion of substrate, oxygen and nitrogen for cell growth.



reaction can be neglected. Other substrates and products can easily be added if appropriate.

In Eq. (4.4), biomass is represented by the formula $CH_\alpha O_\beta N_\delta$. There is no fundamental objection to having a molecular formula for cells, even if it is not widely applied in biology. The formula is a reflection of the biomass composition. As shown in Table 4.2, microorganisms such as *Escherichia coli* contain a wide range of elements; however 90–95% of biomass can be accounted for by four major elements: C, H, O and N. Compositions of several species in terms of these four elements are listed in Table 4.3. The formulae in Table 4.3 refer to dry biomass and are based on one C atom; the total amount of biomass formed during growth can be accounted for by the stoichiometric coefficient c in Eq. (4.4). Bacteria tend to have slightly higher nitrogen contents (11–14%) than fungi (6.3–9.0%) [4]. For a particular species, cell composition depends also on culture conditions and substrate utilised, hence the different entries in Table 4.3 for the same organism. However, the results are remarkably similar for different cells and conditions; $CH_{1.8}O_{0.5}N_{0.2}$ can be used as a general formula when composition analysis is not available. The average 'molecular weight' of biomass based on C, H, O and N content is therefore 24.6, although 5–10% residual ash is often added to account for those elements not included in the formula.

Table 4.2 Elemental composition of *Escherichia coli* bacteria

(From R.Y. Stanier, E.A. Adelberg and J. Ingraham, 1976, The Microbial World, *4th edn, Prentice-Hall, New Jersey)*

Element	*% dry weight*
C	50
O	20
N	14
H	8
P	3
S	1
K	1
Na	1
Ca	0.5
Mg	0.5
Cl	0.5
Fe	0.2
All others	0.3

Eq. (4.4) is not complete unless the stoichiometric coefficients a, b, c, d and e are known. Once a formula for biomass is obtained, these coefficients can be evaluated using normal procedures for balancing equations, i.e. elemental balances and solution of simultaneous equations.

$$\text{C balance:} \quad w = c + d \tag{4.5}$$

$$\text{H balance:} \quad x + bg = c\alpha + 2e \tag{4.6}$$

$$\text{O balance:} \quad y + 2a + bh = c\beta + 2d + e \tag{4.7}$$

$$\text{N balance:} \quad z + bi = c\delta. \tag{4.8}$$

Notice that we have five unknown coefficients (a, b, c, d and e) but only four balance equations. This means that additional information is required before the equations can be solved. Usually this information is obtained from experiments. A useful measurable parameter is the *respiratory quotient* (RQ):

$$\text{respiratory quotient, } RQ = \frac{\text{moles } CO_2 \text{ produced}}{\text{moles } O_2 \text{ consumed}} = \frac{d}{a}. \tag{4.9}$$

Table 4.3 Elemental composition and degree of reduction for selected organisms

(From J.A. Roels, 1980, Application of macroscopic principles to microbial metabolism, Biotechnol. Bioeng. 22, *2457–2514)*

Organism	*Elemental formula*	*Degree of reduction γ (relative to NH_3)*
Escherichia coli	$CH_{1.77}O_{0.49}N_{0.24}$	4.07
Klebsiella aerogenes	$CH_{1.75}O_{0.43}N_{0.22}$	4.23
Kl. aerogenes	$CH_{1.73}O_{0.43}N_{0.24}$	4.15
Kl. aerogenes	$CH_{1.75}O_{0.47}N_{0.17}$	4.30
Kl. aerogenes	$CH_{1.73}O_{0.43}N_{0.24}$	4.15
Pseudomonas $C_{12}B$	$CH_{2.00}O_{0.52}N_{0.23}$	4.27
Aerobacter aerogenes	$CH_{1.83}O_{0.55}N_{0.25}$	3.98
Paracoccus denitrificans	$CH_{1.81}O_{0.51}N_{0.20}$	4.19
P. denitrificans	$CH_{1.51}O_{0.46}N_{0.19}$	3.96
Saccharomyces cerevisiae	$CH_{1.64}O_{0.52}N_{0.16}$	4.12
S. cerevisiae	$CH_{1.83}O_{0.56}N_{0.17}$	4.20
S. cerevisiae	$CH_{1.81}O_{0.51}N_{0.17}$	4.28
Candida utilis	$CH_{1.83}O_{0.54}N_{0.10}$	4.45
C. utilis	$CH_{1.87}O_{0.56}N_{0.20}$	4.15
C. utilis	$CH_{1.83}O_{0.46}N_{0.19}$	4.34
C. utilis	$CH_{1.87}O_{0.56}N_{0.20}$	4.15
Average	$CH_{1.79}O_{0.50}N_{0.20}$	4.19 (standard deviation = 3%)

When an experimental value of *RQ* is available, Eqs (4.5) to (4.9) can be solved to determine the stoichiometric coefficients. The results, however, are sensitive to small errors in *RQ*, which must be measured very accurately. When Eq. (4.4) is completed, the quantities of substrate, nitrogen and oxygen required for production of biomass can be determined directly.

Example 4.7 Stoichiometric coefficients for cell growth

Production of single-cell protein from hexadecane is described by the following reaction equation:

$$C_{16}H_{34} + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.66}O_{0.27}N_{0.20} + d\,CO_2 + e\,H_2O$$

where $CH_{1.66}O_{0.27}N_{0.20}$ represents the biomass. If $RQ = 0.43$, determine the stoichiometric coefficients.

Solution:

C balance: $16 = c + d$ (1)

H balance: $34 + 3\,b = 1.66\,c + 2\,e$ (2)

O balance: $2\,a = 0.27\,c + 2\,d + e$ (3)

N balance: $b = 0.20\,c$ (4)

RQ: $0.43 = {}^{d}/_{a}$. (5)